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Elements of Algebraic Geometry III

Part Two

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6 Local and global Tor functors; Künneth formula

6.1 Introduction

III | 137

(6.1.1) Let $f : X \to Y$ be a morphism of preschemes, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. In studying the higher direct images $\mathrm{R}^n f_*(\mathscr{F})$, one is led to consider the following general problem. Given a base-change morphism $g : Y' \to Y$, put

$$X' = X_{(Y')} = X \times_Y Y', \qquad \mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}, \qquad f' = f_{(Y')} : X' \to Y'.$$

We seek information about the higher direct images $\mathrm{R}^n f'_*(\mathscr{F}')$, assuming that the $\mathrm{R}^n f_*(\mathscr{F})$ are known. One sees readily (cf. for example (III.7.7.2)) that this reduces to studying the variation of

$$\mathrm{R}^n f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}\big)$$

as $\mathscr{G}$ ranges over the quasi-coherent $\mathscr{O}_Y$-modules; in other words, to studying the functor

$$\mathscr{G} \longmapsto \mathrm{R}^n f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}\big).$$

If $\mathscr{F}$ is flat over Y, the functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is exact (0.6.7.4), and consequently the composite functors

$$\mathscr{G} \longmapsto \mathrm{R}^n f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}\big)$$

again form a cohomological functor. This is no longer true in general. In order to apply cohomological methods, one is therefore led to replace $\mathscr{G} \mapsto \mathrm{R}^n f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G})$ by other functors which are always cohomological.

These functors, which generalize the Tor functors of module theory, are defined in Sections 6.3–6.7. There are in fact two such generalizations, one local and the other global, related by spectral sequences discussed in Section 6.7. As an application of those spectral sequences one obtains, under suitable hypotheses, a Künneth formula expressing

$$\mathrm{R}^n(f_1 \times f_2)_*\big(\mathscr{F}_1 \otimes_Y \mathscr{F}_2\big)$$

in terms of the higher direct images $\mathrm{R}^p(f_1)_*(\mathscr{F}_1)$ and $\mathrm{R}^q(f_2)_*(\mathscr{F}_2)$. Other spectral sequences (III.6.8) generalize the associativity spectral sequences for the Tor functor of modules; finally, the base-change problem also leads to spectral sequences (III.6.9).

(6.1.2) Moreover, one observes in (III.6.10) that the cohomological functors

$$\mathscr{G} \longmapsto \mathscr{T}_\bullet(\mathscr{G})$$

so defined are locally on Y of the form

$$\mathscr{G} \longmapsto \mathscr{H}_\bullet\big(\mathscr{L}_\bullet \otimes_Y \mathscr{G}\big),$$

where $\mathscr{L}_\bullet$ is a complex of locally free $\mathscr{O}_Y$-modules, defined up to homotopy, and $\mathscr{H}_\bullet$ denotes homology. It is therefore useful to set aside the particular situation which produced $\mathscr{T}_\bullet$ and to study, in general, functors of the preceding form, imposing when necessary appropriate finiteness hypotheses on the $\mathscr{L}_i$ or on the $\mathscr{H}_i(\mathscr{L}_\bullet)$. This is the subject of Section 7, which can for the most part be read independently of Section 6.

III | 138

The most important properties of these functors concern exactness of a component $\mathscr{T}_i$ of $\mathscr{T}_\bullet$. We shall give various criteria for establishing such properties. As an application, we shall obtain conditions under which, with the notation of (III.6.1.1), the functor

$$\mathscr{G} \longmapsto \mathrm{R}^n f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}\big)$$

is exact; we shall express this by saying that $\mathscr{F}$ is *cohomologically flat over Y in dimension n*. Another important property of the components $\mathscr{T}_i$ of

$$\mathscr{T}_\bullet(\mathscr{G}) = \mathscr{H}_\bullet\big(\mathscr{L}_\bullet \otimes_Y \mathscr{G}\big)$$

is semicontinuity of the function

$$y \longmapsto \dim_{\kappa(y)}\big(T_i(\kappa(y))\big).$$

When $\mathscr{T}_i$ is exact, semicontinuity is replaced by continuity; by a theorem of Grauert the converse also holds when Y is reduced (III.7.8.4).

(6.1.3) In Sections 6 and 7 we systematically use hypercohomology, taking complexes of sheaves rather than sheaves as arguments throughout, although the necessity of this point of view will become apparent only in

later chapters. The cohomological formalism developed here will moreover become more transparent in the chapter of this Treatise devoted to an algebra of cohomological functors of coherent sheaves, including the formalism of duality. This, however, will require developments beyond the scope of the present chapter.

(6.1.4) For brevity, let $K^\bullet$ and $K'^\bullet$ be two complexes in an abelian category C. We shall call a morphism of complexes

$$f : K^\bullet \longrightarrow K'^\bullet$$

a *homotopism* if there is a morphism $g : K'^\bullet \to K^\bullet$ such that both composites $f \circ g$ and $g \circ f$ are homotopic to the identity. By abuse of language, when such a homotopism exists we shall also say that $K^\bullet$ and $K'^\bullet$ are homotopic.

Suppose that the hypercohomology of a covariant additive functor $T : \mathsf{C} \to \mathsf{C}'$, with values in an abelian category C', can be defined relative to a complex of C (0.11.4.3). It is then immediate that a homotopism $K^\bullet \to K'^\bullet$ canonically defines an isomorphism

$$\mathbf{R}^\bullet T(K^\bullet) \xrightarrow{\sim} \mathbf{R}^\bullet T(K'^\bullet)$$

on hypercohomology *(loc. cit.)*.

6.2 Hypercohomology of complexes of modules on a prescheme

(6.2.1) Let X be a prescheme, and let $\mathscr{K}^\bullet = (\mathscr{K}^i)_{i\in\mathbf{Z}}$ be a complex of $\mathscr{O}_X$-modules whose differential has degree $+1$. Recall that, for every morphism $f : X \to Y$ of preschemes, the $\mathscr{O}_Y$-modules of hypercohomology $\mathscr{H}^n(f, \mathscr{K}^\bullet)$, also denoted $\mathscr{H}^n_f(\mathscr{K}^\bullet)$ or $\mathrm{R}^n f_*(\mathscr{K}^\bullet)$, have been defined for every $n \in \mathbf{Z}$ (0.12.4.1). The hypercohomology $\mathscr{H}^\bullet(f, \mathscr{K}^\bullet)$ is the common abutment of the two spectral functors ${}'\mathscr{E}(f, \mathscr{K}^\bullet)$ and ${}''\mathscr{E}(f, \mathscr{K}^\bullet)$ whose E_2-terms are

$${}'\mathscr{E}_2^{pq} = \mathscr{H}^p\big(\mathscr{H}^q(f, \mathscr{K}^\bullet)\big), \tag{6.2.1.1}$$

and

$${}''\mathscr{E}_2^{pq} = \mathscr{H}^p\big(f, \mathscr{H}^q(\mathscr{K}^\bullet)\big) = \mathrm{R}^p f_*\big(\mathscr{H}^q(\mathscr{K}^\bullet)\big). \tag{6.2.1.2}$$

Here $\mathscr{H}^q(f, \mathscr{K}^\bullet)$ is the complex whose component of degree i is III | 139

$$\mathscr{H}^q(f, \mathscr{K}^i) = \mathrm{R}^q f_*(\mathscr{K}^i)$$

(loc. cit.). Recall also that, when Y is reduced to a point, the corresponding hypercohomology is denoted $\mathbf{H}^\bullet(X, \mathscr{K}^\bullet)$; it consists of modules over $\Gamma(X, \mathscr{O}_X)$ and is independent of the one-point prescheme Y chosen. When $Y = X$ and $f = 1_X$,

$$\mathscr{H}^n(f, \mathscr{K}^\bullet) = \mathscr{H}^n(\mathscr{K}^\bullet),$$

the cohomology of the complex $\mathscr{K}^\bullet$. When $\mathscr{K}^i = 0$ except for $i = i_0$, one has

$$\mathscr{H}^n(f, \mathscr{K}^\bullet) = \mathrm{R}^{n-i_0} f_*(\mathscr{K}^{i_0}).$$

The spectral sequence ${}'\mathscr{E}(f, \mathscr{K}^\bullet)$ is always regular; both spectral sequences are biregular when $\mathscr{K}^\bullet$ is bounded below (0.12.4.1). Every homotopism $h : \mathscr{K}^\bullet \to \mathscr{K}'^\bullet$ of complexes of $\mathscr{O}_X$-modules (III.6.1.4) induces an isomorphism

$$\mathscr{H}^\bullet(f, \mathscr{K}^\bullet) \xrightarrow{\sim} \mathscr{H}^\bullet(f, \mathscr{K}'^\bullet)$$

on hypercohomology. The same conclusion holds if one assumes only that

$$\mathscr{H}^\bullet(h) : \mathscr{H}^\bullet(\mathscr{K}^\bullet) \xrightarrow{\sim} \mathscr{H}^\bullet(\mathscr{K}'^\bullet)$$

is an isomorphism and that both complexes are bounded below. This follows immediately from (0.11.1.5), applied to the spectral sequence (III.6.2.1.2) and its analogue for $\mathscr{K}'^\bullet$.

Finally, for every open covering $\mathfrak{U} = (U_\alpha)$ of X, the hypercohomology $\mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$ has also been defined (0.12.4.5) as the cohomology of the bicomplex $\mathrm{C}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$, whose component of bidegree (i, j) is by definition $\mathrm{C}^i(\mathfrak{U}, \mathscr{K}^j)$. The modules $\mathbf{H}^n(\mathfrak{U}, \mathscr{K}^\bullet)$ are again modules over $\Gamma(X, \mathscr{O}_X)$.

Proposition (6.2.2). — *Let X be a scheme, and let $\mathfrak{U} = (U_\alpha)$ be a covering of X by affine open subsets. For every complex $\mathscr{K}^\bullet$ of quasi-coherent $\mathscr{O}_X$-modules, the hypercohomology modules*

$$\mathbf{H}^\bullet(X, \mathscr{K}^\bullet) \quad \textit{and} \quad \mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$$

are canonically isomorphic.

Proof. — Every finite intersection V of members of $\mathfrak{U}$ is affine (I.5.5.6); hence $\mathrm{H}^q(V, \mathscr{K}^i) = 0$ for every i and every $q > 0$ (III.1.3.1). The proposition is therefore a special case of (0.12.4.7).

Proposition (6.2.3). — *Let $f : X \to Y$ be a quasi-compact separated morphism of preschemes. For every complex $\mathscr{K}^\bullet$ of quasi-coherent $\mathscr{O}_X$-modules, the $\mathscr{O}_Y$-modules $\mathscr{H}^n(f, \mathscr{K}^\bullet)$ are quasi-coherent.*

Proof. — The modules

$$\mathscr{H}^q(f, \mathscr{K}^i) = \mathrm{R}^q f_*(\mathscr{K}^i)$$

are quasi-coherent $\mathscr{O}_Y$-modules (III.1.4.10). Hence so is ${}'\mathscr{E}_2^{pq}$: by (III.6.2.1.1), it is the quotient of the kernel of a homomorphism of quasi-coherent modules by the image of such a homomorphism (I.4.1.1). For the same reason, all terms, cycles, and boundaries of the first spectral sequence are quasi-coherent. Regularity implies that $\mathrm{Z}_\infty({}'\mathscr{E}_2^{pq})$ equals one of the $\mathrm{Z}_k({}'\mathscr{E}_2^{pq})$ and is therefore quasi-coherent. Likewise,

$$\mathrm{B}_\infty({}'\mathscr{E}_2^{pq}) = \varinjlim_k \mathrm{B}_k({}'\mathscr{E}_2^{pq})$$

is quasi-coherent (0.11.2.4), (I.4.1.1); thus the ${}'\mathscr{E}_\infty^{pq}$ are quasi-coherent as well.

Since the spectral sequence is regular, the filtration $F^\bullet\mathscr{H}^n(f, \mathscr{K}^\bullet)$ is discrete and exhaustive. Equivalently, $\mathscr{H}^n(f, \mathscr{K}^\bullet)$ is the union of an increasing sequence $(\mathscr{G}_k)_{k\geq 0}$ of $\mathscr{O}_Y$-modules such that $\mathscr{G}_0 = 0$ and every quotient $\mathscr{G}_k/\mathscr{G}_{k-1}$ is one of the ${}'\mathscr{E}_\infty^{pq}$, and hence is quasi-coherent. It follows inductively that the $\mathscr{G}_k$ are quasi-coherent (III.1.4.17). Since

$$\mathscr{H}^n(f, \mathscr{K}^\bullet) = \varinjlim_k \mathscr{G}_k,$$

the proposition follows from (I.4.1.1).

Corollary (6.2.4). — *Under the hypotheses of (III.6.2.3), for every affine open subset V of Y, the canonical homomorphism* III | 140

$$\mathbf{H}^n\big(f^{-1}(V), \mathscr{K}^\bullet\big) \longrightarrow \Gamma\big(V, \mathscr{H}^n(f, \mathscr{K}^\bullet)\big) \tag{6.2.4.1}$$

is bijective for every $n \in \mathbf{Z}$.

Proof. — The proof is the same as that of (III.1.4.11), using (III.6.2.2): one replaces the sheaf there by $\mathscr{K}^\bullet$, uses the Cech bicomplex $f_*\mathrm{C}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$, and replaces ordinary cohomology by $\mathscr{H}^\bullet(f, \mathscr{K}^\bullet)$. The latter is a quasi-coherent $\mathscr{O}_Y$-module by (III.6.2.3).

Proposition (6.2.5). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be proper, and let $\mathscr{K}^\bullet$ be a complex of $\mathscr{O}_X$-modules such that the $\mathscr{O}_X$-modules $\mathscr{H}^q(\mathscr{K}^\bullet)$ are coherent. Then the $\mathscr{O}_Y$-modules $\mathscr{H}^n(f, \mathscr{K}^\bullet)$ are coherent.*

Proof. — The question is local on Y, so we may suppose that Y is Noetherian and affine. By (III.6.2.4), it is enough to prove that the $\mathbf{H}^n(X, \mathscr{K}^\bullet)$ are finite-type modules over $\Gamma(Y, \mathscr{O}_Y)$ (III.3.2.1). By (III.6.2.2),

$$\mathbf{H}^\bullet(X, \mathscr{K}^\bullet) = \mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet),$$

where $\mathfrak{U}$ may be taken finite, since X is quasi-compact. The cochains in each complex $\mathrm{C}^\bullet(\mathfrak{U}, \mathscr{K}^j)$ are alternating by definition. Hence there is an integer $r > 0$ such that

$$\mathrm{C}^i(\mathfrak{U}, \mathscr{K}^j) = 0 \qquad \text{for } i < 0 \text{ and } i > r.$$

It follows from (0.11.3.3) that both spectral sequences of the bicomplex $\mathrm{C}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$ are biregular.

Intersections of members of $\mathfrak{U}$ are affine open subsets (I.5.5.6), so each functor

$$\mathscr{F} \longmapsto \mathrm{C}^i(\mathfrak{U}, \mathscr{F})$$

is exact in the category of quasi-coherent $\mathscr{O}_X$-modules. Consequently the E_2-terms of the second spectral sequence are

$${}''E_2^{pq} = \mathrm{H}^p\big(\mathrm{C}^\bullet(\mathfrak{U}, \mathscr{H}^q(\mathscr{K}^\bullet))\big) = \mathbf{H}^p\big(\mathfrak{U}, \mathscr{H}^q(\mathscr{K}^\bullet)\big) = \mathrm{H}^p\big(X, \mathscr{H}^q(\mathscr{K}^\bullet)\big)$$

(0.11.3.2), (III.1.4.1). Since f is proper, these are finite-type $\Gamma(Y, \mathscr{O}_Y)$-modules (III.3.2.1). Biregularity now implies that the $\mathbf{H}^n(X, \mathscr{K}^\bullet)$ are finite-type $\Gamma(Y, \mathscr{O}_Y)$-modules (0.11.1.8).

A fortiori, if $\mathscr{K}^\bullet$ is a complex of coherent $\mathscr{O}_X$-modules, then the $\mathscr{H}^n(f, \mathscr{K}^\bullet)$ are coherent under the hypotheses on Y and f in (III.6.2.5) (0.5.3.4).

(6.2.6) The hypercohomology $\mathscr{H}^\bullet(f, \mathscr{K}^\bullet)$ is a cohomological functor on the category of bounded-below complexes of $\mathscr{O}_X$-modules (0.12.4.4). It is a cohomological functor on the category of all complexes of $\mathscr{O}_X$-modules when f is quasi-compact and the underlying space of X is locally Noetherian. Indeed, f_* then commutes with inductive limits *(G, II, 3.10.1)*—the question being local on Y—and (0.11.5.2) applies.

Finally, if f is separated, $\mathcal{H}^\bullet(f, \mathscr{K}^\bullet)$ is a cohomological functor on the category of complexes of quasi-coherent $\mathscr{O}_X$-modules. This is immediate when Y is affine, for then X is a scheme and the canonical isomorphism (III.6.2.2) reduces the assertion to the fact that

$$\mathscr{K}^\bullet \longmapsto \mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$$

is cohomological. This in turn is immediate because

$$\mathscr{K}^\bullet \longmapsto \mathrm{C}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$$

is exact in this category (I.1.3.7).

In general, for every affine open subset V of Y, $f^{-1}(V)$ is a scheme. To apply the preceding argument, it is enough to verify that, for an exact sequence **III** | 141

$$0 \longrightarrow \mathscr{K}'^\bullet \longrightarrow \mathscr{K}^\bullet \longrightarrow \mathscr{K}''^\bullet \longrightarrow 0$$

of complexes of quasi-coherent $\mathscr{O}_X$-modules, the connecting homomorphism

$$\partial : \mathcal{H}^n(f, \mathscr{K}''^\bullet)|V \longrightarrow \mathcal{H}^{n+1}(f, \mathscr{K}'^\bullet)|V$$

does not depend on the affine open covering $\mathfrak{U}$ of $f^{-1}(V)$ used to define it. If $\mathfrak{U}'$ is an affine open refinement of $\mathfrak{U}$, this follows from commutativity of the triangle of canonical isomorphisms

$$\begin{array}{ccc} \mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet) & \xrightarrow{\sim} & \\ \downarrow{\scriptstyle \iota} & & \mathbf{H}^\bullet(f^{-1}(V), \mathscr{K}^\bullet) \\ \mathbf{H}^\bullet(\mathfrak{U}', \mathscr{K}^\bullet) & \xrightarrow{\sim} & \end{array}$$

and of the diagram

$$\begin{array}{ccc} \mathbf{H}^n(\mathfrak{U}, \mathscr{K}''^\bullet) & \xrightarrow{\partial} & \mathbf{H}^{n+1}(\mathfrak{U}, \mathscr{K}'^\bullet) \\ \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \iota} \\ \mathbf{H}^n(\mathfrak{U}', \mathscr{K}''^\bullet) & \xrightarrow[\partial]{} & \mathbf{H}^{n+1}(\mathfrak{U}', \mathscr{K}'^\bullet). \end{array}$$

Whenever one of the preceding conditions holds and $\mathscr{K}^\bullet \to \mathscr{K}'^\bullet$ is a homotopism (III.6.1.4), the resulting isomorphism

$$\mathcal{H}^\bullet(f, \mathscr{K}^\bullet) \xrightarrow{\sim} \mathcal{H}^\bullet(f, \mathscr{K}'^\bullet)$$

is an isomorphism of ∂-functors (0.11.4.4).

(6.2.7) Everything above applies naturally, with only a change of notation, to a complex $\mathscr{K}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules whose differential has degree -1: it is enough to consider the complex $\mathscr{K}^\bullet = (\mathscr{K}^i)$ defined by

$$\mathscr{K}^i = \mathscr{K}_{-i} \qquad (i \in \mathbf{Z}).$$

6.3 Hypertor of two complexes of modules

(6.3.1) Let A be a commutative ring, and let $P_\bullet$ and $Q_\bullet$ be two complexes of A-modules whose differentials have degree -1. Let $L_{\bullet\bullet}$ and $M_{\bullet\bullet}$ be projective Cartan–Eilenberg resolutions of $P_\bullet$ and $Q_\bullet$, respectively (0.11.6.1). Taking the sum of the first degrees and the sum of the second degrees,

$$L_{\bullet\bullet} \otimes_A M_{\bullet\bullet}$$

is a bicomplex whose differentials have degree -1. Its homology

$$\mathrm{H}_\bullet(L_{\bullet\bullet} \otimes_A M_{\bullet\bullet})$$

does not depend on the chosen Cartan–Eilenberg resolutions and is, by definition, the hyperhomology of the bifunctor $P_\bullet \otimes_A Q_\bullet$ in $P_\bullet$ and $Q_\bullet$ (0.11.6.5). We put

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) = \mathrm{H}_n(L_{\bullet\bullet} \otimes_A M_{\bullet\bullet}) \tag{6.3.1.1}$$

and call this A-module the *hypertor of index n* of the two complexes $P_\bullet$ and $Q_\bullet$. In the category of bounded-below complexes of A-modules, the modules $\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$ form a homological bifunctor in $P_\bullet$ and $Q_\bullet$ (0.11.6.5). Moreover:

Proposition (6.3.2). — *The bifunctor $\mathbf{Tor}_\bullet^A(P_\bullet, Q_\bullet)$ is the common abutment of two bifunctorial spectral sequences ${}'E(P_\bullet, Q_\bullet)$ and ${}''E(P_\bullet, Q_\bullet)$ whose E_2-terms are*

$$'E^2_{pq} = \mathrm{H}_p\big(\mathrm{Tor}_q^A(P_\bullet, Q_\bullet)\big), \tag{6.3.2.1}$$

and

$$''E^2_{pq} = \bigoplus_{q'+q''=q} \mathrm{Tor}_p^A\big(\mathrm{H}_{q'}(P_\bullet), \mathrm{H}_{q''}(Q_\bullet)\big). \tag{6.3.2.2}$$

In (III.6.3.2.1), $\mathrm{Tor}_q^A(P_\bullet, Q_\bullet)$ denotes the bicomplex formed by the A-modules $\mathrm{Tor}_q^A(P_i, Q_j)$. The spectral sequence (III.6.3.2.2) is always regular. If $P_\bullet$ and $Q_\bullet$ are bounded below, or if A has finite cohomological dimension, then both spectral sequences (III.6.3.2.1) and (III.6.3.2.2) are biregular.

Proof. — This follows from (0.11.6.5). Indeed, if A has finite cohomological dimension n, every A-module has a projective resolution of length n *(M, VI, 2.1).*

Corollary (6.3.3). — *Let $P'_\bullet$ and $Q'_\bullet$ be complexes of A-modules, and let*

$$u : P_\bullet \to P'_\bullet, \qquad v : Q_\bullet \to Q'_\bullet$$

be homomorphisms of complexes. If the induced homomorphisms

$$\mathrm{H}_\bullet(u) : \mathrm{H}_\bullet(P_\bullet) \longrightarrow \mathrm{H}_\bullet(P'_\bullet), \qquad \mathrm{H}_\bullet(v) : \mathrm{H}_\bullet(Q_\bullet) \longrightarrow \mathrm{H}_\bullet(Q'_\bullet)$$

are bijective, then the homomorphism

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}_n^A(P'_\bullet, Q'_\bullet)$$

induced by u and v is bijective.

Proof. — The induced homomorphism

$$''E(P_\bullet, Q_\bullet) \longrightarrow'' E(P'_\bullet, Q'_\bullet)$$

is an isomorphism on the E_2-terms. The conclusion follows from regularity of these spectral sequences (III.6.3.2) and (0.11.1.5).

Proposition (6.3.4). — *Let $P_\bullet$ and $Q_\bullet$ be bounded-below complexes of A-modules. Let $L_{\bullet\bullet}$ and $M_{\bullet\bullet}$ be bicomplexes of flat A-modules such that, for every i, $L_{i,\bullet}$ is a resolution of P_i and $M_{i,\bullet}$ is a resolution of Q_i. Then there are canonical isomorphisms*

$$\begin{aligned} \mathbf{Tor}_n^A(P_\bullet, Q_\bullet) &\xrightarrow{\sim} \mathrm{H}_n(L_{\bullet\bullet} \otimes_A Q_\bullet) \\ &\xrightarrow{\sim} \mathrm{H}_n(P_\bullet \otimes_A M_{\bullet\bullet}) \\ &\xrightarrow{\sim} \mathrm{H}_n(L_{\bullet\bullet} \otimes_A M_{\bullet\bullet}). \end{aligned} \tag{6.3.4.1}$$

Proof. — This follows from (0.11.6.5), parts (ii) and (iii), and from the definition of flat A-modules.

Remarks (6.3.5). — (i) *With the notation of (III.6.3.1), the bicomplexes $L_{\bullet\bullet} \otimes_A M_{\bullet\bullet}$ and $M_{\bullet\bullet} \otimes_A L_{\bullet\bullet}$ are canonically isomorphic. Hence there is a canonical isomorphism*

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \xrightarrow{\sim} \mathbf{Tor}_n^A(Q_\bullet, P_\bullet).$$

(ii) *Let F and G be A-modules, and let $P_\bullet$ and $Q_\bullet$ be the complexes concentrated in degree 0 with values F and G, respectively. Projective resolutions $L_\bullet$ of F and $M_\bullet$ of G may be regarded, after adjoining zeros, as Cartan–Eilenberg resolutions of $P_\bullet$ and $Q_\bullet$. Thus in this case*

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) = \mathrm{Tor}_n^A(F, G).$$

Proposition (6.3.6). — *Let $(P_\bullet^\lambda)$ and $(Q_\bullet^\mu)$ be two filtered inductive systems of complexes of A-modules. There is a canonical isomorphism*

$$\varinjlim_{\lambda,\mu} \mathbf{Tor}_n^A(P_\bullet^\lambda, Q_\bullet^\mu) \xrightarrow{\sim} \mathbf{Tor}_n^A\left(\varinjlim_\lambda P_\bullet^\lambda, \varinjlim_\mu Q_\bullet^\mu\right). \tag{6.3.6.1}$$

Proof. — Put

$$P_\bullet = \varinjlim_{\lambda} P_\bullet^\lambda, \qquad Q_\bullet = \varinjlim_{\mu} Q_\bullet^\mu.$$

By functoriality, the modules $\mathbf{Tor}_n^A(P_\bullet^\lambda, Q_\bullet^\mu)$ form an inductive system, and the canonical maps to $\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$ **III** | 143 give the homomorphism (III.6.3.6.1). More generally, they give a canonical homomorphism of spectral sequences

$$\varinjlim_{\lambda,\mu}{}'' E(P_\bullet^\lambda, Q_\bullet^\mu) \longrightarrow'' E(P_\bullet, Q_\bullet),$$

whose homomorphism on abutments is (III.6.3.6.1).

The spectral sequence $''E(P_\bullet, Q_\bullet)$ is regular by (III.6.3.2). So is $\varinjlim''_{\lambda,\mu} E(P_\bullet^\lambda, Q_\bullet^\mu)$, as follows from the definitions (0.11.1.7) and the proof of (0.11.3.3). By (0.11.1.5), it is therefore enough to prove that

$$\varinjlim_{\lambda,\mu}{}'' E(P_\bullet^\lambda, Q_\bullet^\mu) \longrightarrow'' E(P_\bullet, Q_\bullet) \tag{6.3.6.2}$$

is bijective on the E^2-terms.

Since homology commutes with inductive limits of complexes of modules, it remains only to prove that, for filtered inductive systems (F^λ) and (G^μ) of A-modules, the canonical homomorphism

$$\varinjlim_{\lambda,\mu} \mathrm{Tor}_n^A(F^\lambda, G^\mu) \longrightarrow \mathrm{Tor}_n^A\left(\varinjlim_{\lambda} F^\lambda, \varinjlim_{\mu} G^\mu\right)$$

is bijective.

For each F^λ, consider its canonical free resolution

$$L_\bullet^\lambda : \quad \cdots \longrightarrow L_{i+1}^\lambda \longrightarrow L_i^\lambda \longrightarrow \cdots \longrightarrow L_1^\lambda \longrightarrow L_0^\lambda \longrightarrow 0,$$

where L_0^λ is the A-module of formal linear combinations of elements of F^λ, and L_{i+1}^λ is the A-module of formal linear combinations of elements of $\mathrm{Ker}(L_i^\lambda \to L_{i-1}^\lambda)$. The $L_\bullet^\lambda$ form an inductive system of complexes. Put

$$F = \varinjlim_{\lambda} F^\lambda, \qquad L_i = \varinjlim_{\lambda} L_i^\lambda.$$

Since filtered inductive limits are exact, the L_i form a resolution $L_\bullet$ of F; moreover, each L_i is flat, being an inductive limit of free A-modules (0.6.1.2).

Likewise, let $M_\bullet^\mu$ be the canonical free resolution of G^μ, and put

$$M_\bullet = \varinjlim_{\mu} M_\bullet^\mu, \qquad G = \varinjlim_{\mu} G^\mu.$$

Then $M_\bullet$ is a flat resolution of G. By (III.6.3.5) and (III.6.3.4),

$$\mathrm{Tor}_n^A(F, G) = \mathrm{H}_n(L_\bullet \otimes_A M_\bullet).$$

Since homology commutes with inductive limits of complexes of modules,

$$\mathrm{H}_n(L_\bullet \otimes_A M_\bullet) = \varinjlim_{\lambda,\mu} \mathrm{H}_n(L_\bullet^\lambda \otimes_A M_\bullet^\mu).$$

The last modules are $\mathrm{Tor}_n^A(F^\lambda, G^\mu)$, which proves (III.6.3.6.1).

If, in addition, there is an i_0 such that $P_i^\lambda = Q_i^\mu = 0$ for $i < i_0$, for all λ and μ, the same argument proves that the canonical homomorphism

$$\varinjlim_{\lambda,\mu}{}' E(P_\bullet^\lambda, Q_\bullet^\mu) \longrightarrow' E(P_\bullet, Q_\bullet) \tag{6.3.6.3}$$

is bijective.

Proposition (6.3.7). — *Suppose that $P_\bullet$ and $Q_\bullet$ are bounded below. If $P_\bullet$ consists of flat A-modules, there is a canonical A-linear isomorphism of ∂-functors in $Q_\bullet$*

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \xrightarrow{\sim} \mathrm{H}_n(P_\bullet \otimes_A Q_\bullet). \tag{6.3.7.1}$$

Proof. — The spectral sequence (III.6.3.2.1) is biregular and degenerate, so existence of (III.6.3.7.1) follows from (0.11.1.6). Computing hypertor from a projective Cartan–Eilenberg resolution of $P_\bullet$ as in (III.6.3.4) shows immediately that this isomorphism is an isomorphism of ∂-functors in $Q_\bullet$.

(6.3.8) Let $\rho : A \longrightarrow A'$ be a homomorphism of rings. We shall define a functorial A-linear homomorphism of degree zero, canonically associated with ρ,

$$\rho_{P_\bullet,Q_\bullet} : \mathbf{Tor}_\bullet^A(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}_\bullet^{A'}(P_\bullet \otimes_A A', Q_\bullet \otimes_A A'). \tag{6.3.8.1}$$

For this purpose, let $L_{\bullet\bullet}$ be a projective Cartan–Eilenberg resolution of $P_\bullet$, and let $L'_{\bullet\bullet}$ be a projective Cartan–Eilenberg resolution of $P_\bullet \otimes_A A'$. We first show that there is an A'-linear homomorphism of complexes

$$L_{\bullet\bullet} \otimes_A A' \longrightarrow L'_{\bullet\bullet},$$

determined up to homotopy. Indeed, the construction of $L_{\bullet\bullet}$ is completely determined once one chooses, arbitrarily for each i, a projective resolution $(X^{\mathrm{B}}_{ij})_{j\geq 0}$ of $\mathrm{B}_i(P_\bullet)$ and a projective resolution $(X^{\mathrm{H}}_{ij})_{j\geq 0}$ of $\mathrm{H}_i(P_\bullet)$. These are respectively $\mathrm{B}^{\mathrm{I}}_i(L_{\bullet\bullet})$ and $\mathrm{H}^{\mathrm{I}}_i(L_{\bullet\bullet})$. One then obtains successively

$$\mathrm{Z}^{\mathrm{I}}_i(L_{\bullet\bullet}) = \mathrm{H}^{\mathrm{I}}_i(L_{\bullet\bullet}) \oplus \mathrm{B}^{\mathrm{I}}_i(L_{\bullet\bullet}), \qquad L_{i,\bullet} = \mathrm{Z}^{\mathrm{I}}_i(L_{\bullet\bullet}) \oplus \mathrm{B}^{\mathrm{I}}_{i-1}(L_{\bullet\bullet}).$$

Now $X^{\mathrm{B}}_{i,\bullet} \otimes_A A'$ is not in general a resolution of $\mathrm{B}_i(P_\bullet) \otimes_A A'$, but it is still a complex of projective A'-modules. Hence there is an A'-linear homomorphism

$$X^{\mathrm{B}}_{i,\bullet} \otimes_A A' \longrightarrow \mathrm{B}^{\mathrm{I}}_i(L'_{\bullet\bullet})$$

compatible with the augmentations and determined up to homotopy (M, V, 1.1). Likewise there is an A'-linear homomorphism

$$X^{\mathrm{H}}_{i,\bullet} \otimes_A A' \longrightarrow \mathrm{H}^{\mathrm{I}}_i(L'_{\bullet\bullet})$$

determined up to homotopy. By the construction just recalled, these maps give, for every i, an A'-linear homomorphism

$$L_{i,\bullet} \otimes_A A' \longrightarrow L'_{i,\bullet}.$$

For $i \in \mathbb{Z}$ these homomorphisms are compatible with the differentials $L_{i,\bullet} \longrightarrow L_{i-1,\bullet}$ and their analogues for $L'_{\bullet\bullet}$; together they therefore give the required homomorphism $L_{\bullet\bullet} \otimes_A A' \longrightarrow L'_{\bullet\bullet}$.

To define (III.6.3.8.1), take in the same way projective Cartan–Eilenberg resolutions $M_{\bullet\bullet}$ and $M'_{\bullet\bullet}$ of $Q_\bullet$ and $Q_\bullet \otimes_A A'$, respectively, and an A'-linear homomorphism $M_{\bullet\bullet} \otimes_A A' \longrightarrow M'_{\bullet\bullet}$. The resulting homomorphism

$$(L_{\bullet\bullet} \otimes_A A') \otimes_{A'} (M_{\bullet\bullet} \otimes_A A') \longrightarrow L'_{\bullet\bullet} \otimes_{A'} M'_{\bullet\bullet}$$

gives, by composition, an A-linear homomorphism of bicomplexes

$$L_{\bullet\bullet} \otimes_A M_{\bullet\bullet} \longrightarrow L'_{\bullet\bullet} \otimes_{A'} M'_{\bullet\bullet}.$$

Passing to homology yields (III.6.3.8.1); it is well defined because it is induced by a homomorphism of complexes determined up to homotopy.

If $\rho' : A' \longrightarrow A''$ is a second homomorphism of rings and $\rho'' = \rho' \circ \rho : A \longrightarrow A''$, then

$$\rho''_{P_\bullet,Q_\bullet} = \rho'_{P'_\bullet,Q'_\bullet} \circ \rho_{P_\bullet,Q_\bullet}, \qquad P'_\bullet = P_\bullet \otimes_A A', \quad Q'_\bullet = Q_\bullet \otimes_A A'.$$

The preceding homomorphism of bicomplexes also defines functorial homomorphisms of spectral sequences

$${}'E^r_{pq}(P_\bullet, Q_\bullet) \longrightarrow {}' E^r_{pq}(P_\bullet \otimes_A A', Q_\bullet \otimes_A A')$$

and

$${}''E^r_{pq}(P_\bullet, Q_\bullet) \longrightarrow {}'' E^r_{pq}(P_\bullet \otimes_A A', Q_\bullet \otimes_A A').$$

They are independent of the Cartan–Eilenberg resolutions chosen and satisfy the same transitivity property.

Proposition (6.3.9). — *Let $\rho : A \longrightarrow A'$ be a homomorphism of rings such that A' is flat as an A-module. There are canonical functorial isomorphisms*

$$\mathbf{Tor}_\bullet^{A'}(P_\bullet \otimes_A A', Q_\bullet \otimes_A A') \xrightarrow{\sim} \mathbf{Tor}_\bullet^A(P_\bullet, Q_\bullet) \otimes_A A', \tag{6.3.9.1}$$

$$\begin{cases} {}'E(P_\bullet \otimes_A A', Q_\bullet \otimes_A A') \xrightarrow{\sim}{}' E(P_\bullet, Q_\bullet) \otimes_A A', \\ {}''E(P_\bullet \otimes_A A', Q_\bullet \otimes_A A') \xrightarrow{\sim}{}'' E(P_\bullet, Q_\bullet) \otimes_A A'. \end{cases} \tag{6.3.9.2}$$

Proof. — Since the functor $M \longmapsto M \otimes_A A'$ is exact, $L_{\bullet\bullet} \otimes_A A'$ and $M_{\bullet\bullet} \otimes_A A'$ are projective Cartan–Eilenberg resolutions of $P_\bullet \otimes_A A'$ and $Q_\bullet \otimes_A A'$, respectively. The assertion follows. III | 145

(6.3.10) Let $\rho : A \longrightarrow A'$ be a homomorphism of rings. For every complex $P'_\bullet$ of A'-modules, let $P'_{\bullet[\rho]}$ denote the same complex regarded as a complex of A-modules. The identity map $P'_{\bullet[\rho]} \longrightarrow P'_\bullet$ is the composite of the canonical maps

$$P'_{\bullet[\rho]} \longrightarrow P'_{\bullet[\rho]} \otimes_A A' \xrightarrow{\mu} P'_\bullet,$$

where μ is the A'-linear homomorphism $\mu(x \otimes a') = a'x$. If $Q'_\bullet$ is a second complex of A'-modules, there are therefore canonical functorial homomorphisms of degree zero

$$\begin{aligned} \mathbf{Tor}^A_\bullet(P'_{\bullet[\rho]}, Q'_{\bullet[\rho]}) &\longrightarrow \mathbf{Tor}^{A'}_\bullet(P'_{\bullet[\rho]} \otimes_A A', Q'_{\bullet[\rho]} \otimes_A A') \\ &\longrightarrow \mathbf{Tor}^{A'}_\bullet(P'_\bullet, Q'_\bullet). \end{aligned} \tag{6.3.10.1}$$

The first arrow is the A-linear homomorphism defined in (III.6.3.8); the second follows by functoriality from the A'-linear homomorphisms

$$P'_{\bullet[\rho]} \otimes_A A' \longrightarrow P'_\bullet, \qquad Q'_{\bullet[\rho]} \otimes_A A' \longrightarrow Q'_\bullet.$$

There are analogous homomorphisms for the spectral sequences of (III.6.3.2), together with the evident transitivity properties, whose formulation is left to the reader.

Proposition (6.3.11). — *Let $\rho : A \longrightarrow A'$ be a homomorphism of rings such that A' is flat as an A-module. For every complex $P'_\bullet$ of A'-modules and every bounded-below complex $Q_\bullet$ of A-modules, there is a canonical functorial isomorphism*

$$\mathbf{Tor}^A_\bullet(P'_{\bullet[\rho]}, Q_\bullet) \xrightarrow{\sim} \mathbf{Tor}^{A'}_\bullet(P'_\bullet, Q_\bullet \otimes_A A'). \tag{6.3.11.1}$$

Proof. — If $M_{\bullet\bullet}$ is a projective Cartan–Eilenberg resolution of $Q_\bullet$, then $M_{\bullet\bullet} \otimes_A A'$ is a projective Cartan–Eilenberg resolution of $Q_\bullet \otimes_A A'$. Moreover, up to canonical isomorphism,

$$P'_{\bullet[\rho]} \otimes_A M_{\bullet\bullet} = P'_\bullet \otimes_{A'} (M_{\bullet\bullet} \otimes_A A').$$

The assertion follows from (III.6.3.4).

Remark (6.3.12). — *Let (A^λ) be a filtered inductive system of rings, and let $(P^\lambda_\bullet)$ and $(Q^\lambda_\bullet)$ be two inductive systems of complexes of A^λ-modules. There is a canonical isomorphism generalizing (III.6.3.6.1),*

$$\varinjlim_\lambda \mathbf{Tor}^{A^\lambda}_\bullet(P^\lambda_\bullet, Q^\lambda_\bullet) \xrightarrow{\sim} \mathbf{Tor}^A_\bullet(P_\bullet, Q_\bullet), \tag{6.3.12.1}$$

where

$$A = \varinjlim_\lambda A^\lambda, \qquad P_\bullet = \varinjlim_\lambda P^\lambda_\bullet, \qquad Q_\bullet = \varinjlim_\lambda Q^\lambda_\bullet.$$

Once the homomorphisms

$$\mathbf{Tor}^{A^\lambda}_\bullet(P^\lambda_\bullet, Q^\lambda_\bullet) \longrightarrow \mathbf{Tor}^{A^\mu}_\bullet(P^\mu_\bullet, Q^\mu_\bullet) \qquad (\lambda \leq \mu)$$

have been defined by (III.6.3.10), the proof is the same as that of (III.6.3.6).

Proposition (6.3.13). — *Let S be a multiplicative subset of A, and let $P_\bullet$ and $Q_\bullet$ be complexes of A-modules on which multiplication by every element of S is bijective. Thus, if $A' = S^{-1}A$, both complexes consist of A'-modules. Then there is a canonical isomorphism*

$$\mathbf{Tor}^A_\bullet(P_\bullet, Q_\bullet) \xrightarrow{\sim} \mathbf{Tor}^{A'}_\bullet(P_\bullet, Q_\bullet).$$

Proof. — The hypothesis implies that the canonical homomorphisms

$$P_\bullet \longrightarrow P_\bullet \otimes_A A', \qquad Q_\bullet \longrightarrow Q_\bullet \otimes_A A'$$

are bijective. Functoriality of hypertor also shows that each $s \in S$ acts bijectively on $\mathbf{Tor}^A_\bullet(P_\bullet, Q_\bullet)$; hence

$$\mathbf{Tor}^A_\bullet(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}^A_\bullet(P_\bullet, Q_\bullet) \otimes_A A'$$

is bijective. Since A' is flat over A, the assertion follows from (III.6.3.9). The same argument gives canonical isomorphisms of the associated spectral sequences. III | 146

6.4 Local hypertor functors of complexes of quasi-coherent modules: the affine scheme case

(6.4.1) Let S be an affine scheme with ring A, and let X and Y be affine S-schemes with rings B and C, respectively; thus B and C are A-algebras. Every complex $\mathscr{P}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules is of the form $\widetilde{P_\bullet}$ for a complex $P_\bullet$ of B-modules, and similarly every complex $\mathscr{Q}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules is of the form $\widetilde{Q_\bullet}$ for a complex $Q_\bullet$ of C-modules ((I.1.3.7) and (I.1.3.8)).

Regarded as complexes of A-modules, $P_\bullet$ and $Q_\bullet$ have hypertor modules $\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$. By bifunctoriality, the A-algebras B and C act on this A-module; these actions make it a (B, C)-bimodule, equivalently a module over

$$B \otimes_A C = A(X \times_S Y).$$

We have thus defined a quasi-coherent $\mathscr{O}_{X\times_S Y}$-module

$$\mathscr{T}or_n^{\mathscr{O}_S}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) = \big(\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)\big)^\sim. \tag{6.4.1.1}$$

It is called the *local hypertor of index* n of $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$, and is also denoted $\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$.

Lemma (6.4.2). — *With the notation of (III.6.4.1), suppose that $A = R^{-1}A'$, where A' is a ring and R is a multiplicative subset of A'. Let $S' = \operatorname{Spec}(A')$. Then X and Y may be regarded as S'-schemes and $X \times_{S'} Y = X \times_S Y$ ((I.1.6.2) and (I.3.2.4)); moreover,*

$$\mathscr{T}or_\bullet^{S'}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) = \mathscr{T}or_\bullet^{S}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet).$$

Proof. — This follows from (III.6.4.1.1) and (III.6.3.13).

(6.4.3) With the notation and hypotheses of (III.6.4.1), let $\mathscr{F} = \widetilde{F}$ be a quasi-coherent $\mathscr{O}_X$-module and $\mathscr{G} = \widetilde{G}$ a quasi-coherent $\mathscr{O}_Y$-module. Regarding $\mathscr{F}$ and $\mathscr{G}$ as complexes concentrated in degree zero, we write $\mathscr{T}or_n^{\mathscr{O}_S}(\mathscr{F}, \mathscr{G})$, or $\mathscr{T}or_n^S(\mathscr{F}, \mathscr{G})$, for their local hypertor of index n. By (III.6.3.5)(ii),

$$\mathscr{T}or_n^{\mathscr{O}_S}(\mathscr{F}, \mathscr{G}) = \big(\operatorname{Tor}_n^A(F, G)\big)^\sim. \tag{6.4.3.1}$$

Return now to two arbitrary complexes of quasi-coherent modules $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$. Formulas (III.6.4.1.1) and (III.6.4.3.1), together with (III.6.3.2), show that $\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ is the common abutment of two spectral sequences ${}'\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ and ${}''\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ whose E_2-terms are

$${}'\mathscr{E}^2_{pq} = \mathscr{H}_p\big(\mathscr{T}or_q^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\big) \tag{6.4.3.2}$$

and

$${}''\mathscr{E}^2_{pq} = \bigoplus_{q'+q''=q} \mathscr{T}or_p^S\big(\mathscr{H}_{q'}(\mathscr{P}_\bullet), \mathscr{H}_{q''}(\mathscr{Q}_\bullet)\big). \tag{6.4.3.3}$$

Here $\mathscr{T}or_q^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ in (III.6.4.3.2) denotes the bicomplex of quasi-coherent $\mathscr{O}_{X\times_S Y}$-modules $\mathscr{T}or_q^S(\mathscr{P}_i, \mathscr{Q}_j)$.

(6.4.4) Let

$$X^{(1)} = \operatorname{Spec}(B^{(1)}), \qquad Y^{(1)} = \operatorname{Spec}(C^{(1)})$$

be two further affine schemes, where $B^{(1)}$ and $C^{(1)}$ are A-algebras, and let

$$u : X^{(1)} \longrightarrow X, \qquad v : Y^{(1)} \longrightarrow Y$$

be S-morphisms corresponding to the A-homomorphisms $\phi : B \longrightarrow B^{(1)}$ and $\psi : C \longrightarrow C^{(1)}$. Consider the complexes

$$u^*(\mathscr{P}_\bullet) = (P_\bullet \otimes_B B^{(1)})^\sim, \qquad v^*(\mathscr{Q}_\bullet) = (Q_\bullet \otimes_C C^{(1)})^\sim.$$

The canonical A-linear homomorphisms

$$P_\bullet \longrightarrow P_\bullet \otimes_B B^{(1)}, \qquad Q_\bullet \longrightarrow Q_\bullet \otimes_C C^{(1)}$$

give, by functoriality, an A-linear homomorphism

$$\mathbf{Tor}_\bullet^A(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}_\bullet^A(P_\bullet \otimes_B B^{(1)}, Q_\bullet \otimes_C C^{(1)}).$$

Again by functoriality, this is in fact a homomorphism of $(B \otimes_A C)$-modules. It therefore defines a $(u \times_S v)$-morphism

$$\theta : \mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) \longrightarrow \mathscr{T}or_\bullet^S(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet), \tag{6.4.4.1}$$

and hence a homomorphism of $\mathscr{O}_{X^{(1)}\times_S Y^{(1)}}$-modules

$$\theta^\sharp : (u\times_S v)^*\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet,\mathscr{Q}_\bullet)\longrightarrow \mathscr{T}or_\bullet^S(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet). \tag{6.4.4.2}$$

This is plainly a morphism of bi-∂-functors on the categories of bounded-below quasi-coherent modules. The homomorphism (III.6.4.4.2) need not be bijective; nevertheless:

Lemma (6.4.5). — *With the notation of (III.6.4.4), if u and v are open immersions, then (III.6.4.4.2) is bijective.*

Proof. — Identify $X^{(1)}$ and $Y^{(1)}$ with open subsets of X and Y. They are unions of open subsets of the form $\mathrm{D}(f)$ and $\mathrm{D}(g)$, where $f\in B$ and $g\in C$; the schemes induced on $\mathrm{D}(\phi(f))$ and $\mathrm{D}(f)$, and likewise on $\mathrm{D}(\psi(g))$ and $\mathrm{D}(g)$, are isomorphic.

It is enough to prove the lemma when $X^{(1)}=\mathrm{D}(f)$ and $Y^{(1)}=\mathrm{D}(g)$. Indeed, once this is known, in the general case it suffices to show that the restriction of $\theta^\sharp$ to every $\mathrm{D}(f)\times_S\mathrm{D}(g)$ is an isomorphism. If

$$u_1:\mathrm{D}(f)\longrightarrow X^{(1)},\qquad v_1:\mathrm{D}(g)\longrightarrow Y^{(1)}$$

are the canonical inclusions, this restriction is $(u_1\times_S v_1)^*(\theta^\sharp)$. From (III.6.4.4) and (0.4.4.8), it is immediate that composing it with the canonical homomorphism

$$(u_1\times_S v_1)^*\mathscr{T}or_\bullet^S(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet)\longrightarrow \mathscr{T}or_\bullet^S((u')^*\mathscr{P}_\bullet,(v')^*\mathscr{Q}_\bullet), \tag{6.4.5.1}$$

where $u'=u\circ u_1$ and $v'=v\circ v_1$, gives the canonical homomorphism

$$(u'\times_S v')^*\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet,\mathscr{Q}_\bullet)\longrightarrow \mathscr{T}or_\bullet^S((u')^*\mathscr{P}_\bullet,(v')^*\mathscr{Q}_\bullet). \tag{6.4.5.2}$$

If (III.6.4.5.1) and (III.6.4.5.2) are isomorphisms, then so is $(u_1\times_S v_1)^*(\theta^\sharp)$.

Suppose therefore that $X^{(1)}=\mathrm{D}(f)$ and $Y^{(1)}=\mathrm{D}(g)$, so that $B^{(1)}=B_f$ and $C^{(1)}=C_g$. Then $u^*(\mathscr{P}_\bullet)$ and $v^*(\mathscr{Q}_\bullet)$ identify with $((P_\bullet)_f)^\sim$ and $((Q_\bullet)_g)^\sim$, respectively. Moreover, $X^{(1)}\times_S Y^{(1)}$ identifies with the open subscheme $\mathrm{D}(f\otimes g)$ of

$$X\times_S Y=\mathrm{Spec}(B\otimes_A C)$$

((II.4.3.2.4)). It remains to prove that the homomorphism

$$\big(\mathbf{Tor}_\bullet^A(P_\bullet,Q_\bullet)\big)_{f\otimes g}\longrightarrow \mathbf{Tor}_\bullet^A((P_\bullet)_f,(Q_\bullet)_g), \tag{6.4.5.3}$$

induced by the canonical homomorphisms $P_\bullet\longrightarrow(P_\bullet)_f$ and $Q_\bullet\longrightarrow(Q_\bullet)_g$, is bijective. **III** | 148

By (0.1.6.1),

$$(P_\bullet)_f=\varinjlim_m P_\bullet^{(m)},$$

where every $P_\bullet^{(m)}$ is the same complex of B-modules as $P_\bullet$, and for $m\leq n$ the map $P_\bullet^{(m)}\longrightarrow P_\bullet^{(n)}$ is multiplication by f^{n-m}. The analogous statement holds for $Q_\bullet$, with g in place of f. The homomorphism

$$\mathbf{Tor}_\bullet^A(P_\bullet^{(m)},Q_\bullet^{(m)})\longrightarrow \mathbf{Tor}_\bullet^A(P_\bullet^{(n)},Q_\bullet^{(n)})$$

corresponding to these transition maps is, by definition, multiplication by $(f\otimes g)^{n-m}$. The conclusion follows from (0.1.6.1), applied to the left-hand side of (III.6.4.5.3), and from (III.6.3.6).

(6.4.6) With the notation of (III.6.4.4), one defines in the same way canonical homomorphisms of spectral functors

$$\begin{aligned}(u\times_S v)^*\,{}'\mathscr{E}(\mathscr{P}_\bullet,\mathscr{Q}_\bullet)&\longrightarrow {}'\mathscr{E}(u^*\mathscr{P}_\bullet,v^*\mathscr{Q}_\bullet),\\ (u\times_S v)^*\,{}''\mathscr{E}(\mathscr{P}_\bullet,\mathscr{Q}_\bullet)&\longrightarrow {}''\mathscr{E}(u^*\mathscr{P}_\bullet,v^*\mathscr{Q}_\bullet).\end{aligned} \tag{6.4.6.1}$$

The argument of (III.6.4.5) shows that when u and v are open immersions, these homomorphisms are bijective. Taking (III.6.3.6.2) and (III.6.3.6.3) into account, the argument proves that they are isomorphisms on the E^2-terms, while (III.6.4.5) proves that they are isomorphisms on the abutments. The assertion therefore follows from (0.11.1.2) and (0.11.2.4).

6.5 Local hypertor functors of complexes of quasi-coherent modules: the general case

(6.5.1) Let S now be an arbitrary prescheme and let X and Y be two S-preschemes. Let $\mathscr{P}_\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-modules and $\mathscr{Q}_\bullet$ a complex of quasi-coherent $\mathscr{O}_Y$-modules. Set $Z = X \times_S Y$. We shall define quasi-coherent $\mathscr{O}_Z$-modules

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet),$$

called the *local hypertors* of $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$, which reduce to those of (III.6.4) when S, X, and Y are affine.

When $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are concentrated in degree zero, with respective terms $\mathscr{F}$ and $\mathscr{G}$, we write $\mathscr{T}or_n^S(\mathscr{F}, \mathscr{G})$ instead of $\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$.

(6.5.2) Suppose first that S is affine, and let (X_λ) and (Y_μ) be coverings of X and Y by affine open subsets. Then

$$Z_{\lambda\mu} = X_\lambda \times_S Y_\mu$$

form an affine open covering of Z. Put

$$\mathscr{P}_{\lambda,\bullet} = \mathscr{P}_\bullet|X_\lambda, \qquad \mathscr{Q}_{\mu,\bullet} = \mathscr{Q}_\bullet|Y_\mu.$$

For each pair (λ, μ) we thus have a quasi-coherent $\mathscr{O}_{Z_{\lambda\mu}}$-module

$$\mathscr{F}_{\lambda\mu} = \mathscr{T}or_n^S(\mathscr{P}_{\lambda,\bullet}, \mathscr{Q}_{\mu,\bullet}).$$

We must show that the $\mathscr{F}_{\lambda\mu}$ satisfy the gluing condition (0.3.3.1). It is enough to verify that, for every affine open

$$U \subset X_\lambda \cap X_{\lambda'}, \qquad V \subset Y_\mu \cap Y_{\mu'},$$

the restrictions of $\mathscr{F}_{\lambda\mu}$ and $\mathscr{F}_{\lambda'\mu'}$ to $U \times_S V$ are canonically isomorphic. This follows at once from their canonical isomorphisms with

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V)$$

given by (III.6.4.5).

The definition and (III.6.4.5) also show that the resulting $\mathscr{O}_Z$-module is independent, up to a unique isomorphism compatible with the local identifications, of the chosen open coverings. We denote it by **III** | 149

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet).$$

Finally, for every open subset U of X and every open subset V of Y, its restriction to $U \times_S V$ is canonically isomorphic to

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V).$$

(6.5.3) Pass now to the general case, where S is arbitrary. Let (S_α) be a covering of S by affine open subsets, and let X_α and Y_α be the inverse images of S_α in X and Y. We must show that the sheaves

$$\mathscr{G}_\alpha = \mathscr{T}or_n^{S_\alpha}(\mathscr{P}_\bullet|X_\alpha, \mathscr{Q}_\bullet|Y_\alpha)$$

satisfy the gluing condition.

For every affine open subset $T \subset S_\alpha \cap S_\beta$, let U and V be its inverse images in X and Y. It is enough to define canonical isomorphisms between the restrictions of $\mathscr{G}_\alpha$ and $\mathscr{G}_\beta$ to $U \times_S V$ and

$$\mathscr{T}or_n^T(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V).$$

We may further restrict to the case in which $T = \mathrm{D}(f_\alpha) = \mathrm{D}(f_\beta)$, where f_α and f_β are sections of $\mathscr{O}_S$ over S_α and S_β, respectively. By (III.6.4.2),

$$\mathscr{T}or_n^T(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V)$$

is canonically isomorphic both to $\mathscr{T}or_n^{S_\alpha}(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V)$ and to $\mathscr{T}or_n^{S_\beta}(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V)$. Combining these isomorphisms with the local identifications of (III.6.5.2) completes the definition of the $\mathscr{O}_Z$-module

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet).$$

For every open subset U of X and every open subset V of Y, the restriction of this module to $U \times_S V$ is canonically isomorphic to

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V).$$

This construction plainly defines, on the categories of bounded-below complexes of quasi-coherent modules, a bi-∂-functor

$$\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$$

with values in $\mathscr{O}_Z$-modules. Indeed the assertion is local on X, Y, and S, by (III.6.4.5) and by the fact that (III.6.4.4.2) is a morphism of bi-∂-functors.

If $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are concentrated in degree zero, with respective terms $\mathscr{F}$ and $\mathscr{G}$, then

$$\mathscr{T}or_0^S(\mathscr{F}, \mathscr{G}) = \mathscr{F} \otimes_S \mathscr{G},$$

the external tensor product defined in (I.9.1.2); this follows from (III.6.4.1.1) and (I.9.1.3).

(6.5.4) The preceding construction and (III.6.4.6) show that $\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ is the common abutment of two spectral functors ${}'\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ and ${}''\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$, whose E_2-terms are

$${}'\mathscr{E}^2_{pq} = \mathscr{H}_p\big(\mathscr{T}or_q^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\big) \tag{6.5.4.1}$$

and

$${}''\mathscr{E}^2_{pq} = \bigoplus_{q'+q''=q} \mathscr{T}or_p^S\big(\mathscr{H}_{q'}(\mathscr{P}_\bullet), \mathscr{H}_{q''}(\mathscr{Q}_\bullet)\big). \tag{6.5.4.2}$$

The spectral sequence (III.6.5.4.2) is always regular. Both spectral sequences are biregular when $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are bounded below. A further case in which they are biregular is described next.

(6.5.5) Let T be a topological space. A sheaf of rings $\mathscr{A}$ on T is said to have *cohomological dimension at most* n if the ring $\mathscr{A}_t$ has cohomological dimension at most n for every $t \in T$. In this case the ringed space $(T, \mathscr{A})$ is also said to have cohomological dimension at most n. A sheaf of rings, or a ringed space, has *finite cohomological dimension* if this holds for some integer n.

If the $\mathscr{A}_t$ are Noetherian commutative local rings, saying that they have cohomological dimension at most n means that they are regular and have Krull dimension at most n ((0.17.3.1)). In the terminology of dimension theory to be introduced in Chapter IV, a locally Noetherian prescheme T has cohomological dimension at most n exactly when it is regular ((0.4.1.4)) and has dimension at most n. Equivalently, for every affine open subset U of T, the ring $\Gamma(U, \mathscr{O}_T)$ has cohomological dimension at most n ((0.17.2.6)). **III** | 150

Consequently, if S is locally Noetherian and has finite cohomological dimension, then (III.6.3.2) shows that both ${}'\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ and ${}''\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ are biregular.

Finally, under the canonical isomorphism $X \times_S Y \xrightarrow{\sim} Y \times_S X$, $\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ corresponds canonically to $\mathscr{T}or_\bullet^S(\mathscr{Q}_\bullet, \mathscr{P}_\bullet)$.

Proposition (6.5.6). — *Let $(\mathscr{P}_{\alpha\bullet})$ be a filtered direct system of complexes of quasi-coherent $\mathscr{O}_X$-modules. There is a canonical isomorphism*

$$\varinjlim_\alpha \mathscr{T}or_\bullet^S(\mathscr{P}_{\alpha\bullet}, \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{T}or_\bullet^S\left(\varinjlim_\alpha \mathscr{P}_{\alpha\bullet}, \mathscr{Q}_\bullet\right). \tag{6.5.6.1}$$

Proof. — The question is local on S, X, and Y. We may therefore suppose that S, X, and Y are affine, in which case the proposition reduces to (III.6.3.6).

(6.5.7) *Remarks.*

(i) Consider in particular the case $S = X = Y$, with $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ two complexes of quasi-coherent $\mathscr{O}_S$-modules. Then the $\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ are quasi-coherent $\mathscr{O}_S$-modules. Moreover, for every point $z \in S$, (III.6.5.6) gives a canonical isomorphism

$$\big(\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\big)_z \xrightarrow{\sim} \mathbf{Tor}_n^{\mathscr{O}_z}\big((\mathscr{P}_\bullet)_z, (\mathscr{Q}_\bullet)_z\big), \tag{6.5.7.1}$$

because the question is local and, by (III.6.4.1.1), reduces to the case of modules.

(ii) The definition of hypertor can be generalized to two complexes $\mathscr{P}_\bullet, \mathscr{Q}_\bullet$ of $\mathscr{O}_X$-modules on the same ringed space $(X, \mathscr{O}_X)$. For every open subset U of X, put

$$A(U) = \Gamma(U, \mathscr{O}_X), \qquad P_\bullet(U) = \Gamma(U, \mathscr{P}_\bullet), \qquad Q_\bullet(U) = \Gamma(U, \mathscr{Q}_\bullet).$$

The $A(U)$-modules

$$\mathbf{Tor}_n^{A(U)}\big(P_\bullet(U), Q_\bullet(U)\big)$$

then form a presheaf on X; denote the associated $\mathscr{O}_X$-module by $\mathscr{T}or_n^X(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$. When X is a prescheme, (III.6.3.12) shows that this module is canonically isomorphic to the hypertor defined above. We shall not pursue this generalization further.

Proposition (6.5.8). — *Let X and Y be two S-preschemes, and let $\mathscr{F}$ (respectively $\mathscr{G}$) be a quasi-coherent $\mathscr{O}_X$-module (respectively $\mathscr{O}_Y$-module). If $\mathscr{F}$ or $\mathscr{G}$ is S-flat, then*

$$\mathscr{T}or_n^S(\mathscr{F},\mathscr{G}) = 0 \qquad (n \neq 0).$$

Proof. — The question is local on X and Y, so we may suppose that X, Y, and S are affine, with respective rings B, C, and A, and write $\mathscr{F} = \widetilde{M}$ and $\mathscr{G} = \widetilde{N}$, where M is a B-module and N a C-module. Suppose, for example, that $\mathscr{F}$ is S-flat. This means that, for every $s \in S$, M_s is a flat A_s-module ((0.6.7.1)); hence M is a flat A-module ((0.6.3.3)). Thus $\mathrm{Tor}_n^A(M,N) = 0$ for $n > 0$ and every C-module N ((0.6.1.1)), and the conclusion follows from (III.6.4.1.1).

Corollary (6.5.9). — *Let X and Y be two S-preschemes, and let $\mathscr{P}_\bullet$ (respectively $\mathscr{Q}_\bullet$) be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules (respectively $\mathscr{O}_Y$-modules). Suppose that every $\mathscr{P}_i$ is S-flat. Then there is a canonical isomorphism of ∂-functors in $\mathscr{Q}_\bullet$* III | 151

$$\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{P}_\bullet \otimes_S \mathscr{Q}_\bullet). \tag{6.5.9.1}$$

Proof. — When S, X, and Y are affine, this is precisely (III.6.3.7); the general case follows by the arguments of (III.6.5.2) and (III.6.5.3).

(6.5.10) *Corollary.* Suppose that X is flat over S ((0.6.7.1)), that $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are bounded below, and that every $\mathscr{P}_i$ is a locally free $\mathscr{O}_X$-module, not necessarily of finite type. Then the homomorphism (III.6.5.9.1) is bijective.

Indeed, the hypothesis of (III.6.5.9) is satisfied: flatness is, by definition, a pointwise property on X, and every direct sum of flat modules is flat ((0.6.1.2)).

Proposition (6.5.11). — *Let X' and Y' be two S-preschemes, and let $f : X \to X'$ and $g : Y \to Y'$ be two affine S-morphisms. For a complex $\mathscr{P}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules and a complex $\mathscr{Q}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules, there is a canonical functorial isomorphism*

$$(f \times_S g)_*\big(\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\big) \xrightarrow{\sim} \mathscr{T}or_\bullet^S(f_*\mathscr{P}_\bullet, g_*\mathscr{Q}_\bullet). \tag{6.5.11.1}$$

Proof. — Since f and g are affine, $f_*\mathscr{P}_\bullet$ and $g_*\mathscr{Q}_\bullet$ are complexes of quasi-coherent modules ((II.1.2.6)). If $Z' = X' \times_S Y'$, both sides of (III.6.5.11.1) are quasi-coherent $\mathscr{O}_{Z'}$-modules by (III.6.5.1). We readily reduce to the case in which S, X', and Y' are affine; then X and Y are affine as well, and the assertion follows immediately from (III.6.4.1.1) and (I.1.6.3).

(6.5.12) *Remark.* Let X' and Y' be two S-preschemes, with X' flat over S. Let X be a closed subprescheme of X', and let $i : X \to X'$ be the canonical immersion. Let $\mathscr{P}_\bullet$ (respectively $\mathscr{Q}_\bullet$) be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules (respectively $\mathscr{O}_{Y'}$-modules). Finally, let $\mathscr{L}'_{\bullet\bullet}$ be a resolution of $i_*\mathscr{P}_\bullet$ consisting of locally free $\mathscr{O}_{X'}$-modules, such that every point of X' has an affine open neighborhood U for which $\mathscr{L}'_j|U$ is a free resolution of $i_*\mathscr{P}_j|U$, for every j. There is then a canonical isomorphism

$$(i \times_S 1)_*\big(\mathscr{T}or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\big) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}'_{\bullet\bullet} \otimes_S \mathscr{Q}_\bullet). \tag{6.5.12.1}$$

If S, X', and Y' are affine, and if $\mathscr{L}'_{j\bullet}$ is a free resolution of $i_*\mathscr{P}_j$ for every j, then (III.6.5.11) reduces us to the case $X' = X$, with X flat over S, and it is enough to apply (III.6.3.4). In general, (III.6.5.12.1) is defined locally, and it remains to verify that these local definitions give a global isomorphism.

For this, recall the definition of the first isomorphism (III.6.3.4.1). It comes from an isomorphism of spectral sequences ((0.11.6.5) and (0.11.5.3)), itself obtained from a morphism of bicomplexes

$$L_{\bullet\bullet} \longrightarrow L''_{\bullet\bullet},$$

where $L_{\bullet\bullet}$ is the given resolution of $P_\bullet$, and $L''_{\bullet\bullet}$ is a projective resolution of $P_\bullet$ in the category of bounded- III | 152
below complexes of A-modules ((0.11.5.2.2)). The assertion follows from the fact that (III.6.3.4.1) is independent of the chosen projective resolution $L''_{\bullet\bullet}$, since any two such resolutions are related by a homotopism (*M*, V, 1.2).

Proposition (6.5.13). — *Let X and Y be two S-preschemes, and suppose that one of the following conditions holds:*

(i) X and $Z = X \times_S Y$ are locally Noetherian, and X is flat over S;

(ii) S and X are locally Noetherian, and Y is of finite type over S.

Let $\mathscr{P}_\bullet$ (respectively $\mathscr{Q}_\bullet$) be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules (respectively $\mathscr{O}_Y$-modules). Suppose in addition that, for every n, $\mathscr{H}_n(\mathscr{P}_\bullet)$ (respectively $\mathscr{H}_n(\mathscr{Q}_\bullet)$) is an $\mathscr{O}_X$-module (respectively $\mathscr{O}_Y$-module) of finite type. Then

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$$

is a coherent $\mathscr{O}_Z$-module for every n.

Proof. — Since $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are bounded below, the spectral sequence ${}''\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ is biregular by (III.6.5.4). In either case (i) or (ii), Z is locally Noetherian, so (0.11.1.8) reduces the assertion to the coherence of the terms ${}''\mathscr{E}^2_{pq}$. The finite-type hypothesis on the homology modules and formula (III.6.5.4.2) reduce the proposition to the special case in which $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are concentrated in degree zero, namely the following corollary.

Corollary (6.5.14). — Suppose that one of conditions (i) *and* (ii) *of (III.6.5.13) holds, and let $\mathscr{F}$ (respectively $\mathscr{G}$) be a quasi-coherent $\mathscr{O}_X$-module (respectively $\mathscr{O}_Y$-module) of finite type. Then $\mathscr{T}or_n^S(\mathscr{F}, \mathscr{G})$ is a coherent $\mathscr{O}_Z$-module for every n.*

Proof. — The question is local on X and Y, so we may suppose that S, X, and Y are affine.

(i) Under the hypotheses of (i), X and Z are Noetherian. There is therefore a locally free resolution $\mathscr{L}_\bullet$ of $\mathscr{F}$ consisting of $\mathscr{O}_X$-modules of finite type ((I.1.3.7)). Since X is flat over S, (III.6.3.4) gives

$$\mathscr{T}or_n^S(\mathscr{F}, \mathscr{G}) = \mathscr{H}_n(\mathscr{L}_\bullet \otimes_S \mathscr{G}).$$

The modules $\mathscr{L}_i \otimes_S \mathscr{G}$ are quasi-coherent $\mathscr{O}_Z$-modules of finite type ((I.9.1.1)), hence are coherent. It follows that $\mathscr{H}_n(\mathscr{L}_\bullet \otimes_S \mathscr{G})$ is coherent ((0.5.3.4)).

(ii) Now suppose that condition (ii) holds. Since $A(Y)$ is a quotient of a polynomial $A(S)$-algebra in finitely many indeterminates ((I.6.3.3)), Y is a closed subprescheme of an affine S-scheme Y' that is flat and of finite type over S. The scheme Y' is Noetherian ((I.6.3.7)), so there exists a locally free resolution $\mathscr{M}_\bullet$ of $j_*\mathscr{G}$ by $\mathscr{O}_{Y'}$-modules of finite type, where $j : Y \to Y'$ is the canonical immersion. By (III.6.5.12), $\mathscr{T}or_n^S(\mathscr{F}, \mathscr{G})$ is the inverse image under $1 \times j$ of the $\mathscr{O}_{Z'}$-module

$$\mathscr{H}_n(\mathscr{F} \otimes_S \mathscr{M}_\bullet), \qquad Z' = X \times_S Y'.$$

As in (i), the $\mathscr{F} \otimes_S \mathscr{M}_i$ are coherent $\mathscr{O}_{Z'}$-modules, and (0.5.3.4) again gives the conclusion.

(6.5.15) The theory developed above for two complexes $\mathscr{P}_\bullet, \mathscr{Q}_\bullet$ of quasi-coherent sheaves on two S-preschemes X and Y extends without difficulty to an arbitrary finite family of S-preschemes $X^{(i)}$ $(1 \leq i \leq m)$, each equipped with a complex $\mathscr{P}_\bullet^{(i)}$ of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. If

$$Z = X^{(1)} \times_S X^{(2)} \times_S \cdots \times_S X^{(m)},$$

one thereby defines a quasi-coherent $\mathscr{O}_Z$-module

$$\mathscr{T}or_q^S\big(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}, \ldots, \mathscr{P}_\bullet^{(m)}\big).$$

We leave the development of the theory in this general case to the reader, and record only, for later reference, the E_2-term of the second, regular spectral sequence whose abutment is this hypertor: III | 153

$$''\mathscr{E}^2_{pq} = \bigoplus_{q_1+q_2+\cdots+q_m=q} \mathscr{T}or_p^S\big(\mathscr{H}_{q_1}(\mathscr{P}_\bullet^{(1)}), \ldots, \mathscr{H}_{q_m}(\mathscr{P}_\bullet^{(m)})\big). \tag{6.5.15.1}$$

In (III.6.8) we shall study the associativity spectral sequences to which these hypertor functors of an arbitrary finite number of complexes give rise.

6.6 Global hypertor functors of complexes of quasi-coherent modules and Künneth spectral sequences: the affine-base case

(6.6.1) Let $S = \mathrm{Spec}(A)$ be an affine scheme, and let $X^{(i)}$ $(i = 1, 2)$ be two quasi-compact S-schemes. For $i = 1, 2$, let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules whose differential has degree -1. Choose a finite covering

$$\mathfrak{U}^{(i)} = (U_\alpha^{(i)})$$

of $X^{(i)}$ by affine open subsets. Put

$$X = X^{(1)} \times_S X^{(2)}.$$

This is a quasi-compact S-scheme ((I.5.5.1) and (I.6.6.4)), and $\mathfrak{U} = \mathfrak{U}^{(1)} \times_S \mathfrak{U}^{(2)}$ is the covering of X by the affine open subsets $U_\alpha^{(1)} \times_S U_\beta^{(2)}$.

For $p \leq 0$ and $q \in \mathbf{Z}$, the group of $(-p)$-alternating cochains

$$C^{-p}(\mathfrak{U}^{(i)}, \mathscr{P}_q^{(i)})$$

of the covering $\mathfrak{U}^{(i)}$ with coefficients in $\mathscr{P}_q^{(i)}$ (G, II, 5.1) is an A-module. For $p > 0$, put $C^p(\mathfrak{U}^{(i)}, \mathscr{P}_q^{(i)}) = 0$. This defines a bicomplex of A-modules

$$C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)})$$

whose two differentials have degree -1.

It follows from the definitions ((0.12.1.2)) that the homology module

$$H_n\big(C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)})\big)$$

of this bicomplex, viewed in the usual way as a simple complex for the total degree, is the hypercohomology module

$$\mathrm{H}^{-n}\big(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)\bullet}\big),$$

where $\mathscr{P}^{(i)\bullet}$ is the complex with differential of degree $+1$ whose component of degree q is $\mathscr{P}_{-q}^{(i)}$. By abuse of notation we write this as

$$\mathrm{H}^{-n}\big(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}\big).$$

By (III.6.2.2), this module is canonically isomorphic to

$$\mathrm{H}^{-n}\big(X^{(i)}, \mathscr{P}^{(i)\bullet}\big),$$

also written $\mathrm{H}^{-n}(X^{(i)}, \mathscr{P}_\bullet^{(i)})$. It is therefore independent of the chosen finite covering $\mathfrak{U}^{(i)}$.

(6.6.2) Apply the general theory of hyperhomology of functors with respect to bicomplexes ((0.11.7.4)) to the two bicomplexes of A-modules

$$L_{\bullet\bullet}^{(i)} = C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}) \qquad (i = 1, 2)$$

and to the covariant bifunctor $L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}$. Because the cochains are alternating and the coverings $\mathfrak{U}^{(i)}$ are finite, the modules

$$C^{-p}(\mathfrak{U}^{(i)}, \mathscr{P}_q^{(i)})$$

are nonzero for only finitely many values of p, independently of q. In particular, both degrees of each $L_{\bullet\bullet}^{(i)}$ are bounded below.

Denote by

$$\mathbf{Tor}_n^A(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})$$

or by

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

the n-th hyperhomology module of $L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}$. We call it the hypertor of index n of $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$ relative to the coverings $\mathfrak{U}^{(1)}$ and $\mathfrak{U}^{(2)}$. When these complexes are concentrated in degree zero, with terms $\mathscr{F}^{(1)}$ and $\mathscr{F}^{(2)}$, write

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)})$$

for their hypertor. Following the general conventions, the same notation

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

also denotes the bicomplex whose component of bidegree (j, k) is

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_j^{(1)}, \mathscr{P}_k^{(2)}).$$

The construction $L_{\bullet\bullet}^{(i)}$ is exact in $\mathscr{P}_\bullet^{(i)}$, since all intersections of members of $\mathfrak{U}^{(i)}$ are affine ((I.5.5.6) and (I.1.3.11)). Consequently

$$\mathbf{Tor}_\bullet^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

is a covariant bi-∂-functor in $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$, with values in the category of A-modules ((0.11.7.3)). By (0.11.7.2), it is the common abutment of six biregular spectral functors, denoted

$$^{(t)}E^S(\mathfrak{U}^{(1)},\mathfrak{U}^{(2)};\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}),$$

or simply

$$^{(t)}E(\mathfrak{U}^{(1)},\mathfrak{U}^{(2)};\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}),$$

where t is one of a, b, a', b', c, d. Their E_2-terms are

$$\begin{aligned}
^{(a)}E^2_{pq} &= \bigoplus_{q_1+q_2=q} \operatorname{Tor}_p^A\big(H_{q_1}(L^{(1)}_{\bullet\bullet}), H_{q_2}(L^{(2)}_{\bullet\bullet})\big),\\
^{(b)}E^2_{pq} &= H_p\big(\mathbf{Tor}_q^{A,\mathrm{II}}(L^{(1)}_{\bullet\bullet}, L^{(2)}_{\bullet\bullet})\big),\\
^{(a')}E^2_{pq} &= \bigoplus_{q_1+q_2=q} \operatorname{Tor}_p^A\big(H^{\mathrm{I}}_{q_1}(L^{(1)}_{\bullet\bullet}), H^{\mathrm{I}}_{q_2}(L^{(2)}_{\bullet\bullet})\big),\\
^{(b')}E^2_{pq} &= H_p\big(\mathbf{Tor}_q^{A}(L^{(1)}_{\bullet\bullet}, L^{(2)}_{\bullet\bullet})\big),\\
^{(c)}E^2_{pq} &= \bigoplus_{q_1+q_2=q} \operatorname{Tor}_p^A\big(H^{\mathrm{II}}_{q_1}(L^{(1)}_{\bullet\bullet}), H^{\mathrm{II}}_{q_2}(L^{(2)}_{\bullet\bullet})\big),\\
^{(d)}E^2_{pq} &= H_p\big(\mathbf{Tor}_q^{A,\mathrm{I}}(L^{(1)}_{\bullet\bullet}, L^{(2)}_{\bullet\bullet})\big).
\end{aligned}$$

These notations agree with those of the general theory of hyperhomology. We now make the initial terms more explicit.

(6.6.3) *Spectral sequences* (a) *and* (a′). By (III.6.6.1),

$$H_n(L^{(i)}_{\bullet\bullet}) = \mathrm{H}^{-n}(X^{(i)}, \mathscr{P}_\bullet^{(i)}),$$

and hence

$$^{(a)}E^2_{pq} = \bigoplus_{q_1+q_2=q} \operatorname{Tor}_p^A\big(\mathrm{H}^{-q_1}(X^{(1)}, \mathscr{P}_\bullet^{(1)}), \mathrm{H}^{-q_2}(X^{(2)}, \mathscr{P}_\bullet^{(2)})\big).$$

By definition, the component of degree k of the complex $H^{\mathrm{I}}_n(L^{(i)}_{\bullet\bullet})$ is

$$H_n\big(C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_k^{(i)})\big) = \mathrm{H}^{-n}(\mathfrak{U}^{(i)}, \mathscr{P}_k^{(i)}).$$

By (III.1.4.1), this module is canonically isomorphic to $\mathrm{H}^{-n}(X^{(i)}, \mathscr{P}_k^{(i)})$. Therefore

$$^{(a')}E^2_{pq} = \bigoplus_{q_1+q_2=q} \operatorname{Tor}_p^A\big(\mathrm{H}^{-q_1}(X^{(1)}, \mathscr{P}_\bullet^{(1)}), \mathrm{H}^{-q_2}(X^{(2)}, \mathscr{P}_\bullet^{(2)})\big),$$

where each $\mathrm{H}^{-q_i}(X^{(i)}, \mathscr{P}_\bullet^{(i)})$ in this last formula denotes the complex obtained by taking cohomology term by term.

(6.6.4) *Spectral sequences* (b) *and* (b′). By definition,

$$\mathbf{Tor}_q^{A,\mathrm{II}}(L^{(1)}_{\bullet\bullet}, L^{(2)}_{\bullet\bullet})$$

is a bicomplex whose component of bidegree (h, k) is

$$\operatorname{Tor}_q^A\big(C^{-h}(\mathfrak{U}^{(1)}, \mathscr{P}_\bullet^{(1)}), C^{-k}(\mathfrak{U}^{(2)}, \mathscr{P}_\bullet^{(2)})\big).$$

Let $\Phi^{(i)}$ be the index set of $\mathfrak{U}^{(i)}$. For $r \geq 0$, the complex $C^r(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)})$ is the direct sum of the complexes

$$\Gamma(U^{(i)}_\rho, \mathscr{P}_\bullet^{(i)}),$$

where $U^{(i)}_\rho$ is the intersection of the $U^{(i)}_\xi$ for $\xi \in \rho$, and ρ ranges over the appropriate elements of $\mathfrak{P}(\Phi^{(i)})$. Consequently

$$\operatorname{Tor}_q^A\big(C^{-h}(\mathfrak{U}^{(1)}, \mathscr{P}_\bullet^{(1)}), C^{-k}(\mathfrak{U}^{(2)}, \mathscr{P}_\bullet^{(2)})\big)$$

is the direct sum of the A-modules

$$\operatorname{Tor}_q^A\big(\Gamma(U^{(1)}_\sigma, \mathscr{P}_\bullet^{(1)}), \Gamma(U^{(2)}_\tau, \mathscr{P}_\bullet^{(2)})\big),$$

where σ and τ range over $\mathfrak{P}(\Phi^{(1)})$ and $\mathfrak{P}(\Phi^{(2)})$ with

$$\mathrm{Card}(\sigma) = 1-h, \qquad \mathrm{Card}(\tau) = 1-k,$$

respectively. Since $X^{(1)}$ and $X^{(2)}$ are schemes, the $U_\rho^{(i)}$ are affine. Thus (III.6.4.1.1) gives

$$\mathrm{Tor}_q^A\big(\Gamma(U_\sigma^{(1)}, \mathscr{P}_\bullet^{(1)}), \Gamma(U_\tau^{(2)}, \mathscr{P}_\bullet^{(2)})\big) = \Gamma\big(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \mathscr{T}\!or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big).$$

It follows that ${}^{(b)}E_{pq}^2$ is the $(-p)$-th cohomology module of the complex

$$L^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathcal{S})$$

of bialternating cochains on $\Phi^{(1)}$ and $\Phi^{(2)}$ with values in the coefficient system

$$\mathcal{S} : (\sigma, \tau) \longmapsto \Gamma\big(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \mathscr{T}\!or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big) \tag{6.6.4.1}$$

((0.11.8.4)). By (0.11.8.5) and (0.11.8.6), this complex has the same cohomology as the complex of all cochains

$$C^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathcal{S})$$

and as the complex

$$P^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathcal{S}),$$

whose elements are linear combinations of

$$\lambda(\sigma, \tau) \in \Gamma\big(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \mathscr{T}\!or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big),$$

where

$$\sigma = (\alpha_0, \ldots, \alpha_h), \qquad \tau = (\beta_0, \ldots, \beta_h)$$

are sequences with the same number of terms. Now

$$U_\sigma^{(1)} \times_S U_\tau^{(2)} = (U_{\alpha_0}^{(1)} \times_S U_{\beta_0}^{(2)}) \cap \cdots \cap (U_{\alpha_h}^{(1)} \times_S U_{\beta_h}^{(2)})$$

by (I.3.2.7). Let $\mathfrak{U}$ be the covering of $Z = X^{(1)} \times_S X^{(2)}$ by the affine open subsets $U_\alpha^{(1)} \times_S U_\beta^{(2)}$. Since Z is a scheme, (III.1.3.1) yields

$${}^{(b)}E_{pq}^2 = \mathrm{H}^{-p}\big(Z, \mathscr{T}\!or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big). \tag{6.6.4.2}$$

Next,

$$\mathbf{Tor}_q^A(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})$$

is a bicomplex whose component of bidegree (h, k) is the direct sum of the A-modules

$$\mathrm{Tor}_q^A\big(C^{-h_1}(\mathfrak{U}^{(1)}, \mathscr{P}_{k_1}^{(1)}), C^{-h_2}(\mathfrak{U}^{(2)}, \mathscr{P}_{k_2}^{(2)})\big),$$

where $h_1 + h_2 = h$ and $k_1 + k_2 = k$. Expanding the cochain modules as above shows that this component is also the direct sum of

$$\Gamma\big(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \mathscr{T}\!or_q^S(\mathscr{P}_{k_1}^{(1)}, \mathscr{P}_{k_2}^{(2)})\big),$$

where $k_1 + k_2 = k$ and

$$\mathrm{Card}(\sigma) + \mathrm{Card}(\tau) = 2 - h.$$

The term ${}^{(b')}E_{pq}^2$ is the $(-p)$-th cohomology module of a bicomplex $N^{\bullet\bullet} = (N^{hk})$. For fixed k, the simple complex $N^{\bullet k}$ consists of bialternating cochains on $\Phi^{(1)}$ and $\Phi^{(2)}$ with values in the coefficient system

$$\mathcal{S}_k : (\sigma, \tau) \longmapsto \Gamma\left(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \bigoplus_{k_1+k_2=k} \mathscr{T}\!or_q^S(\mathscr{P}_{-k_1}^{(1)}, \mathscr{P}_{-k_2}^{(2)})\right).$$

These coefficient systems form a complex $\mathcal{S}^\bullet$, whose differential is induced by that of the simple complex associated to the bicomplex

$$\mathscr{T}\!or_q^S(\mathscr{P}^{(1)\bullet}, \mathscr{P}^{(2)\bullet}).$$

By (0.11.8.9), $N^{\bullet\bullet}$ has the same cohomology as

$$C^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathcal{S}^\bullet),$$

and also the same cohomology as

$$P^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathcal{S}^\bullet),$$

whose elements of bidegree (h,k) are linear combinations of

$$\lambda(\sigma,\tau) \in \Gamma\left(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \bigoplus_{k_1+k_2=k} \mathscr{T}or_q^S(\mathscr{P}_{-k_1}^{(1)}, \mathscr{P}_{-k_2}^{(2)})\right),$$

with $\sigma = (\alpha_0, \dots, \alpha_h)$ and $\tau = (\beta_0, \dots, \beta_h)$ having the same number of terms ((0.11.8.10)). As above, ${}^{(b')}E_{pq}^2$ is therefore the $(-p)$-th cohomology module of

$$C^\bullet(\mathfrak{U}, \mathcal{Q}^\bullet),$$

where $\mathcal{Q}^\bullet$ is the simple complex associated to the bicomplex

$$\mathscr{T}or_q^S(\mathscr{P}^{(1)\bullet}, \mathscr{P}^{(2)\bullet})$$

of $\mathscr{O}_Z$-modules. With the conventions of (III.6.6.1), we obtain

$${}^{(b')}E_{pq}^2 = \mathrm{H}^{-p}\big(Z, \mathscr{T}or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big). \tag{6.6.4.3}$$

III | 156

(6.6.5) *Spectral sequences* (c) *and* (d). By definition, the component of degree h of the complex $H_n^{\mathrm{II}}(L_{\bullet\bullet}^{(i)})$ is the A-module

$$C^{-h}\big(\mathfrak{U}^{(i)}, \mathscr{H}_n(\mathscr{P}_\bullet^{(i)})\big),$$

because the functor C^{-h} is exact. Hence, by the definition of the hypertor of two modules relative to two coverings ((III.6.6.2)),

$${}^{(c)}E_{pq}^2 = \bigoplus_{q_1+q_2=q} \mathbf{Tor}_p^S\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{H}_{q_1}(\mathscr{P}_\bullet^{(1)}), \mathscr{H}_{q_2}(\mathscr{P}_\bullet^{(2)})\big).$$

Finally, by definition,

$$\mathbf{Tor}_q^{A,\mathrm{I}}(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})$$

is a bicomplex whose component of bidegree (h,k) is the A-module

$$\mathbf{Tor}_q^S\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_h^{(1)}, \mathscr{P}_k^{(2)}\big).$$

Consequently

$${}^{(d)}E_{pq}^2 = H_p\left(\mathbf{Tor}_q^S\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)\right).$$

(6.6.6) The theory of hyperhomology for functors of bicomplexes ((0.11.7.3)) shows, as in (III.6.3.4), that for every *flat* Cartan–Eilenberg resolution $M_{\bullet\bullet}^{(i)}$ of $L_{\bullet\bullet}^{(i)}$ in the category of complexes of modules bounded below ($i = 1, 2$), there are canonical isomorphisms of bi-∂-functors

(6.6.6.1)

$$\begin{aligned}\mathbf{Tor}_\bullet^S\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) &\xrightarrow{\sim} H_\bullet\big(M_{\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet}^{(2)}\big)\\ &\xrightarrow{\sim} H_\bullet\big(M_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}\big)\\ &\xrightarrow{\sim} H_\bullet\big(L_{\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet}^{(2)}\big).\end{aligned} \tag{6.6.6.1}$$

(6.6.7) We shall now show that the global hypertor defined in (III.6.6.2), and the six corresponding spectral sequences, do not depend, up to canonical isomorphisms, on the finite affine open coverings $\mathfrak{U}^{(i)}$ used to define them. It is enough to show that if $\mathfrak{V}^{(i)}$ are two other coverings of the same kind, with $\mathfrak{V}^{(i)}$ finer than $\mathfrak{U}^{(i)}$ for $i = 1, 2$, then there are canonical isomorphisms of spectral functors

(6.6.7, t)

$${}^{(t)}E\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \xrightarrow{\sim} {}^{(t)}E\big(\mathfrak{V}^{(1)}, \mathfrak{V}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big), \tag{6.6.7, t}$$

where t is one of a, b, a', b', c, d.

For $i = 1, 2$ there are homomorphisms of bicomplexes

$$C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}) \longrightarrow C^\bullet(\mathfrak{V}^{(i)}, \mathscr{P}_\bullet^{(i)})$$

that are well defined up to homotopy ((G.II.5.7.1)). It follows at once that the homomorphisms (6.6.7, t) are canonically defined and compatible with the boundary operators in their abutments ((0.11.3.2)). Moreover, the calculation of the E_2-terms of the spectral sequences (a), (b), (a′), and (b′) shows that (6.6.7, t) is an isomorphism on their E_2-terms. Since these spectral sequences are biregular, it follows that (6.6.7, t) is an isomorphism for these spectral functors, and hence an isomorphism of bi-∂-functors on their common abutment ((0.11.1.5)).

In particular, if $\mathscr{F}^{(i)}$ is a quasi-coherent $\mathscr{O}_{X^{(i)}}$-module ($i = 1, 2$), the canonical homomorphism

$$\mathbf{Tor}_\bullet^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)}) \longrightarrow \mathbf{Tor}_\bullet^S(\mathfrak{V}^{(1)}, \mathfrak{V}^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)})$$

is bijective. In view of the calculation in (III.6.6.5), the homomorphism (6.6.7, t) is therefore also an isomorphism on E_2-terms for $t = c$ and $t = d$. As above, we conclude that it is an isomorphism of spectral sequences for these two values of t. III | 157

The isomorphisms (6.6.7, t) may be regarded as defining inductive systems of spectral functors on the filtering set of pairs $(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)})$ of finite affine open coverings of $X^{(1)}$ and $X^{(2)}$. We denote the inductive limit by

$${}^{(t)}E(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \quad \text{or} \quad {}^{(t)}E^S(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}).$$

Its abutment is denoted

$$\mathbf{Tor}_\bullet^S(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

and is called the *global hypertor* of the two complexes $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$. If the two complexes are concentrated in degree zero, with terms $\mathscr{F}^{(1)}$ and $\mathscr{F}^{(2)}$, we write

$$\mathbf{Tor}_\bullet^S(X^{(1)}, X^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)}).$$

In accordance with the general conventions,

$$\mathbf{Tor}_q^S(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

will denote the bicomplex whose component of bidegree (h, k) is

$$\mathbf{Tor}_q^S(X^{(1)}, X^{(2)}; \mathscr{P}_h^{(1)}, \mathscr{P}_k^{(2)}).$$

(6.6.8) Under the hypotheses of (III.6.6.1), consider two S-morphisms

$$f_i : X^{(i)} \longrightarrow Y^{(i)},$$

where $Y^{(i)} = \mathrm{Spec}(B_i)$ is an affine S-scheme, so that B_i is an A-algebra ($i = 1, 2$). This defines an A-homomorphism

$$B_i \longrightarrow \Gamma(X^{(i)}, \mathscr{O}_{X^{(i)}})$$

((I.2.2.4)), and consequently each $L_{\bullet\bullet}^{(i)}$ defined in (III.6.6.2) is a bicomplex of B_i-modules. Thus

$$L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}$$

is a quadricomplex of $(B_1 \otimes_A B_2)$-modules, and its six hyperhomology spectral functors may be regarded as taking values in the category of spectral sequences of $(B_1 \otimes_A B_2)$-modules.

Put

$$Y = Y^{(1)} \times_S Y^{(2)} = \mathrm{Spec}(B_1 \otimes_A B_2).$$

We may then consider the quasi-coherent $\mathscr{O}_Y$-modules associated with these modules ((I.1.3.4)). For $t = a, b, a', b', c, d$, write

$${}^{(t)}\mathscr{E}(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

for the six spectral sequences of $\mathscr{O}_Y$-modules

$$\big({}^{(t)}E(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big)^\sim,$$

and write

$$\mathfrak{Tor}_\bullet^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

for their common abutment

$$\big(\mathbf{Tor}_\bullet^S(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big)^\sim.$$

When $\mathscr{P}_\bullet^{(i)}$ is concentrated in degree zero, with term $\mathscr{F}^{(i)}$ ($i = 1, 2$), we write this abutment as

$$\mathfrak{Tor}_\bullet^S(f_1, f_2; \mathscr{F}^{(1)}, \mathscr{F}^{(2)}).$$

6.7 Global hypertor functors of complexes of quasi-coherent modules and Künneth spectral sequences: the general case

(6.7.1) We now generalize the definitions of (III.6.6.8) to the case where S is an arbitrary prescheme, $X^{(i)}$ and $Y^{(i)}$ are S-preschemes, and

$$f_i : X^{(i)} \longrightarrow Y^{(i)}$$

are separated and quasi-compact S-morphisms. For every pair of bounded-below complexes $\mathscr{P}_\bullet^{(i)}$ of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules $(i = 1, 2)$, the aim is to define a quasi-coherent $\mathscr{O}_Y$-module

$$\mathfrak{Tor}_n^S\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)$$

for every n, together with six spectral functors, in such a way that these definitions reduce to those of (III.6.6.8) when S, $Y^{(1)}$, and $Y^{(2)}$ are affine; here $Y = Y^{(1)} \times_S Y^{(2)}$.

First suppose that $S = \operatorname{Spec}(A)$ is affine, while $Y^{(1)}$ and $Y^{(2)}$ are arbitrary. Let $W^{(i)}$ be an affine open subset of $Y^{(i)}$. Then $f_i^{-1}(W^{(i)})$ is a quasi-compact S-scheme, and

$$W = W^{(1)} \times_S W^{(2)}$$

is an affine open subset of Y. Let

$$f_i' : f_i^{-1}(W^{(i)}) \longrightarrow W^{(i)}$$

be the restriction of f_i, and let

$$\mathscr{P}_\bullet'^{(i)} = \mathscr{P}_\bullet^{(i)}\,\big|_{f_i^{-1}(W^{(i)})} \qquad (i = 1, 2).$$

By (III.6.6.8) we then have the spectral sequences of quasi-coherent $(\mathscr{O}_Y|W)$-modules

$${}^{(t)}\mathscr{E}\big(f_1', f_2'; \mathscr{P}_\bullet'^{(1)}, \mathscr{P}_\bullet'^{(2)}\big),$$

and it remains to verify that they satisfy the gluing conditions of (0.3.3.1).

We are immediately reduced to the case in which

$$Y^{(i)} = \operatorname{Spec}(B_i), \qquad W^{(i)} = D(g_i), \qquad g_i \in B_i,$$

so that

$$W = D(g_1 \otimes g_2)$$

in $Y = \operatorname{Spec}(B_1 \otimes_A B_2)$ ((II.4.3.2.1)). Put $X'^{(i)} = f_i^{-1}(W^{(i)})$. We must establish an isomorphism of spectral functors

(6.7.1.1)

$$ {}^{(t)}E\big(X'^{(1)}, X'^{(2)}; \mathscr{P}_\bullet'^{(1)}, \mathscr{P}_\bullet'^{(2)}\big) \xrightarrow{\sim} {}^{(t)}E\big(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \otimes_B B_g, \tag{6.7.1.1}$$

where

$$B = B_1 \otimes_A B_2, \qquad g = g_1 \otimes g_2.$$

III | 158

Choose finite affine open coverings $\mathfrak{U}^{(i)}$ of $X^{(i)}$, and let $\mathfrak{U}'^{(i)}$ be their traces on $X'^{(i)}$. The latter are again coverings by affine open subsets ((I.5.5.10)), and more precisely

$$C^\bullet\big(\mathfrak{U}'^{(i)}, \mathscr{P}_\bullet'^{(i)}\big) = C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}\big) \otimes_{B_i} (B_i)_{g_i}.$$

Put

$$L_{\bullet\bullet}^{(i)} = C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}\big), \qquad L_{\bullet\bullet}'^{(i)} = C^\bullet\big(\mathfrak{U}'^{(i)}, \mathscr{P}_\bullet'^{(i)}\big).$$

Then

$$L_{\bullet\bullet}'^{(1)} \otimes_A L_{\bullet\bullet}'^{(2)} = \big(L_{\bullet\bullet}^{(1)} \otimes_{B_1} (B_1)_{g_1}\big) \otimes_A \big(L_{\bullet\bullet}^{(2)} \otimes_{B_2} (B_2)_{g_2}\big).$$

Since

$$B_g = (B_1)_{g_1} \otimes_A (B_2)_{g_2}$$

up to canonical isomorphism, we likewise have, up to canonical isomorphism,

$$L_{\bullet\bullet}'^{(1)} \otimes_A L_{\bullet\bullet}'^{(2)} = \big(L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}\big) \otimes_B B_g.$$

Let $M^{(i)}_{\bullet\bullet\bullet}$ be a projective Cartan–Eilenberg resolution of $L^{(i)}_{\bullet\bullet}$, chosen to consist of B_i-modules. Because $(B_i)_{g_i}$ is flat over B_i,

$$M'^{(i)}_{\bullet\bullet\bullet} = M^{(i)}_{\bullet\bullet\bullet} \otimes_{B_i} (B_i)_{g_i}$$

is a projective Cartan–Eilenberg resolution of $L'^{(i)}_{\bullet\bullet}$. Moreover,

$$M'^{(1)}_{\bullet\bullet\bullet} \otimes_A M'^{(2)}_{\bullet\bullet\bullet} = \left(M^{(1)}_{\bullet\bullet\bullet} \otimes_A M^{(2)}_{\bullet\bullet\bullet}\right) \otimes_B B_g.$$

The desired isomorphism (6.7.1.1) now follows immediately from the definitions of the hyperhomology of a bicomplex and from the exactness of $G \longmapsto G \otimes_B B_g$ on B-modules.

(6.7.2) Now suppose that S is arbitrary, and let

$$u_i : Y^{(i)} \longrightarrow S \qquad (i = 1, 2)$$

be the structural morphisms. Choose an affine open covering (S_α) of S, put

$$Y^{(i)}_\alpha = u_i^{-1}(S_\alpha), \qquad X^{(i)}_\alpha = f_i^{-1}(Y^{(i)}_\alpha),$$

and let

$$f_{i\alpha} : X^{(i)}_\alpha \longrightarrow Y^{(i)}_\alpha$$

be the restriction of f_i, which is separated and quasi-compact. The schemes

$$Y_\alpha = Y^{(1)}_\alpha \times_{S_\alpha} Y^{(2)}_\alpha$$

form an open covering of Y, and on every Y_α, (III.6.7.1) defines the spectral functors

$${}^{(t)}\mathscr{E}^{S_\alpha}\left(f_{1\alpha}, f_{2\alpha}; \mathscr{P}^{(1)}_\bullet | X^{(1)}_\alpha, \mathscr{P}^{(2)}_\bullet | X^{(2)}_\alpha\right).$$

It remains again to show that these functors satisfy the gluing conditions.

We reduce immediately to the following situation: $S = \mathrm{Spec}(A)$ is affine, $S' = D(h)$ with $h \in A$, and $u_i(Y^{(i)}) \subset S'$. We may moreover assume that $Y^{(i)} = \mathrm{Spec}(B_i)$ is affine. It is then enough to define canonical isomorphisms

(6.7.2.1)

$${}^{(t)}E^S\left(X^{(1)}, X^{(2)}; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right) \xrightarrow{\sim} {}^{(t)}E^{S'}\left(X^{(1)}, X^{(2)}; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right). \tag{6.7.2.1}$$

With the notation of (III.6.6.2), the $L^{(i)}_{\bullet\bullet}$ consist of A_h-modules, and hence, up to canonical isomorphism,

$$L^{(1)}_{\bullet\bullet} \otimes_{A_h} L^{(2)}_{\bullet\bullet} = L^{(1)}_{\bullet\bullet} \otimes_A L^{(2)}_{\bullet\bullet}.$$

Since a projective Cartan–Eilenberg resolution $M^{(i)}_{\bullet\bullet\bullet}$ of $L^{(i)}_{\bullet\bullet}$ may be chosen to consist of A_h-modules, the desired canonical isomorphism follows immediately.

Theorem (6.7.3). — *Let S be a prescheme, let $f_i : X^{(i)} \to Y^{(i)}$ be a separated and quasi-compact S-morphism of S-preschemes, and let $\mathscr{P}^{(i)}_\bullet$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules $(i = 1, 2)$. Write $u_i : Y^{(i)} \to S$ for the structural morphisms, and put*

$$Y = Y^{(1)} \times_S Y^{(2)}.$$

There exists a bi-∂-functor

$$\mathfrak{Tor}^S_\bullet\left(f_1, f_2; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right)$$

with values in the category of quasi-coherent $\mathscr{O}_Y$-modules, such that if S_α is an affine open subset of S, $V^{(i)}$ is an affine open subset of $Y^{(i)}_\alpha = u_i^{-1}(S_\alpha)$ $(i = 1, 2)$, and $V = V^{(1)} \times_{S_\alpha} V^{(2)}$, then

$$\begin{aligned}&\mathfrak{Tor}^S_\bullet\left(f_1, f_2; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right)\Big|_V \\ &\quad = \left(\mathbf{Tor}^{S_\alpha}_\bullet\left(f_1^{-1}(V^{(1)}), f_2^{-1}(V^{(2)}); \mathscr{P}^{(1)}_\bullet | f_1^{-1}(V^{(1)}), \mathscr{P}^{(2)}_\bullet | f_2^{-1}(V^{(2)})\right)\right)^\sim.\end{aligned}$$

This bifunctor is the common abutment of six biregular spectral functors

$${}^{(t)}\mathscr{E}\left(f_1, f_2; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right) \qquad (t = a, b, a', b', c, d),$$

whose E_2-terms are

$$
\begin{aligned}
{}^{(a)}\mathscr{E}_{pq}^2 &= \bigoplus_{q_1+q_2=q} \mathscr{T}or_p^S\big(\mathscr{H}^{-q_1}(f_1,\mathscr{P}_\bullet^{(1)}),\mathscr{H}^{-q_2}(f_2,\mathscr{P}_\bullet^{(2)})\big),\\
{}^{(b)}\mathscr{E}_{pq}^2 &= \mathscr{H}^{-p}\big(f_1\times_S f_2,\mathfrak{Tor}_q^S(\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)})\big),\\
{}^{(a')}\mathscr{E}_{pq}^2 &= \bigoplus_{q_1+q_2=q} \mathfrak{Tor}_p^S\big(\mathscr{H}^{-q_1}(f_1,\mathscr{P}_\bullet^{(1)}),\mathscr{H}^{-q_2}(f_2,\mathscr{P}_\bullet^{(2)})\big),\\
{}^{(b')}\mathscr{E}_{pq}^2 &= \mathscr{H}^{-p}\big(f_1\times_S f_2,\mathscr{T}or_q^S(\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)})\big),\\
{}^{(c)}\mathscr{E}_{pq}^2 &= \bigoplus_{q_1+q_2=q} \mathfrak{Tor}_p^S\big(f_1,f_2;\mathscr{H}_{q_1}(\mathscr{P}_\bullet^{(1)}),\mathscr{H}_{q_2}(\mathscr{P}_\bullet^{(2)})\big),\\
{}^{(d)}\mathscr{E}_{pq}^2 &= \mathscr{H}_p\Big(\mathfrak{Tor}_q^S\big(f_1,f_2;\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}\big)\Big).
\end{aligned}
$$

The spectral sequences (a) *and* (b) *are called the* Künneth spectral sequences. *Notice that* (a) *and* (a′), *respectively* (b) *and* (b′), *are identical when $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$ are concentrated in degree zero. In that case,* (c) *and* (d) *are degenerate and are therefore of no interest.*

Remark (6.7.4). — *The global hypertors just defined include as special cases both the hypercohomology modules defined in (III.6.2.1) and the local hypertors defined in (III.6.5.3). Indeed, for every quasi-compact and separated morphism $f : X \to Y$ and every bounded-below complex $\mathscr{P}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules, there is a canonical isomorphism of ∂-functors in $\mathscr{P}_\bullet$:*

(6.7.4.1)

$$\mathfrak{Tor}_n^Y\big(f,1_Y;\mathscr{P}_\bullet,\mathscr{O}_Y\big)\xrightarrow{\sim}\mathscr{H}^{-n}(f,\mathscr{P}_\bullet)\qquad(n\in\mathbf{Z}). \tag{6.7.4.1}$$

The gluing methods of (III.6.7.2) immediately reduce the proof to the case in which Y is affine. By (III.6.2.2), both sides of (6.7.4.1) *may then be computed using the same finite covering $\mathfrak{U}$ of X by affine open subsets and, for the first side, the covering of Y consisting of Y itself. With the notation of (III.6.6.2), the bicomplex $L_{\bullet\bullet}^{(2)}$ is then concentrated in bidegree $(0,0)$, where it is equal to A, and the conclusion follows from (0.11.7.5).*

For a generalization, see (III.6.7.7). Notice, however, that if $\mathscr{O}_Y$ on the left side of (6.7.4.1) *is replaced by an arbitrary quasi-coherent $\mathscr{O}_Y$-module $\mathscr{F}$, there is in general no longer an isomorphism with*

$$\mathscr{H}^{-n}\big(f,\mathscr{P}_\bullet\otimes_{\mathscr{O}_Y}\mathscr{F}\big),$$

although the bicomplex $L_{\bullet\bullet}^{(1)}\otimes_A L_{\bullet\bullet}^{(2)}$ occurring in the preceding calculation still identifies with

$$C^\bullet\big(\mathfrak{U},\mathscr{P}_\bullet\otimes_Y\mathscr{F}\big).$$

On the other hand, there is a canonical isomorphism of bi-∂-functors

(6.7.4.2)

$$\mathfrak{Tor}_n^S\big(1_{X^{(1)}},1_{X^{(2)}};\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}\big)\xrightarrow{\sim}\mathscr{T}or_n^S\big(\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}\big). \tag{6.7.4.2}$$

Indeed, (III.6.7.1) and (III.6.7.2) reduce us again to the case in which S and the $X^{(i)}$ are affine. In computing the first side of (6.7.4.2), *we may then take $\mathfrak{U}^{(i)}$ to consist of the single member $X^{(i)}$.* **III** | 160

With the notation of (III.6.6.2), $L_{\bullet\bullet}^{(i)}$ then reduces to

$$\Gamma(X^{(i)},\mathscr{P}_\bullet^{(i)}),$$

regarded as a bicomplex whose components of first degree different from zero vanish. The equality of the two sides of (6.7.4.2) *follows from (III.6.4.1.1) and (III.6.3.1).*

Proposition (6.7.5). — *Let*

$$u:\mathscr{P}_\bullet^{(1)}\longrightarrow\mathscr{Q}_\bullet^{(1)}$$

be a homomorphism of bounded-below complexes of quasi-coherent $\mathscr{O}_{X^{(1)}}$-modules such that the induced homomorphism

$$\mathscr{H}_\bullet(u):\mathscr{H}_\bullet(\mathscr{P}_\bullet^{(1)})\longrightarrow\mathscr{H}_\bullet(\mathscr{Q}_\bullet^{(1)})$$

is an isomorphism. Then the homomorphisms

$${}^{(t)}\mathscr{E}\big(f_1,f_2;\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}\big)\longrightarrow{}^{(t)}\mathscr{E}\big(f_1,f_2;\mathscr{Q}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}\big)$$

induced by u are isomorphisms for $t = a, b, c$.

Proof. — For the spectral sequence (c), the assertion follows from biregularity and from the fact that, by hypothesis, the homomorphism under consideration is an isomorphism on E_2-terms ((0.11.1.5)). This already shows that

$$\mathfrak{Tor}_n^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \longrightarrow \mathfrak{Tor}_n^S(f_1, f_2; \mathscr{Q}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

is an isomorphism. Applying (6.7.4.1) and (6.7.4.2), we first see that the homomorphisms

$$\mathscr{H}^{-n}(f_1, \mathscr{P}_\bullet^{(1)}) \longrightarrow \mathscr{H}^{-n}(f_1, \mathscr{Q}_\bullet^{(1)})$$

and

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \longrightarrow \mathscr{T}or_n^S(\mathscr{Q}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

induced by u are isomorphisms. The assertion for (a) and (b) now follows from the biregularity of these spectral sequences ((III.6.7.3)) and the bijectivity of the induced homomorphisms on E_2-terms ((0.11.1.5)).

Notice also that if $u : \mathscr{P}_\bullet^{(1)} \to \mathscr{Q}_\bullet^{(1)}$ is a homotopy equivalence, then it induces canonical isomorphisms

$${}^{(t)}\mathscr{E}(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \xrightarrow{\sim} {}^{(t)}\mathscr{E}(f_1, f_2; \mathscr{Q}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

for all six spectral sequences. Indeed, when S and the $Y^{(i)}$ are affine, u induces a homotopy equivalence of bicomplexes

$$C^\bullet(\mathfrak{U}^{(1)}, \mathscr{P}_\bullet^{(1)}) \longrightarrow C^\bullet(\mathfrak{U}^{(1)}, \mathscr{Q}_\bullet^{(1)}),$$

and the assertion follows from the general theory of hyperhomology ((0.11.3.2)). The passage to the general case is made by gluing, using the fact that a homotopy equivalence of complexes induces one between projective Cartan–Eilenberg resolutions of those complexes (M, XVII, 1.2).

Proposition (6.7.6). — *Suppose that either $\mathscr{P}_\bullet^{(1)}$ or $\mathscr{P}_\bullet^{(2)}$ consists of S-flat modules, both complexes being bounded below. There is then a canonical isomorphism of bi-∂-functors*

(6.7.6.1)

$$\mathfrak{Tor}_n^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \xrightarrow{\sim} \mathscr{H}^{-n}(f_1 \times_S f_2, \mathscr{P}_\bullet^{(1)} \otimes_S \mathscr{P}_\bullet^{(2)}). \tag{6.7.6.1}$$

Proof. — First suppose that S, $Y^{(1)}$, and $Y^{(2)}$ are affine, so that we are in the situation of (III.6.6.2), whose notation we retain. Suppose, for example, that $\mathscr{P}_\bullet^{(1)}$ consists of S-flat modules. Using (III.6.6.6), the hypertor is the homology of

$$L_{\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet\bullet}^{(2)},$$

where $M_{\bullet\bullet\bullet}^{(2)}$ is a projective Cartan–Eilenberg resolution of $L_{\bullet\bullet}^{(2)}$ in the sense of (0.11.7.1). On the other hand, the modules $L_{ij}^{(1)}$ are flat over A by the hypothesis ((I.1.4.15.1)). It follows from (0.11.7.5) that there is a canonical isomorphism

(6.7.6.2)

$$\mathbf{Tor}_\bullet^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \xrightarrow{\sim} H_\bullet(L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}). \tag{6.7.6.2}$$

There is also a natural homomorphism of bicomplexes

$$L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)} \longrightarrow C^\bullet(\mathfrak{U}, \mathscr{Q}_\bullet),$$

where $\mathfrak{U}$ is the covering of

$$Z = X^{(1)} \times_S X^{(2)}$$

by the affine open subsets $U_\alpha^{(1)} \times_S U_\beta^{(2)}$, and

$$\mathscr{Q}_\bullet = \mathscr{P}_\bullet^{(1)} \otimes_S \mathscr{P}_\bullet^{(2)}$$

is regarded as the simple complex for the total degree.

The substance of this homomorphism was already given in the calculation of spectral sequence (b′) in (III.6.6.4), with $q = 0$. Retaining the notation used there, one need only observe that there is, first, a natural homomorphism from $N^{\bullet k}$ to

$$C^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathcal{S}_k)$$

((0.11.8.5)); second, a natural homomorphism from the latter complex to

$$P^{\bullet}\big(\Phi^{(1)}, \Phi^{(2)}; \mathcal{S}_k\big)$$

((0.11.8.6)); and finally, a natural homomorphism from this last cochain complex to its subcomplex of alternating cochains ((0.11.8.7)). The resulting homomorphism of bicomplexes induces an isomorphism in homology, as was seen in (III.6.6.4). Composing with (6.7.6.2) therefore gives an isomorphism

(6.7.6.3)

$$\mathbf{Tor}_n^S\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \xrightarrow{\sim} \mathrm{H}^{-n}\big(\mathfrak{U}, \mathscr{P}_\bullet^{(1)} \otimes_S \mathscr{P}_\bullet^{(2)}\big). \tag{6.7.6.3}$$

It remains to prove that the isomorphism just defined is independent of the chosen open coverings. The right side of (6.7.6.3) is canonically isomorphic, by (III.6.2.2), to

$$\mathrm{H}^{-n}\big(X^{(1)} \times_S X^{(2)}, \mathscr{P}_\bullet^{(1)} \otimes_S \mathscr{P}_\bullet^{(2)}\big).$$

For the left side, use (III.6.6.7) and note, with the notation used there, that the following diagram commutes up to homotopy:

$$\begin{array}{ccc} C^{\bullet}(\mathfrak{U}^{(1)}, \mathscr{P}_\bullet^{(1)}) \otimes_A C^{\bullet}(\mathfrak{U}^{(2)}, \mathscr{P}_\bullet^{(2)}) & \longrightarrow & C^{\bullet}(\mathfrak{U}, \mathscr{Q}_\bullet) \\ \downarrow & & \downarrow \\ C^{\bullet}(\mathfrak{V}^{(1)}, \mathscr{P}_\bullet^{(1)}) \otimes_A C^{\bullet}(\mathfrak{V}^{(2)}, \mathscr{P}_\bullet^{(2)}) & \longrightarrow & C^{\bullet}(\mathfrak{V}, \mathscr{Q}_\bullet). \end{array}$$

The horizontal arrows are those defined above. Finally, one passes to the general case by gluing, exactly as in (III.6.7.1) and (III.6.7.2); the details are left to the reader.

Proposition (6.7.7). — *Suppose that $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$ are bounded below, and that either all the modules*

$$\mathscr{H}^{-n}(f_1, \mathscr{P}_\bullet^{(1)})$$

or all the modules

$$\mathscr{H}^{-n}(f_2, \mathscr{P}_\bullet^{(2)})$$

are S-flat. There is then a canonical isomorphism of bi-∂-functors, for $n \in \mathbf{Z}$,

(6.7.7.1)

$$\mathfrak{Tor}_n^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \xrightarrow{\sim} \bigoplus_{q_1+q_2=n} \mathscr{H}^{-q_1}(f_1, \mathscr{P}_\bullet^{(1)}) \otimes_S \mathscr{H}^{-q_2}(f_2, \mathscr{P}_\bullet^{(2)}). \tag{6.7.7.1}$$

Proof. — By (III.6.5.8), spectral sequence (a) of (III.6.7.3) degenerates. The assertion therefore follows immediately from (0.11.1.6), since that spectral sequence is biregular ((III.6.7.3)).

Theorem (6.7.8). — *Suppose that the following conditions hold:*

1° *The complexes $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$ are bounded below.*

2° *Either $\mathscr{P}_\bullet^{(1)}$ or $\mathscr{P}_\bullet^{(2)}$ consists of S-flat modules.* **III** | 162

3° *Either all the modules $\mathscr{H}^{-n}(f_1, \mathscr{P}_\bullet^{(1)})$ or all the modules $\mathscr{H}^{-n}(f_2, \mathscr{P}_\bullet^{(2)})$ are S-flat.*

There is then a canonical isomorphism of bi-∂-functors, for $n \in \mathbf{Z}$,

(6.7.8.1)

$$\mathscr{H}^n\big(f_1 \times_S f_2, \mathscr{P}_\bullet^{(1)} \otimes_S \mathscr{P}_\bullet^{(2)}\big) \xrightarrow{\sim} \bigoplus_{n_1+n_2=n} \mathscr{H}^{n_1}(f_1, \mathscr{P}_\bullet^{(1)}) \otimes_S \mathscr{H}^{n_2}(f_2, \mathscr{P}_\bullet^{(2)}). \tag{6.7.8.1}$$

This is the Künneth formula.

Proof. — The formula follows from (III.6.7.6) and (III.6.7.7).

When S, $Y^{(1)}$, and $Y^{(2)}$ are affine, the inverse of (6.7.8.1) is obtained, with the notation of (III.6.7.6), from the homomorphism of bicomplexes

$$C^{\bullet}(\mathfrak{U}^{(1)}, \mathscr{P}_\bullet^{(1)}) \otimes_A C^{\bullet}(\mathfrak{U}^{(2)}, \mathscr{P}_\bullet^{(2)}) \longrightarrow C^{\bullet}(\mathfrak{U}, \mathscr{Q}_\bullet)$$

by the procedure defined in (G, I, 2.7), as follows from (G, I, 5.5).

Proposition (6.7.9). — *Suppose that the following three conditions hold:*

1° *The schemes S, $Y^{(1)}$, and $Y^{(2)}$ are locally noetherian; f_1 and f_2 are proper; and either $Y^{(1)}$ or $Y^{(2)}$ is of finite type over S.*

2° *The complexes $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$ are bounded below.*

3° *For every $n \in \mathbf{Z}$, the module $\mathscr{H}_n(\mathscr{P}_\bullet^{(i)})$ is coherent, for $i = 1, 2$.*

Then

$$\mathfrak{Tor}_n^S\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)$$

is a coherent $\mathscr{O}_Y$-module, where $Y = Y^{(1)} \times_S Y^{(2)}$.

Proof. — It follows from (III.6.5.13) that the local hypertor modules

$$\mathscr{T}\mathscr{o}\mathscr{r}_n^S\big(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)$$

are coherent $\mathscr{O}_X$-modules, where $X = X^{(1)} \times_S X^{(2)}$. Indeed, each $X^{(i)}$ is locally noetherian, and one of them is, by hypothesis, of finite type over S ((I.6.3.4) and (I.6.3.8)); hence X is locally noetherian.

Since Y is locally noetherian and $f_1 \times_S f_2$ is proper ((II.5.4.2)), it follows from (III.6.2.5) that the terms ${}^{(b)}\mathscr{E}_{pq}^2$ of (III.6.7.3) are coherent $\mathscr{O}_Y$-modules. All the spectral sequences of (III.6.7.3) are biregular by hypothesis 2°, so the conclusion follows from (0.11.1.8).

(6.7.10) Let $Y'^{(i)}$ be two S-preschemes and let

$$v_i : Y'^{(i)} \longrightarrow Y^{(i)} \qquad (i = 1, 2)$$

be two S-morphisms. Write

$$v = v_1 \times_S v_2 : Y' \longrightarrow Y, \qquad Y' = Y'^{(1)} \times_S Y'^{(2)}.$$

For $i = 1, 2$, let $X'^{(i)}$ be an S-prescheme, and let

$$u_i : X'^{(i)} \longrightarrow X^{(i)}, \qquad f_i' : X'^{(i)} \longrightarrow Y'^{(i)}$$

be S-morphisms such that the diagrams

(6.7.10.1)

$$\begin{array}{ccc} X'^{(i)} & \xrightarrow{u_i} & X^{(i)} \\ {\scriptstyle f_i'}\downarrow & & \downarrow{\scriptstyle f_i} \\ Y'^{(i)} & \xrightarrow[v_i]{} & Y^{(i)} \end{array} \tag{6.7.10.1}$$

commute, the morphisms f_i' being separated and quasi-compact. There are then canonical $\mathscr{O}_{Y'}$-linear homomorphisms of spectral functors

(6.7.10.2)

$$v^{*(t)}\mathscr{E}\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow {}^{(t)}\mathscr{E}\big(f_1', f_2'; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big) \tag{6.7.10.2}$$

for $t = a, a', b, b', c, d$.

To define them, first suppose that

$$S = \mathrm{Spec}(A), \qquad Y^{(i)} = \mathrm{Spec}(B_i), \qquad Y'^{(i)} = \mathrm{Spec}(B_i')$$

are affine. The schemes $X^{(i)}$ and $X'^{(i)}$ are then quasi-compact. To compute

$${}^{(t)}\mathscr{E}\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big),$$

choose, as in (III.6.6.1), finite affine open coverings $\mathfrak{U}^{(i)}$ of $X^{(i)}$. To compute

$${}^{(t)}\mathscr{E}\big(f_1', f_2'; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big),$$

choose finite affine open coverings $\mathfrak{U}'^{(i)}$ of $X'^{(i)}$, respectively finer than the inverse-image coverings $u_i^{-1}(\mathfrak{U}^{(i)})$.

There is a canonical homomorphism from the bicomplex

$$C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}\big) = L_{\bullet\bullet}^{(i)}$$

to

$$C^\bullet\big(u_i^{-1}(\mathfrak{U}^{(i)}), u_i^*(\mathscr{P}_\bullet^{(i)})\big)$$

((0.4.4.3.2)). A simplicial map (G, II, 5.7) from $\mathfrak{U}'^{(i)}$ to $u_i^{-1}(\mathfrak{U}^{(i)})$ defines a homomorphism of bicomplexes

$$C^\bullet\big(u_i^{-1}(\mathfrak{U}^{(i)}), u_i^*(\mathscr{P}_\bullet^{(i)})\big) \longrightarrow C^\bullet\big(\mathfrak{U}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big).$$

By composition one obtains a homomorphism

$$L_{\bullet\bullet}^{(i)} \longrightarrow L_{\bullet\bullet}'^{(i)} = C^\bullet\big(\mathfrak{U}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big).$$

Changing the simplicial map replaces this homomorphism by a homotopic one (G, II, 5.7.1). Thus one obtains a well-defined homomorphism of spectral functors

(6.7.10.3)

$$^{(t)}\mathscr{E}\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow {}^{(t)}\mathscr{E}\big(\mathfrak{U}'^{(1)}, \mathfrak{U}'^{(2)}; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big). \tag{6.7.10.3}$$

Suppose that $\mathfrak{V}^{(i)}$ is a finite affine open covering of $X^{(i)}$ finer than $\mathfrak{U}^{(i)}$, and that $\mathfrak{V}'^{(i)}$ is a finite affine open covering of $X'^{(i)}$ finer than both $\mathfrak{U}'^{(i)}$ and $u_i^{-1}(\mathfrak{V}^{(i)})$. One verifies immediately that the diagram

$$\begin{array}{ccc} C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}) & \longrightarrow & C^\bullet(\mathfrak{V}^{(i)}, \mathscr{P}_\bullet^{(i)}) \\ \downarrow & & \downarrow \\ C^\bullet\big(\mathfrak{U}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big) & \longrightarrow & C^\bullet\big(\mathfrak{V}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big) \end{array}$$

commutes. Consequently, (6.7.10.3) is essentially independent of the coverings used. We have therefore defined a homomorphism of A-modules

(6.7.10.4)

$$^{(t)}E\big(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow {}^{(t)}E\big(X'^{(1)}, X'^{(2)}; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big). \tag{6.7.10.4}$$

By the definition of the $u_i^*(\mathscr{P}_\bullet^{(i)})$ and the commutativity of (6.7.10.1), this is also a homomorphism of $(B_1 \otimes_A B_2)$-modules. Since the right-hand side of (6.7.10.4) consists of $(B_1' \otimes_A B_2')$-modules, it canonically induces a homomorphism of $(B_1' \otimes_A B_2')$-modules

(6.7.10.5)

$$^{(t)}E\big(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \otimes_{B_1 \otimes_A B_2} (B_1' \otimes_A B_2') \longrightarrow {}^{(t)}E\big(X'^{(1)}, X'^{(2)}; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big). \tag{6.7.10.5}$$

In view of (I.1.6.5), this is precisely (6.7.10.2) in the affine case.

It remains to pass to the general case by the gluing procedures of (III.6.7.1) and (III.6.7.2). The second passage is immediate. For the first, take, as in (III.6.7.1), elements $g_i \in B_i$ and their images $g_i' \in B_i'$, and set

$$g = g_1 \otimes g_2 \in B = B_1 \otimes_A B_2, \qquad g' = g_1' \otimes g_2' \in B' = B_1' \otimes_A B_2'.$$

Everything then reduces to the canonical isomorphism

$$(M \otimes_B B')_{g'} \xrightarrow{\sim} M_g \otimes_{B_g} B'_{g'}$$

((0.1.5.4)). The details are left to the reader.

(6.7.11) The theory of global hypertor developed above for two S-morphisms $X^{(i)} \to Y^{(i)}$ and two bounded-below complexes $\mathscr{P}_\bullet^{(i)}$ of quasi-coherent modules extends immediately to the following general situation. Let S be a prescheme, let $(Y^{(i)})_{i\in I}$ be a finite family of S-preschemes, let

$$f_i : X^{(i)} \longrightarrow Y^{(i)} \qquad (i \in I)$$

be a finite family of separated and quasi-compact S-morphisms, and for each i let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. If Y is the product of the S-preschemes $Y^{(i)}$, then for every $n \in \mathbf{Z}$ one defines a quasi-coherent $\mathscr{O}_Y$-module

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I}; (\mathscr{P}_\bullet^{(i)})_{i\in I}\big).$$

These modules form a covariant ∂-functor in each complex $\mathscr{P}_\bullet^{(i)}$; moreover, this functor is the common abutment of six spectral functors

$$^{(t)}\mathscr{E}\big((f_i)_{i\in I};(\mathscr{P}_\bullet^{(i)})_{i\in I}\big).$$

We leave to the reader the repetition, in this general case, of the definitions and arguments given above for $I=\{1,2\}$.

When I has a single element, this construction recovers the hypercohomology $\mathscr{H}^\bullet(f,\mathscr{P}_\bullet)$ defined in (III.6.2.7), as already noted in (III.6.7.4). When I is the interval $1\leq i\leq m$ in $\mathbf{N}$, we write

$$\mathfrak{Tor}_n^S\big(f_1,\dots,f_m;\mathscr{P}_\bullet^{(1)},\dots,\mathscr{P}_\bullet^{(m)}\big)$$

for

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I};(\mathscr{P}_\bullet^{(i)})_{i\in I}\big).$$

Proposition (6.7.12). — *With the notation of (III.6.7.11), let J be a subset of I such that, for $i\in I-J$,*

$$X^{(i)}=Y^{(i)}=S,$$

the morphism f_i is the identity, and $\mathscr{P}_\bullet^{(i)}$ is the complex concentrated in degree 0 with value $\mathscr{O}_S$. There is then a canonical isomorphism of ∂-functors

(6.7.12.1)

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I};(\mathscr{P}_\bullet^{(i)})_{i\in I}\big)\xrightarrow{\sim}\mathfrak{Tor}_n^S\big((f_i)_{i\in J};(\mathscr{P}_\bullet^{(i)})_{i\in J}\big). \tag{6.7.12.1}$$

Proof. — It is enough to define this isomorphism when S and the $Y^{(i)}$ are affine, since the gluing is carried out as usual. For $i\in I-J$, take $\mathfrak{U}^{(i)}$ to be the covering consisting of the single set S. Then

$$L_{\bullet\bullet}^{(i)}=C^\bullet\big(\mathfrak{U}^{(i)},\mathscr{P}_\bullet^{(i)}\big)$$

has only one nonzero term, in bidegree $(0,0)$, equal to

$$\Gamma(S,\mathscr{O}_S)=A(S).$$

The isomorphism (6.7.12.1) is therefore immediate.

Remark (6.7.13). — *With the notation of (III.6.7.3), consider the canonical S-isomorphism*

$$Y^{(1)}\times_S Y^{(2)}\longrightarrow Y^{(2)}\times_S Y^{(1)}$$

of (I.3.3.5). Under this isomorphism, the image of

$$\mathfrak{Tor}_n^S\big(f_1,f_2;\mathscr{P}_\bullet^{(1)},\mathscr{P}_\bullet^{(2)}\big)$$

is

$$\mathfrak{Tor}_n^S\big(f_2,f_1;\mathscr{P}_\bullet^{(2)},\mathscr{P}_\bullet^{(1)}\big).$$

This is a local assertion, so it reduces to the case considered in (III.6.6.2). If $M_{\bullet\bullet\bullet}^{(i)}$ is a projective Cartan–Eilenberg resolution of $L_{\bullet\bullet}^{(i)}$ for $i=1,2$, the isomorphism in question interchanges

$$M_{\bullet\bullet\bullet}^{(1)}\otimes_A M_{\bullet\bullet\bullet}^{(2)}\qquad\text{and}\qquad M_{\bullet\bullet\bullet}^{(2)}\otimes_A M_{\bullet\bullet\bullet}^{(1)}.$$

The assertion follows by taking the homology of the simple complexes associated with these tricomplexes.

6.8 The associativity spectral sequences of global hypertor

(6.8.1) Retain the hypotheses and notation of (III.6.7.11); in particular, the complexes $\mathscr{P}_\bullet^{(i)}$ are bounded below. Let $(I_j)_{j\in J}$ be a partition of the index set I. We seek an "associativity" relation between

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I};(\mathscr{P}_\bullet^{(i)})_{i\in I}\big)$$

and the "partial" hypertors

$$\mathcal{T}_{\bullet j}=\mathfrak{Tor}_\bullet^S\big((f_i)_{i\in I_j};(\mathscr{P}_\bullet^{(i)})_{i\in I_j}\big).$$

To simplify the notation, we shall restrict attention to the case where I is the interval $1 \leq i \leq m$ and the partition consists of the two intervals

$$\{1, 2, \ldots, r\} \qquad \text{and} \qquad \{r+1, \ldots, m\}.$$

Proposition (6.8.2). — *There is a canonical biregular spectral functor, called the* associativity spectral functor*, denoted*

$$^{(e)}\mathscr{E}^S\big(f_1, \ldots, f_m; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(m)}\big),$$

or simply $^{(e)}\mathscr{E}$ *with the same arguments, whose abutment is*

$$\mathfrak{Tor}_\bullet^S\big(f_1, \ldots, f_m; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(m)}\big)$$

and whose E_2*-term is*

$$^{(e)}\mathscr{E}_{pq}^2 = \bigoplus_{q_1+q_2=q} \mathscr{T}or_p^S\Big(\mathfrak{Tor}_{q_1}^S\big(f_1, \ldots, f_r; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(r)}\big), \mathfrak{Tor}_{q_2}^S\big(f_{r+1}, \ldots, f_m; \mathscr{P}_\bullet^{(r+1)}, \ldots, \mathscr{P}_\bullet^{(m)}\big)\Big).$$

Proof. — Here Y is canonically identified with $Z^{(1)} \times_S Z^{(2)}$, where

$$Z^{(1)} = Y^{(1)} \times_S \cdots \times_S Y^{(r)} \quad \text{and} \quad Z^{(2)} = Y^{(r+1)} \times_S \cdots \times_S Y^{(m)}.$$

It is enough to treat the case in which S and all the $Y^{(i)}$ are affine. The passage from this case to the general one uses the methods of (III.6.7.1) and (III.6.7.2); the details are left to the reader. Thus it is enough to prove the following affine statement.

Corollary (6.8.3). — *Let A be a ring, let $S = \mathrm{Spec}(A)$, and let $X^{(i)}$, $1 \leq i \leq m$, be quasi-compact S-schemes. For each i, let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. There is a canonical biregular spectral functor with abutment*

$$\mathbf{Tor}_\bullet^S\big(X^{(1)}, \ldots, X^{(m)}; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(m)}\big)$$

and E_2*-term*

$$^{(e)}E_{pq}^2 = \bigoplus_{q_1+q_2=q} \mathrm{Tor}_p^A\Big(\mathbf{Tor}_{q_1}^S\big(X^{(1)}, \ldots, X^{(r)}; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(r)}\big), \mathbf{Tor}_{q_2}^S\big(X^{(r+1)}, \ldots, X^{(m)}; \mathscr{P}_\bullet^{(r+1)}, \ldots, \mathscr{P}_\bullet^{(m)}\big)\Big).$$

Proof. — Following the definition of (III.6.6.2), choose for each i a finite affine open covering $\mathfrak{U}^{(i)}$ of $X^{(i)}$, form the bicomplex

$$L_{\bullet\bullet}^{(i)} = C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}\big),$$

and choose a projective Cartan–Eilenberg resolution $M_{\bullet\bullet\bullet}^{(i)}$ of this bicomplex in the sense of (0.11.7.1). The hypertor in question is the homology of the tensor product

$$M_{\bullet\bullet\bullet} = \bigotimes_{i=1}^m M_{\bullet\bullet\bullet}^{(i)}.$$

Regard $M_{\bullet\bullet\bullet}$ as a simple complex $N_\bullet$, the tensor product of the two simple complexes

$$N'_\bullet = \bigotimes_{i=1}^r M_{\bullet\bullet\bullet}^{(i)}, \qquad N''_\bullet = \bigotimes_{i=r+1}^m M_{\bullet\bullet\bullet}^{(i)},$$

where $N'_\bullet$ and $N''_\bullet$ are graded by the sums of the total degrees of the $M_{\bullet\bullet\bullet}^{(i)}$.

The A-modules forming $N'_\bullet$ and $N''_\bullet$ are projective. Hence (III.6.5.9) gives

$$H_\bullet(M_{\bullet\bullet\bullet}) = \mathbf{Tor}_\bullet^A(N'_\bullet, N''_\bullet).$$

The required spectral sequence is therefore precisely (III.6.3.2.2) applied to $N'_\bullet$ and $N''_\bullet$, using the preceding interpretation of their homology modules for the two partial products

$$X^{(1)} \times_S \cdots \times_S X^{(r)} \quad \text{and} \quad X^{(r+1)} \times_S \cdots \times_S X^{(m)}.$$

Finally, the regularity assertions follow from (III.6.3.2). Indeed, since the $\mathscr{P}_\bullet^{(i)}$ are bounded below, so are the $M_{\bullet\bullet\bullet}^{(i)}$, and therefore so are $N'_\bullet$ and $N''_\bullet$.

6.9 The base change spectral sequences of global hypertor

(6.9.1) Retain the hypotheses and notation of (III.6.7.11); in particular, the complexes $\mathscr{P}_\bullet^{(i)}$ are bounded below. Let $g : S' \to S$ be a morphism of preschemes, and set

$$Y'^{(i)} = Y^{(i)}_{(S')}, \qquad X'^{(i)} = X^{(i)}_{(S')}, \qquad \mathscr{P}'^{(i)}_\bullet = \mathscr{P}^{(i)}_\bullet \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}.$$

Thus $\mathscr{P}'^{(i)}_\bullet$ is a complex of quasi-coherent $\mathscr{O}_{X'^{(i)}}$-modules. Let

$$f'_i = (f_i)_{(S')} : X'^{(i)} \longrightarrow Y'^{(i)};$$

this morphism is separated and quasi-compact ((I.5.5.1) and (III.6.6.4)).

We shall study the relation between the quasi-coherent $\mathscr{O}_{Y'}$-modules

$$\mathfrak{Tor}_n^{S'}\big((f'_i)_{i\in I}; (\mathscr{P}'^{(i)}_\bullet)_{i\in I}\big)$$

and

$$\mathfrak{Tor}_n^{S}\big((f_i)_{i\in I}; (\mathscr{P}^{(i)}_\bullet)_{i\in I}\big) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'},$$

where

$$Y' = Y \times_S S' = Y_{(S')}.$$

The following particularly simple case reduces to (I.1.4.15) when I has a single element and $\mathscr{P}_\bullet$ consists of a single module.

Proposition (6.9.2). — *If $g : S' \to S$ is flat, there is a canonical isomorphism of ∂-functors in the complexes $\mathscr{P}_\bullet^{(i)}$:*

(6.9.2.1)

$$\mathfrak{Tor}_n^{S}\big((f_i)_{i\in I}; (\mathscr{P}^{(i)}_\bullet)_{i\in I}\big) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} \xrightarrow{\sim} \mathfrak{Tor}_n^{S'}\big((f'_i)_{i\in I}; (\mathscr{P}'^{(i)}_\bullet)_{i\in I}\big). \tag{6.9.2.1}$$

Proof. — It is enough to consider the case where S, S', and all the $Y^{(i)}$ are affine; the gluing is carried out by the methods of (III.6.7.1) and (III.6.7.2). Write

$$S = \operatorname{Spec}(A), \qquad S' = \operatorname{Spec}(A'),$$

and choose for each i a finite affine open covering $\mathfrak{U}^{(i)}$ of $X^{(i)}$. If $u_i : X'^{(i)} \to X^{(i)}$ is the canonical projection, then

$$\mathfrak{U}'^{(i)} = u_i^{-1}(\mathfrak{U}^{(i)})$$

is a finite affine open covering of $X'^{(i)}$ ((II.1.5.5)), and

$$C^\bullet\big(\mathfrak{U}'^{(i)}, \mathscr{P}'^{(i)}_\bullet\big) = C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)}_\bullet\big) \otimes_A A'.$$

Let $M^{(i)}_{\bullet\bullet\bullet}$ be a projective Cartan–Eilenberg resolution, in the sense of (0.11.7.1), of

$$L^{(i)}_{\bullet\bullet} = C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)}_\bullet\big)$$

by A-modules. Since A' is flat over A,

$$M^{(i)}_{\bullet\bullet\bullet} \otimes_A A'$$

is a projective Cartan–Eilenberg resolution, over A', of $L^{(i)}_{\bullet\bullet} \otimes_A A'$. The same flatness gives

$$H_\bullet\left(\bigotimes_{i=1}^{m}(M^{(i)}_{\bullet\bullet\bullet} \otimes_A A')\right) = H_\bullet\left(\bigotimes_{i=1}^{m} M^{(i)}_{\bullet\bullet\bullet}\right) \otimes_A A',$$

which proves (6.9.2.1).

(6.9.2.2) When I consists of the single element 1, (6.9.2.1) also follows directly from (III.6.7.7). Indeed, take

$$Y_2 = X_2 = S', \qquad f_2 = 1_{S'},$$

and let $\mathscr{P}^{(2)}_\bullet$ be concentrated in degree 0 with value $\mathscr{O}_{S'}$. Then

$$\mathscr{H}^n(1_{S'}, \mathscr{O}_{S'})$$

vanishes for $n \neq 0$ and equals $\mathscr{O}_{S'}$ for $n = 0$ ((III.6.2.1)).

In the general case, replace $\mathscr{O}_{S'}$ by a bounded-below complex $\mathscr{Q}'_\bullet$ of quasi-coherent $\mathscr{O}_{S'}$-modules. Taking, for simplicity, $I = \{1, 2, \dots, m\}$, one can then consider the ∂-functor

$$\mathfrak{Tor}^S_\bullet(f_1, \dots, f_m, 1_{S'}; \mathscr{P}^{(1)}_\bullet, \dots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet).$$

Proposition (6.9.3). — *There are three canonical biregular spectral functors, denoted*

$${}^{(t)}\mathscr{E}(f_1, \dots, f_m; \mathscr{P}^{(1)}_\bullet, \dots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet), \qquad t = e, f, f',$$

with common abutment

$$\mathfrak{Tor}^S_\bullet(f_1, \dots, f_m, 1_{S'}; \mathscr{P}^{(1)}_\bullet, \dots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet),$$

and whose E_2-terms are respectively as follows.

(6.9.3.1)

$$ {}^{(e)}\mathscr{E}^2_{pq} = \bigoplus_{q'+q''=q} \mathscr{T}or^S_p\Big(\mathfrak{Tor}^S_{q'}(f_1, \dots, f_m; \mathscr{P}^{(1)}_\bullet, \dots, \mathscr{P}^{(m)}_\bullet), \mathscr{H}_{q''}(\mathscr{Q}'_\bullet)\Big). \tag{6.9.3.1}$$

(6.9.3.2)

$$ {}^{(f)}\mathscr{E}^2_{pq} = \bigoplus_{q_1+\cdots+q_{m+1}=q} \mathscr{T}or^{S'}_p\Big(\mathfrak{Tor}^S_{q_1}(f_1, 1_{S'}; \mathscr{P}^{(1)}_\bullet, \mathscr{O}_{S'}), \dots, \mathfrak{Tor}^S_{q_m}(f_m, 1_{S'}; \mathscr{P}^{(m)}_\bullet, \mathscr{O}_{S'}), \mathscr{H}_{q_{m+1}}(\mathscr{Q}'_\bullet)\Big). \tag{6.9.3.2}$$

(6.9.3.3)

$$ {}^{(f')}\mathscr{E}^2_{pq} = \bigoplus_{q_1+\cdots+q_m=q} \mathfrak{Tor}^{S'}_p\Big(f'_1, \dots, f'_m, 1_{S'}; \mathscr{T}or^S_{q_1}(\mathscr{P}^{(1)}_\bullet, \mathscr{O}_{S'}), \dots, \mathscr{T}or^S_{q_m}(\mathscr{P}^{(m)}_\bullet, \mathscr{O}_{S'}), \mathscr{Q}'_\bullet\Big). \tag{6.9.3.3}$$

Proof. — The sequence (e) is precisely the associativity sequence of (III.6.8.2), with $r = m$. To define the other two spectral sequences, it is again enough to treat the case in which S, S', and the $Y^{(i)}$ are affine. The passage to the general case uses the methods of (III.6.7.1) and (III.6.7.2) and is left to the reader. Thus it remains to prove the following affine statement.

Corollary (6.9.4). — *Let A be a ring, let A' be an A-algebra, and put*

$$S = \mathrm{Spec}(A), \qquad S' = \mathrm{Spec}(A').$$

Let $X^{(i)}$, $1 \leq i \leq m$, be quasi-compact S-schemes, and for each i put

$$X'^{(i)} = X^{(i)}_{(S')};$$

this is a quasi-compact S'-scheme. For each i, let $\mathscr{P}^{(i)}_\bullet$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. Finally, let $Q'_\bullet$ be a bounded-below complex of A'-modules, and let $\widetilde{Q}'_\bullet$ be the associated complex.

There are three biregular spectral functors in the $\mathscr{P}^{(i)}_\bullet$ and in $Q'_\bullet$, with common abutment

$$\mathbf{Tor}^S_\bullet(X^{(1)}, \dots, X^{(m)}, S'; \mathscr{P}^{(1)}_\bullet, \dots, \mathscr{P}^{(m)}_\bullet, \widetilde{Q}'_\bullet),$$

and whose E_2-terms are respectively

$$ {}^{(e)}E^2_{pq} = \bigoplus_{q'+q''=q} \mathrm{Tor}^A_p\Big(\mathbf{Tor}^S_{q'}(X^{(1)}, \dots, X^{(m)}; \mathscr{P}^{(1)}_\bullet, \dots, \mathscr{P}^{(m)}_\bullet), H_{q''}(Q'_\bullet)\Big),$$

$$ {}^{(f)}E_{pq}^2 = \bigoplus_{q_1+\cdots+q_{m+1}=q} \operatorname{Tor}_p^{A'}\Big(\mathbf{Tor}_{q_1}^S\big(X^{(1)}, S'; \mathscr{P}_\bullet^{(1)}, \mathscr{O}_{S'}\big), \ldots, \mathbf{Tor}_{q_m}^S\big(X^{(m)}, S'; \mathscr{P}_\bullet^{(m)}, \mathscr{O}_{S'}\big), H_{q_{m+1}}(Q'_\bullet)\Big), $$

and

$$ {}^{(f')}E_{pq}^2 = \bigoplus_{q_1+\cdots+q_m=q} \mathbf{Tor}_p^{S'}\Big(X'^{(1)}, \ldots, X'^{(m)}, S'; \mathscr{Tor}_{q_1}^S\big(\mathscr{P}_\bullet^{(1)}, \mathscr{O}_{S'}\big), \ldots, \mathscr{Tor}_{q_m}^S\big(\mathscr{P}_\bullet^{(m)}, \mathscr{O}_{S'}\big), \widetilde{Q}'_\bullet\Big). $$

Proof. — We do not return to the first of these spectral functors, which was treated in (III.6.8.3) and is included here only for reference. To define the other two, choose for each i a finite affine open covering $\mathfrak{U}^{(i)}$ of $X^{(i)}$. If

$$ u_i : X'^{(i)} \longrightarrow X^{(i)} $$

is the canonical projection, put

$$ \mathfrak{U}'^{(i)} = u_i^{-1}(\mathfrak{U}^{(i)}). $$

By (III.6.6.6), the common abutment is obtained by choosing, for $1 \leq i \leq m$, a projective Cartan–Eilenberg resolution $M_{\bullet\bullet\bullet}^{(i)}$ of

$$ L_{\bullet\bullet}^{(i)} = C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}\big) $$

in the sense of (0.11.7.1), and then taking the homology of the tricomplex

$$ M_{\bullet\bullet\bullet} = M_{\bullet\bullet\bullet}^{(1)} \otimes_A \cdots \otimes_A M_{\bullet\bullet\bullet}^{(m)} \otimes_A Q'_\bullet, $$

where $Q'_\bullet$ is regarded as a tricomplex whose last two degrees are zero. Set

$$ M'^{(i)}_{\bullet\bullet\bullet} = M^{(i)}_{\bullet\bullet\bullet} \otimes_A A'. $$

Since $Q'_\bullet$ is a complex of A'-modules, we may rewrite

$$ M_{\bullet\bullet\bullet} = M'^{(1)}_{\bullet\bullet\bullet} \otimes_{A'} \cdots \otimes_{A'} M'^{(m)}_{\bullet\bullet\bullet} \otimes_{A'} Q'_\bullet. $$

Regard each $M'^{(i)}_{\bullet\bullet\bullet}$ as a simple complex for its total degree. Its terms are projective A'-modules; hence (III.6.3.7), extended to any finite number of complexes, identifies the homology of $M_{\bullet\bullet\bullet}$ with

$$ \operatorname{Tor}_\bullet^{A'}\big(M'^{(1)}_{\bullet\bullet\bullet}, \ldots, M'^{(m)}_{\bullet\bullet\bullet}, Q'_\bullet\big). $$

By (III.6.5.15), this is the abutment of a spectral sequence having the required regularity properties and E_2-term

$$ E_{pq}^2 = \bigoplus_{q_1+\cdots+q_{m+1}=q} \operatorname{Tor}_p^{A'}\Big(H_{q_1}(M'^{(1)}_{\bullet\bullet\bullet}), \ldots, H_{q_m}(M'^{(m)}_{\bullet\bullet\bullet}), H_{q_{m+1}}(Q'_\bullet)\Big). $$

III | 168

By the definition of global hypertors,

$$ H_{q_i}(M'^{(i)}_{\bullet\bullet\bullet}) = \mathbf{Tor}_{q_i}^S\big(X^{(i)}, S'; \mathscr{P}_\bullet^{(i)}, \mathscr{O}_{S'}\big), $$

which gives the sequence (f).

On the other hand, regard $M'^{(i)}_{\bullet\bullet\bullet}$ as a bicomplex whose first degree is the sum of the first two degrees of the tricomplex and whose second degree is its third degree. Since its terms are projective A'-modules, the general theory of hyperhomology shows that the homology of

$$ M'^{(1)}_{\bullet\bullet\bullet} \otimes_{A'} \cdots \otimes_{A'} M'^{(m)}_{\bullet\bullet\bullet} \otimes_{A'} Q'_\bullet $$

is canonically isomorphic to its hyperhomology ((0.11.6.5)). It is therefore the abutment of a spectral sequence with E_2-term

$$ E_{pq}^2 = \bigoplus_{q_1+\cdots+q_{m+1}=q} \operatorname{Tor}_p^{A'}\Big(H^{\mathrm{II}}_{q_1}(M'^{(1)}_{\bullet\bullet\bullet}), \ldots, H^{\mathrm{II}}_{q_m}(M'^{(m)}_{\bullet\bullet\bullet}), H^{\mathrm{II}}_{q_{m+1}}(Q'_\bullet)\Big). $$

The second degree of $Q'_\bullet$ is zero, so

$$H_n^{\mathrm{II}}(Q'_\bullet) = 0 \quad (n \neq 0), \qquad H_0^{\mathrm{II}}(Q'_\bullet) = Q'_\bullet,$$

and the preceding formula becomes

$$E^2_{pq} = \bigoplus_{q_1+\cdots+q_m=q} \mathrm{Tor}_p^{A'}\Big(H_{q_1}^{\mathrm{II}}(M'^{(1)}_{\bullet\bullet\bullet}), \ldots, H_{q_m}^{\mathrm{II}}(M'^{(m)}_{\bullet\bullet\bullet}), Q'_\bullet\Big).$$

Moreover, by (III.6.3.4),

$$\begin{aligned} H_{q_i}^{\mathrm{II}}(M'^{(i)}_{\bullet\bullet\bullet}) &= H_{q_i}^{\mathrm{II}}(M^{(i)}_{\bullet\bullet\bullet} \otimes_A A') \\ &= \mathrm{Tor}_{q_i}^{A}(L^{(i)}_{\bullet\bullet}, A'). \end{aligned}$$

Now $L^{(i)}_{-j,k}$ is the direct sum of the modules $\Gamma(V, \mathscr{P}^{(i)}_k)$, where V runs through the affine intersections of $j+1$ members of $\mathfrak{U}^{(i)}$. If $V' = u_i^{-1}(V)$, then V' is affine in $X'^{(i)}$, and (III.6.4.1.1) gives

$$\Gamma\big(V', \mathscr{T}\!or_{q_i}^S(\mathscr{P}^{(i)}_k, \mathscr{O}_{S'})\big) = \mathrm{Tor}_{q_i}^A\big(\Gamma(V, \mathscr{P}^{(i)}_k), A'\big).$$

Consequently, the bicomplex $H_{q_i}^{\mathrm{II}}(M'^{(i)}_{\bullet\bullet\bullet})$ is

$$C^\bullet\big(\mathfrak{U}'^{(i)}, \mathscr{T}\!or_{q_i}^S(\mathscr{P}^{(i)}_\bullet, \mathscr{O}_{S'})\big).$$

This gives the asserted E_2-term of the sequence (f'). Its biregularity under the stated hypotheses follows in the usual way, since, when $\mathscr{P}^{(i)}_\bullet$ is bounded below, every degree of $M'^{(i)}_{\bullet\bullet\bullet}$ is bounded below.

Remark (6.9.5). — *As in (III.6.7.6), replacing the complexes $\mathscr{P}^{(i)}_\bullet$ and $\mathscr{Q}'_\bullet$ by respectively homotopic complexes does not change the spectral sequences (e), (f), and (f'), up to canonical isomorphism.*

Moreover, for the sequence (f), homomorphisms of complexes

$$\mathscr{P}^{(i)}_\bullet \longrightarrow \mathscr{R}^{(i)}_\bullet, \qquad \mathscr{Q}'_\bullet \longrightarrow \mathscr{T}'_\bullet$$

which induce isomorphisms in homology

$$\mathscr{H}_\bullet(\mathscr{P}^{(i)}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}^{(i)}_\bullet), \qquad \mathscr{H}_\bullet(\mathscr{Q}'_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{T}'_\bullet)$$

also induce an isomorphism of spectral sequences

$${}^{(f)}\mathscr{E}\big(f_1, \ldots, f_m; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet\big) \xrightarrow{\sim} {}^{(f)}\mathscr{E}\big(f_1, \ldots, f_m; \mathscr{R}^{(1)}_\bullet, \ldots, \mathscr{R}^{(m)}_\bullet, \mathscr{T}'_\bullet\big).$$

The proof is the same as that of (III.6.7.6), taking into account the result of that section and the regularity of the sequence (f).

Corollary (6.9.6). — *Under the hypotheses of (III.6.9.1), suppose that:*

1° *the complexes $\mathscr{P}^{(i)}_\bullet$ consist of S-flat modules, and the $\mathscr{O}_Y$-modules*

$$\mathfrak{Tor}_n^S\big(f_1, \ldots, f_m; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet\big)$$

are S-flat;

2° *the complexes $\mathscr{P}^{(i)}_\bullet$ and $\mathscr{Q}'_\bullet$ are bounded below.*

Put

$$\mathscr{P}'^{(i)}_\bullet = \mathscr{P}^{(i)}_\bullet \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}.$$

There are canonical functorial isomorphisms

(6.9.6.1)

$$\mathfrak{Tor}_n^{S'}\big(f'_1, \ldots, f'_m, 1_{S'}; \mathscr{P}'^{(1)}_\bullet, \ldots, \mathscr{P}'^{(m)}_\bullet, \mathscr{Q}'_\bullet\big) \xrightarrow{\sim} \bigoplus_{n'+n''=n} \mathfrak{Tor}_{n'}^S\big(f_1, \ldots, f_m; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet\big) \otimes_{\mathscr{O}_S} \mathscr{H}_{n''}(\mathscr{Q}'_\bullet). \tag{6.9.6.1}$$

In particular, when $\mathscr{Q}'_\bullet$ consists of a single term $\mathscr{F}'$ in degree 0, there are canonical functorial isomorphisms

(6.9.6.2)

$$\begin{aligned}\mathfrak{Tor}_n^{S'}(f'_1,\dots,f'_m,1_{S'};\mathscr{P}'^{(1)}_\bullet,\dots,\mathscr{P}'^{(m)}_\bullet,\mathscr{F}') \xrightarrow{\sim}\\ \mathfrak{Tor}_n^{S}(f_1,\dots,f_m;\mathscr{P}^{(1)}_\bullet,\dots,\mathscr{P}^{(m)}_\bullet)\otimes_{\mathscr{O}_S}\mathscr{F}'.\end{aligned} \tag{6.9.6.2}$$

More particularly, for $\mathscr{F}'=\mathscr{O}_{S'}$,

(6.9.6.3)

$$\begin{aligned}\mathfrak{Tor}_n^{S'}(f'_1,\dots,f'_m;\mathscr{P}'^{(1)}_\bullet,\dots,\mathscr{P}'^{(m)}_\bullet) \xrightarrow{\sim}\\ \mathfrak{Tor}_n^{S}(f_1,\dots,f_m;\mathscr{P}^{(1)}_\bullet,\dots,\mathscr{P}^{(m)}_\bullet)\otimes_{\mathscr{O}_S}\mathscr{O}_{S'}.\end{aligned} \tag{6.9.6.3}$$

Proof. — The flatness of the terms of $\mathscr{P}^{(i)}_\bullet$ implies that

$$\mathscr{T}\!or_{q_i}^S(\mathscr{P}^{(i)}_\bullet,\mathscr{O}_{S'})=0 \qquad (q_i\neq 0)$$

by (III.6.5.8). Thus the sequence (f') degenerates. Hypothesis 2° also makes it biregular ((III.6.9.3)), so the edge homomorphism

$$\begin{aligned}\mathfrak{Tor}_n^{S}(f_1,\dots,f_m,1_{S'};\mathscr{P}^{(1)}_\bullet,\dots,\mathscr{P}^{(m)}_\bullet,\mathscr{Q}'_\bullet)\longrightarrow {}^{(f')}\mathscr{E}^2_{n0}\\ =\mathfrak{Tor}_n^{S'}(f'_1,\dots,f'_m,1_{S'};\mathscr{P}'^{(1)}_\bullet,\dots,\mathscr{P}'^{(m)}_\bullet,\mathscr{Q}'_\bullet)\end{aligned} \tag{6.9.6.4}$$

is bijective by (0.11.1.6).

The flatness of the modules

$$\mathfrak{Tor}_n^{S}(f_1,\dots,f_m;\mathscr{P}^{(1)}_\bullet,\dots,\mathscr{P}^{(m)}_\bullet)$$

implies, by (III.6.5.8), that ${}^{(e)}\mathscr{E}^2_{pq}=0$ for $p\neq 0$. Hence (e) also degenerates, and its biregularity makes the edge homomorphism

$$\begin{aligned}\mathfrak{Tor}_n^{S}(f_1,\dots,f_m,1_{S'};\mathscr{P}^{(1)}_\bullet,\dots,\mathscr{P}^{(m)}_\bullet,\mathscr{Q}'_\bullet)\longrightarrow {}^{(e)}\mathscr{E}^2_{0n}\\ =\bigoplus_{n'+n''=n}\mathfrak{Tor}_{n'}^{S}(f_1,\dots,f_m;\mathscr{P}^{(1)}_\bullet,\dots,\mathscr{P}^{(m)}_\bullet)\otimes_{\mathscr{O}_S}\mathscr{H}_{n''}(\mathscr{Q}'_\bullet)\end{aligned} \tag{6.9.6.5}$$

bijective. Combining these two edge isomorphisms gives (III.6.9.6.1). Formula (III.6.9.6.2) follows because $\mathscr{H}_q(\mathscr{Q}'_\bullet)=0$ for $q\neq 0$ and $\mathscr{H}_0(\mathscr{Q}'_\bullet)=\mathscr{F}'$. Finally, taking $\mathscr{F}'=\mathscr{O}_{S'}$ in (III.6.9.6.2) and using (III.6.7.12) gives (III.6.9.6.3).

Corollary (6.9.7). — *Under the hypotheses of (III.6.9.1), suppose that S and S' are affine, and choose an integer d_i for each $1\leq i\leq m$. There is an integer N, depending only on S, the $X^{(i)}$, and the d_i, with the following property. For every integer n_0, the canonical isomorphisms (III.6.9.6.3) hold for $n\leq n_0$ and for every system of complexes $\mathscr{P}^{(i)}_\bullet$ satisfying:*

1° *$\mathscr{P}^{(i)}_k=0$ for $k<d_i$;*

2° *$\mathscr{P}^{(i)}_k$ is S-flat for $k<n_0+N$;*

3°
$$\mathfrak{Tor}_q^{S}(f_1,\dots,f_m;\mathscr{P}^{(1)}_\bullet,\dots,\mathscr{P}^{(m)}_\bullet)$$
is S-flat for $q<n_0+N$.

Proof. — Suppose that $\mathscr{P}^{(i)}_k$ is S-flat for $k<r$. Then

$$\mathscr{T}\!or_{q_i}^S(\mathscr{P}^{(i)}_k,\mathscr{O}_{S'})=0 \qquad (k<r,\ q_i\neq 0).$$

Consider

$$\begin{aligned}\mathfrak{Tor}_p^{S'}\Big(f'_1,\dots,f'_m;\mathscr{T}\!or_{q_1}^S(\mathscr{P}^{(1)}_\bullet,\mathscr{O}_{S'}),\dots,\\ \mathscr{T}\!or_{q_m}^S(\mathscr{P}^{(m)}_\bullet,\mathscr{O}_{S'})\Big).\end{aligned} \tag{6.9.7.1}$$

Compute (6.9.7.1) by the method of (III.6.6.2), using the inverse images of fixed finite affine open coverings $\mathfrak{U}^{(i)}$ of $X^{(i)}$. These coverings are independent of S' and of the complexes $\mathscr{P}_\bullet^{(i)}$. The terms $L_{jk}^{(i)}$ vanish for $j < -N_i$, where N_i depends only on $\mathfrak{U}^{(i)}$.

If some q_i is nonzero, the simple complex whose homology in degree p is (6.9.7.1) has zero terms in every degree

$$< r + \sum_{j\neq i} d_j - \sum_{\ell=1}^{m} N_\ell.$$

Consequently (6.9.7.1) vanishes for $p < r - N$, where

$$N = \max_i \left(\sum_{\ell=1}^{m} N_\ell - \sum_{j\neq i} d_j \right).$$

It follows that ${}^{(f')}\mathscr{E}_{pq}^2 = 0$ for $q \neq 0$ and $p < r - N$. Since also ${}^{(f')}\mathscr{E}_{pq}^2 = 0$ for $q < 0$, the edge homomorphism (6.9.6.4) is bijective for $n < r - N$ (M, XV, 5.6), when $\mathscr{Q}'_\bullet = \mathscr{O}_{S'}$.

Secondly, if

$$\mathfrak{Tor}_q^S\big(f_1, \ldots, f_m; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(m)}\big)$$

is S-flat for $q < r$, then ${}^{(e)}\mathscr{E}_{pq}^2 = 0$ for $p \neq 0$ and $q < r$; moreover, ${}^{(e)}\mathscr{E}_{pq}^2 = 0$ for $p < 0$. Thus the edge homomorphism (6.9.6.5) is bijective for $n < r$, which completes the proof.

The case of (III.6.9.3) most important in applications is $m = 1$, with $\mathscr{Q}'_\bullet$ reduced to a single term $\mathscr{F}'$ in degree 0. We state it again for later reference.[1]

Proposition (6.9.8). — *Let S be a prescheme, let $g : S' \to S$ be a morphism, and let $f : X \to Y$ be a separated and quasi-compact S-morphism of S-preschemes. Let $\mathscr{P}_\bullet$ be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules, and let $\mathscr{F}'$ be a quasi-coherent $\mathscr{O}_{S'}$-module. There are two biregular spectral functors in $\mathscr{P}_\bullet$ and $\mathscr{F}'$, with values in the category of quasi-coherent $\mathscr{O}_{Y_{(S')}}$-modules, having the same abutment*

$$\mathfrak{Tor}_\bullet^S\big(f, 1_{S'}; \mathscr{P}_\bullet, \mathscr{F}'\big),$$

and whose E_2-terms are

(6.9.8.1)

$$'\mathscr{E}_{pq}^2 = \mathscr{T}or_p^S\big(\mathscr{H}^{-q}(f, \mathscr{P}_\bullet), \mathscr{F}'\big) \tag{6.9.8.1}$$

and

(6.9.8.2)

$$''\mathscr{E}_{pq}^2 = \mathscr{H}^{-p}\big(f', \mathscr{T}or_q^S(\mathscr{P}_\bullet, \mathscr{F}')\big), \tag{6.9.8.2}$$

where

$$f' = f_{(S')} : X_{(S')} \longrightarrow Y_{(S')}.$$

Proof. — These spectral sequences may also be obtained, without starting from (III.6.9.3), from the sequences (a) and (b') of (III.6.7.3), by taking

$$X^{(1)} = X, \quad Y^{(1)} = Y, \quad X^{(2)} = Y^{(2)} = S', \quad f_1 = f, \quad f_2 = 1_{S'}.$$

When $S = S' = Y$ and Y is affine, one obtains two spectral sequences with E_2-terms

$$'\mathscr{E}_{pq}^2 = \mathscr{T}or_p^Y\big(\mathscr{H}^{-q}(f, \mathscr{P}_\bullet), \mathscr{F}\big) \tag{6.9.8.3}$$

and

$$''\mathscr{E}_{pq}^2 = \mathscr{H}^{-p}\big(f, \mathscr{T}or_q^Y(\mathscr{P}_\bullet, \mathscr{F})\big). \tag{6.9.8.4}$$

By (III.6.7.6), both abut to the hypercohomology

$$\mathscr{H}^\bullet\big(f, \mathscr{P}_\bullet \otimes_Y \mathscr{F}\big)$$

of f_* with respect to the complex $\mathscr{P}_\bullet \otimes_Y \mathscr{F}$ of $\mathscr{O}_X$-modules, for every quasi-coherent Y-flat $\mathscr{O}_Y$-module $\mathscr{F}$; if the terms of $\mathscr{P}_\bullet$ are Y-flat, $\mathscr{F}$ may be any quasi-coherent $\mathscr{O}_Y$-module. These spectral sequences are distinct from those of (III.6.2.1).

Corollary (6.9.9). — *Under the hypotheses of (III.6.9.8), suppose that $\mathscr{P}_\bullet$ is bounded below, consists of S-flat modules, and that the $\mathscr{O}_Y$-modules $\mathscr{H}^n(f, \mathscr{P}_\bullet)$ are S-flat.* **III** | 171

Put

$$\mathscr{P}'_\bullet = \mathscr{P}_\bullet \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}.$$

There are canonical functorial isomorphisms

(6.9.9.1)

$$\begin{aligned}\mathfrak{Tor}_n^{S'}\big(f', 1_{S'}; \mathscr{P}'_\bullet, \mathscr{F}'\big) \xrightarrow{\sim}\\ \mathscr{H}^{-n}(f, \mathscr{P}_\bullet) \otimes_{\mathscr{O}_S} \mathscr{F}'.\end{aligned} \tag{6.9.9.1}$$

In particular, for $\mathscr{F}' = \mathscr{O}_{S'}$,

(6.9.9.2)

$$\mathscr{H}^n(f', \mathscr{P}'_\bullet) \xrightarrow{\sim} \mathscr{H}^n(f, \mathscr{P}_\bullet) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}. \tag{6.9.9.2}$$

This is the case $m = 1$ of (III.6.9.6).

Corollary (6.9.10). — *Let S be a prescheme, let $f : X \to Y$ be a separated and quasi-compact S-morphism of S-preschemes, and let $\mathscr{P}_\bullet$ be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules that are flat over S. Suppose further that the $\mathscr{O}_Y$-modules $\mathscr{H}^n(f, \mathscr{P}_\bullet)$ are flat over S.*

For $s \in S$, write X_s and Y_s for the fibres

$$X \times_S \operatorname{Spec}(\kappa(s)), \qquad Y \times_S \operatorname{Spec}(\kappa(s)),$$

write

$$f_s : X_s \longrightarrow Y_s$$

for the morphism $f \times_S 1$, and put

$$\mathscr{P}^s_\bullet = \mathscr{P}_\bullet \otimes_{\mathscr{O}_S} \kappa(s).$$

Then there are canonical functorial isomorphisms

(6.9.10.1)

$$\mathscr{H}^n(f_s, \mathscr{P}^s_\bullet) \xrightarrow{\sim} \mathscr{H}^n(f, \mathscr{P}_\bullet) \otimes_{\mathscr{O}_S} \kappa(s). \tag{6.9.10.1}$$

Thus, under suitable flatness hypotheses, the formation of the derived functors $R^n f_*(\mathscr{F})$ "commutes with passage to the fibres." We shall obtain this result again by another method in § 7.

6.10 Local structure of certain cohomological functors

Proposition (6.10.1). — *Let $S = \operatorname{Spec}(A)$ be an affine scheme, and let $Y^{(i)}$ $(1 \leq i \leq n)$ be a finite family of affine S-schemes, flat over S. For each i, let*

$$f_i : X^{(i)} \longrightarrow Y^{(i)}$$

be a separated and quasi-compact S-morphism, and let $\mathscr{P}^{(i)}_\bullet$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. Let Y be the product over S of the schemes $Y^{(i)}$.

There exists a complex $\mathscr{R}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules, flat over S, with the following property. For every affine S-scheme S' and every bounded-below complex $\mathscr{Q}'_\bullet$ of quasi-coherent $\mathscr{O}_{S'}$-modules, there is an isomorphism

(6.10.1.1)

$$\mathbf{Tor}^S_\bullet\big(f_1, \ldots, f_n, 1_{S'}; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(n)}_\bullet, \mathscr{Q}'_\bullet\big) \xrightarrow{\sim} \mathscr{H}_\bullet\big(\mathscr{R}_\bullet \otimes_{\mathscr{O}_S} \mathscr{Q}'_\bullet\big), \tag{6.10.1.1}$$

which is an isomorphism of ∂-functors in $\mathscr{Q}'_\bullet$.

Moreover, for every S-morphism $u : S'' \to S'$ of affine S-schemes, the diagram

[1] The case treated in (III.6.9.8), and in particular the spectral sequences (III.6.9.8.3) and (III.6.9.8.4), was pointed out to us by J.-P. Serre in 1957.

(6.10.1.2)

$$\begin{array}{ccc} \mathbf{Tor}_\bullet^S(f_1,\dots,f_n,1_{S'};\\ \mathscr{P}_\bullet^{(1)},\dots,\mathscr{P}_\bullet^{(n)},\mathscr{Q}'_\bullet) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{R}_\bullet\otimes_{\mathscr{O}_S}\mathscr{Q}'_\bullet) \\ \downarrow & & \downarrow \\ \mathbf{Tor}_\bullet^S(f_1,\dots,f_n,1_{S''};\\ \mathscr{P}_\bullet^{(1)},\dots,\mathscr{P}_\bullet^{(n)},u^*(\mathscr{Q}'_\bullet)) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{R}_\bullet\otimes_{\mathscr{O}_S}u^*(\mathscr{Q}'_\bullet)) \end{array} \tag{6.10.1.2}$$

commutes. Here the vertical arrows are the canonical $(1_Y\times_S u)$-morphisms defined in (III.6.7.10).

Proof. — Compute the hypertor by the method of (III.6.6.2), taking account of Remark (III.6.6.6). With the notation used there, each $L_{\bullet\bullet}^{(i)}$ is a bicomplex of A_i-modules, where A_i denotes the ring of $Y^{(i)}$. It therefore has a projective Cartan–Eilenberg resolution $M_{\bullet\bullet\bullet}^{(i)}$ (in the sense of (0.11.7.1)) by projective A_i-modules. By (III.6.6.6), the left side of (6.10.1.1) is canonically isomorphic to **III** | 172

$$\mathrm{H}_\bullet\big(M_{\bullet\bullet\bullet}\otimes_A Q'_\bullet\big),$$

where

$$M_{\bullet\bullet\bullet}=M_{\bullet\bullet\bullet}^{(1)}\otimes_A M_{\bullet\bullet\bullet}^{(2)}\otimes_A\cdots\otimes_A M_{\bullet\bullet\bullet}^{(n)}$$

and $\mathscr{Q}'_\bullet=\widetilde{Q'_\bullet}$.

By hypothesis, the rings A_i are flat A-modules. Hence the $M_{\bullet\bullet\bullet}^{(i)}$ are tricomplexes of flat A-modules (0.6.2.1), and so is $M_{\bullet\bullet\bullet}$. Furthermore, if B is the ring of Y, that is, the tensor product over A of the A_i, then $M_{\bullet\bullet\bullet}$ is a tricomplex of B-modules. Consequently the complex

$$\mathscr{R}_\bullet=\widetilde{M_{\bullet\bullet\bullet}},$$

where $M_{\bullet\bullet\bullet}$ is regarded as a simple complex, has the required properties, as follows immediately from (III.6.7.10).

Corollary (6.10.2). — *In Proposition (III.6.10.1), the complex $\mathscr{R}_\bullet$ may be taken to be bounded below. If the $\mathscr{P}_\bullet^{(i)}$ are bounded above and the $Y^{(i)}$ have finite cohomological dimension, then $\mathscr{R}_\bullet$ may be taken to be bounded above.*

Proof. — The first assertion follows because all three degrees of each $M_{\bullet\bullet\bullet}^{(i)}$ are bounded below. On the other hand, if the rings A_i have finite cohomological dimension, the third degree of each $M_{\bullet\bullet\bullet}^{(i)}$ assumes only finitely many values, and by its construction the same is true of its first degree (III.6.6.2). Its second degree is bounded above precisely when the degree of $\mathscr{P}_\bullet^{(i)}$ is bounded above (III.6.6.2), which proves the second assertion.

Remarks (6.10.3). — (i) *With the notation of (III.6.10.1),*

$$\mathscr{H}_\bullet(\mathscr{R}_\bullet\otimes_S\mathscr{Q}'_\bullet)\simeq\mathbf{Tor}_\bullet(\mathscr{R}_\bullet,\mathscr{Q}'_\bullet),$$

because $\mathscr{R}_\bullet$ consists of $\mathscr{O}_Y$-modules that are flat over S (III.6.5.9). It is therefore (III.6.5.4) the abutment of a regular spectral sequence whose E^2-term is

(6.10.3.1)

$$E^2_{pq}=\bigoplus_{q'+q''=q}\mathscr{T}or_p^S\big(\mathscr{H}_{q'}(\mathscr{R}_\bullet),\mathscr{H}_{q''}(\mathscr{Q}'_\bullet)\big). \tag{6.10.3.1}$$

This is precisely the spectral sequence (e) for base change in (III.6.9.3).

(ii) *Let $\mathscr{R}'_\bullet$ be another complex of quasi-coherent $\mathscr{O}_Y$-modules, flat over S, and let*

$$g:\mathscr{R}'_\bullet\longrightarrow\mathscr{R}_\bullet$$

be a homomorphism of complexes for which

$$\mathscr{H}_\bullet(g):\mathscr{H}_\bullet(\mathscr{R}'_\bullet)\longrightarrow\mathscr{H}_\bullet(\mathscr{R}_\bullet)$$

is an isomorphism. By (III.6.3.3) and (III.6.5.9), g induces an isomorphism of ∂-functors in $\mathscr{Q}'_\bullet$,

$$\mathscr{H}_\bullet(\mathscr{R}'_\bullet\otimes_S\mathscr{Q}'_\bullet)\xrightarrow{\sim}\mathscr{H}_\bullet(\mathscr{R}_\bullet\otimes_S\mathscr{Q}'_\bullet),$$

such that the diagram

$$\begin{array}{ccc} \mathscr{H}_\bullet(\mathscr{R}'_\bullet \otimes_S \mathscr{Q}'_\bullet) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S \mathscr{Q}'_\bullet) \\ \downarrow & & \downarrow \\ \mathscr{H}_\bullet(\mathscr{R}'_\bullet \otimes_S u^*(\mathscr{Q}'_\bullet)) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S u^*(\mathscr{Q}'_\bullet)) \end{array}$$

commutes. Thus the properties in (III.6.10.1) do not determine the complex $\mathscr{R}_\bullet$ completely.

(iii) *In the proof of (III.6.10.1), the $M^{(i)}_{\bullet\bullet\bullet}$ may be chosen to consist of free A_i-modules, as follows immediately by dualising the proof of (0.11.5.2.1). The*

$$M^{(i)}_{\bullet\bullet\bullet} \otimes_{A_i} B$$

then consist of free B-modules; since $M_{\bullet\bullet\bullet}$ is their tensor product over B, one may in addition require in (III.6.10.1) that $\mathscr{R}_\bullet$ be associated with a complex of free B-modules.

Moreover, by (M, XVII, 1.2), the tricomplex $M_{\bullet\bullet\bullet}$ depends functorially on each of the bicomplexes $L^{(i)}_{\bullet\bullet}$, hence on each $\mathscr{P}^{(i)}_\bullet$ once a finite covering of each $X^{(i)}$ has been fixed. Here "morphisms" of tricomplexes are understood as homomorphism classes modulo homotopy. Replacing a covering of $X^{(i)}$ by a finer one gives, for $L^{(i)}_{\bullet\bullet}$, homomorphisms defined up to homotopy by (III.6.6.8); thus, with the preceding convention on morphisms, $M_{\bullet\bullet\bullet}$ is a functor in each $\mathscr{P}^{(i)}_\bullet$.

We shall make this functorial dependence more precise, in particular the behaviour of $\mathscr{R}_\bullet$ with respect to short exact sequences

$$0 \longrightarrow \mathscr{P}'^{(i)}_\bullet \longrightarrow \mathscr{P}^{(i)}_\bullet \longrightarrow \mathscr{P}''^{(i)}_\bullet \longrightarrow 0,$$

in the chapter devoted to a general algebra of cohomological functors mentioned in (III.6.1.3).

Scholium (6.10.4). — The fact that $\mathscr{R}_\bullet$ consists of $\mathscr{O}_Y$-modules flat over S immediately implies that

$$\mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S \mathscr{Q}'_\bullet)$$

is a homological functor in $\mathscr{Q}'_\bullet$; compare the argument of (III.7.7.1). As noted in (III.6.1.1), this property motivates the introduction of hypertor.

Put

$$X = X^{(1)} \times_S X^{(2)} \times_S \cdots \times_S X^{(n)}, \qquad f = f_1 \times_S f_2 \times_S \cdots \times_S f_n,$$
$$\mathscr{P}_\bullet = \mathscr{P}^{(1)}_\bullet \otimes_S \mathscr{P}^{(2)}_\bullet \otimes_S \cdots \otimes_S \mathscr{P}^{(n)}_\bullet,$$

and

$$X' = X \times_S S', \qquad Y' = Y \times_S S', \qquad f' = f \times_S 1_{S'}.$$

Problems of base change lead one to study the hypercohomology

$$\mathscr{H}^*_{f'}(\mathscr{P}_\bullet \otimes_S \mathscr{N}')$$

as a functor of the quasi-coherent $\mathscr{O}_{S'}$-module $\mathscr{N}'$, or equivalently the hypercohomology

$$\mathscr{H}^*_f(\mathscr{P}_\bullet \otimes_S \mathscr{N})$$

as a functor of the quasi-coherent $\mathscr{O}_S$-module $\mathscr{N}$. When the $\mathscr{P}^{(i)}_\bullet$, and therefore $\mathscr{P}_\bullet$, are flat over S, the preceding results and (III.6.7.6) show that this is indeed a cohomological functor in $\mathscr{N}$. This is no longer true when the flatness of the $\mathscr{P}^{(i)}_\bullet$ is not assumed, and the study of

$$\mathscr{H}^*_f(\mathscr{P}_\bullet \otimes_S \mathscr{N})$$

can then no longer be approached by the usual methods of homological algebra.

We shall chiefly use the case $n = 1$, $Y = S$, in which $\mathscr{P}_\bullet$ consists of $\mathscr{O}_X$-modules flat over Y. In this case we have the following theorem.

Theorem (6.10.5). — *Let $Y = \mathrm{Spec}(A)$ be an affine Noetherian scheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{P}_\bullet$ be a bounded-below complex of coherent $\mathscr{O}_X$-modules flat over Y. There exists a bounded-below complex $\mathscr{L}_\bullet$ of $\mathscr{O}_Y$-modules, whose terms have the form*

$$\mathscr{L}_i = \mathscr{O}_Y^{m_i},$$

and an isomorphism

(6.10.5.1)

$$\mathscr{H}^*(f, \mathscr{P}_\bullet \otimes_Y \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{Q}_\bullet) \tag{6.10.5.1}$$

of ∂-functors in the bounded-below complex $\mathscr{Q}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules.

Moreover, for every morphism $u : Y' \to Y$, put

$$X' = X_{(Y')}, \qquad f' = f_{(Y')}, \qquad \mathscr{P}'_\bullet = \mathscr{P}_\bullet \otimes_Y \mathscr{O}_{Y'}, \qquad \mathscr{L}'_\bullet = u^*(\mathscr{L}_\bullet).$$

Then $\mathscr{L}'_\bullet$ is a complex of locally free $\mathscr{O}_{Y'}$-modules of finite type, and there is an isomorphism **III** | 174

(6.10.5.2)

$$\mathscr{H}^*(f', \mathscr{P}'_\bullet \otimes_{Y'} \mathscr{Q}'_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}'_\bullet \otimes_{Y'} \mathscr{Q}'_\bullet) \tag{6.10.5.2}$$

of ∂-functors in the bounded-below complex $\mathscr{Q}'_\bullet$ of quasi-coherent $\mathscr{O}_{Y'}$-modules. These isomorphisms may be chosen so that the diagram

(6.10.5.3)

$$\begin{array}{ccc} \mathscr{H}^*(f, \mathscr{P}_\bullet \otimes_Y \mathscr{Q}_\bullet) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{Q}_\bullet) \\ \downarrow & & \downarrow \\ \mathscr{H}^*(f', \mathscr{P}'_\bullet \otimes_{Y'} u^*(\mathscr{Q}_\bullet)) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{L}'_\bullet \otimes_{Y'} u^*(\mathscr{Q}_\bullet)) \end{array} \tag{6.10.5.3}$$

commutes.

Proof. — Proposition (III.6.10.1) first gives a bounded-below complex $\mathscr{R}_\bullet$ (III.6.10.2) of quasi-coherent $\mathscr{O}_Y$-modules that are free over Y (III.6.10.3) (iii), together with an isomorphism

(6.10.5.4)

$$\mathscr{H}^*(f, \mathscr{P}_\bullet \otimes_Y \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_Y \mathscr{Q}_\bullet) \tag{6.10.5.4}$$

of ∂-functors in $\mathscr{Q}_\bullet$. The terms of $\mathscr{R}_\bullet$ need not, *a priori*, be $\mathscr{O}_Y$-modules of finite type.

Apply (6.10.5.4) when $\mathscr{Q}_\bullet$ is the complex concentrated in one degree with sole term $\mathscr{O}_Y$. It follows that $\mathscr{H}_\bullet(\mathscr{R}_\bullet)$ is isomorphic to $\mathscr{H}^*(f, \mathscr{P}_\bullet)$ and hence consists of coherent $\mathscr{O}_Y$-modules (III.6.2.5). By (0.11.9.2), there is therefore a bounded-below complex $\mathscr{L}_\bullet$ whose terms are associated with free A-modules of finite type, and a homomorphism

$$\mathscr{L}_\bullet \longrightarrow \mathscr{R}_\bullet$$

which induces an isomorphism

$$\mathscr{H}_\bullet(\mathscr{L}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}_\bullet).$$

Formula (6.10.5.1) now follows from (III.6.10.3) (ii).

The remaining assertions of (III.6.10.5) follow from (III.6.10.1) and (III.6.10.3) (ii) when Y' is affine. In the general case, cover Y' by affine open subsets (V_α). If (6.10.5.2) is formed over each V_α, its restrictions from V_α and V_β to an affine open subset

$$W \subset V_\alpha \cap V_\beta$$

both coincide with the isomorphism formed over W. This follows from the commutativity of (6.10.1.2), applied to the canonical inclusions $W \to V_\alpha$ and $W \to V_\beta$, and proves that the local isomorphisms glue.

Remark (6.10.6). — *In the following chapters we shall apply (III.6.10.5) chiefly when $\mathscr{P}_\bullet$ is concentrated in one degree and its sole term is a coherent $\mathscr{O}_X$-module $\mathscr{F}$ flat over Y. Since*

$$\mathscr{H}_n(\mathscr{L}_\bullet) = \mathscr{H}^{-n}(f, \mathscr{F}) = R^{-n} f_*(\mathscr{F}) \qquad \textit{(III.6.2.1)},$$

the $\mathscr{H}_n(\mathscr{L}_\bullet)$ vanish for $n > 0$. We shall see later (III.7.7.12) (i) that $\mathscr{L}_\bullet$ may then be chosen to have terms only in degrees ≤ 0, and hence only finitely many terms, provided the requirement that its terms be associated with free A-modules of finite type is replaced by the requirement that they be locally free of finite type.

The complex $\mathscr{L}_\bullet$ corresponding to such an $\mathscr{O}_X$-module $\mathscr{F}$ seems to have no special property apart from **III** | 175 *the preceding restriction on its degrees. One may therefore ask conversely whether, given a bounded-below complex $\mathscr{L}_\bullet$ whose terms are $\mathscr{O}_Y$-modules associated with projective A-modules of finite type and whose terms of degree > 0 vanish, there exist a Y-scheme X, projective and flat over Y, and a locally free $\mathscr{O}_X$-module $\mathscr{F}$ such that there is an isomorphism*

$$\mathscr{H}^*(f, \mathscr{F} \otimes_Y \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{Q}_\bullet)$$

functorial in $\mathscr{Q}_\bullet$. Such a result would reduce the cohomological theory of coherent modules flat over Y on proper Y-schemes entirely to the theory, "up to homotopy," of complexes of projective A-modules of finite type over a Noetherian ring A.

7 Base change for homological functors of sheaves of modules

7.1 Functors of A-modules

(7.1.1) Given a ring A, not necessarily commutative, denote by $\mathtt{Ab}_A$ the category of left A-modules, and simply by $\mathtt{Ab}$ the category of $\mathbf{Z}$-modules, which is the same as the category of commutative groups. Let

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

be an additive covariant functor, and let M be an (A, A)-bimodule. Then $T(M)$ is naturally endowed with the structure of a right A-module.

Indeed, for every $a \in A$, let $h_{a,M}$, or simply h_a, denote the endomorphism

$$x \longmapsto xa$$

of the left A-module M. By hypothesis, $T(h_a)$ is an endomorphism of the $\mathbf{Z}$-module $T(M)$. Since T is covariant and additive, for $a, b \in A$ one has

$$T(h_{ab}) = T(h_b \circ h_a) = T(h_b) \circ T(h_a)$$

and

$$T(h_{a+b}) = T(h_a + h_b) = T(h_a) + T(h_b).$$

Thus

$$(y, a) \longmapsto T(h_a)(y)$$

defines a right A-module scalar multiplication on $T(M)$. In particular, $T(A_s)$ is a right A-module.

(7.1.2) Suppose that A is commutative. It follows from (III.7.1.1) that, for every A-module M, the group $T(M)$ is naturally an A-module. Moreover, if $u : M \to N$ is an A-linear homomorphism, then, for every $a \in A$,

$$u \circ h_{a,M} = h_{a,N} \circ u,$$

and hence

$$T(u) \circ T(h_{a,M}) = T(h_{a,N}) \circ T(u).$$

Consequently $T(u) : T(M) \to T(N)$ is A-linear, and T may be regarded as an additive covariant functor from $\mathtt{Ab}_A$ to itself.

More precisely, this defines a canonical equivalence between the category of additive covariant functors

$$\mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

and the category of A-linear covariant functors

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}_A,$$

that is, functors satisfying

$$T(h_{a,M}) = h_{a,T(M)} \qquad (a \in A).$$

The forgetful functor

$$I : \mathtt{Ab}_A \longrightarrow \mathtt{Ab},$$

which sends an A-module to its underlying $\mathbf{Z}$-module, is exact and faithful. The two functors corresponding under the preceding equivalence therefore have the same exactness properties.

(7.1.3) Continue to suppose that A is commutative. Let B be an A-algebra, not necessarily commutative, and let

$$\rho : A \longrightarrow B$$

be the ring homomorphism defining the algebra structure. It defines an additive covariant functor **III** | 176

$$\rho_* : M \longmapsto M_{[\rho]}$$

from the category $\mathtt{Ab}_B$ of left B-modules to $\mathtt{Ab}_A$. By composition one obtains the additive covariant functor

$$T_{(B)} : \mathtt{Ab}_B \xrightarrow{\rho_*} \mathtt{Ab}_A \xrightarrow{T} \mathtt{Ab}.$$

For typographical reasons this functor will also be denoted $T^{(B)}$ or $T \otimes_A B$, and is said to be obtained from T by extension of scalars from A to B.

If B is commutative, $T_{(B)}$ may of course be regarded as a functor from $\mathtt{Ab}_B$ to itself (III.7.1.2). If B is commutative and C is a B-algebra, then

$$T_{(C)} = (T_{(B)})_{(C)}.$$

Extension of scalars is plainly functorial and additive in T. Furthermore, if T commutes with direct limits or direct sums, then so does $T_{(B)}$; and if T is left exact, right exact, or exact, respectively, the same is true of $T_{(B)}$. Indeed, ρ_* is exact and commutes with direct limits and direct sums.

(7.1.4) Suppose again that A is commutative, and let

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}_A$$

be an additive covariant A-linear functor that commutes with direct limits. For every multiplicative subset S of A and every A-module M, there is a canonical functorial isomorphism of A-modules

(7.1.4.1)

$$T(S^{-1}M) \xrightarrow{\sim} S^{-1}T(M). \tag{7.1.4.1}$$

First suppose that S is the set of powers f^n, $n \geq 0$, of an element $f \in A$. Then

$$M_f = \varinjlim M_n,$$

where (M_n, ϕ_{nm}) is the direct system of A-modules with $M_n = M$ and

$$\phi_{nm} : z \longmapsto f^{n-m}z \qquad (0.1.6.1).$$

Formula (7.1.4.1) follows in this case from the hypothesis on T. For arbitrary S, the module $S^{-1}M$ is the direct limit of the M_f, for $f \in S$ (0.1.4.5), and the same argument applies.

The functoriality of (7.1.4.1) also shows that it is an isomorphism of $S^{-1}A$-modules. Thus, up to canonical isomorphism, one may write

(7.1.4.2)

$$T_{(S^{-1}A)}(S^{-1}M) = S^{-1}T(M) = T(S^{-1}M). \tag{7.1.4.2}$$

If $S = A - \mathfrak{p}$ is the complement of a prime ideal $\mathfrak{p}$ of A, write $T_{\mathfrak{p}}$ instead of $T_{(A_{\mathfrak{p}})}$.

Proposition (7.1.5). — *Under the hypotheses of (III.7.1.4), suppose that $T_{\mathfrak{m}}$ is left exact, respectively right exact or exact, for every maximal ideal $\mathfrak{m}$ of A. Then T is left exact, respectively right exact or exact.*

Proof. — If two submodules N and P of an A-module M satisfy

$$N_{\mathfrak{m}} = P_{\mathfrak{m}}$$

for every maximal ideal $\mathfrak{m}$ of A, then $N = P$ (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 3, th. 1). Applying this fact to the kernels and images in question proves the assertion.

7.2 Characterisation of the tensor product functor

(7.2.1) Let A be a ring, not necessarily commutative; let M (respectively N) be a left (respectively right) A-module, and let P be a $\mathbf{Z}$-module. Recall that giving a $\mathbf{Z}$-homomorphism

$$v : N \otimes_A M \longrightarrow P$$

is equivalent to giving a $\mathbf{Z}$-bilinear map

$$u : N \times M \longrightarrow P$$

such that

$$u(ta, x) = u(t, ax) \qquad (a \in A,\ t \in N,\ x \in M),$$

the two maps being related by

$$v(t \otimes x) = u(t, x).$$

On the other hand, giving u is equivalent to giving a $\mathbf{Z}$-homomorphism

$$x \longmapsto f_x$$

from M to $\mathrm{Hom}_{\mathbf{Z}}(N, P)$ such that

$$f_{ax}(t) = f_x(ta) \qquad (a \in A,\ t \in N,\ x \in M),$$

the two maps being related by $u(t, x) = f_x(t)$.

(7.2.2) Let

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

be an additive covariant functor. For every left A-module M we shall define a canonical homomorphism of $\mathbf{Z}$-modules, functorial in M,

(7.2.2.1)

$$t_M : T(A_s) \otimes_A M \longrightarrow T(M). \tag{7.2.2.1}$$

By (III.7.2.1), it is enough to define a $\mathbf{Z}$-homomorphism

$$x \longmapsto t'_M(x)$$

from M to $\mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M))$ such that

$$t'_M(ax)(y) = t'_M(x)(ya) \qquad (a \in A,\ x \in M,\ y \in T(A_s)).$$

For this purpose, note that $\mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M))$ is canonically a left A-module, by virtue of the right A-module structure on $T(A_s)$, with scalar multiplication defined by

$$(a \cdot w)(y) = w(ya)$$

for $a \in A$, $w \in \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M))$, and $y \in T(A_s)$. Define t'_M to be the composite of the two canonical homomorphisms

$$M \xrightarrow{\sim} \mathrm{Hom}_A(A_s, M) \xrightarrow{T} \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M)).$$

The second arrow sends w to $T(w)$; the first is the canonical isomorphism of A-modules sending x to the map $\theta_x : A_s \to M$ defined by

$$\theta_x(\xi) = \xi x \qquad (\xi \in A,\ x \in M).$$

Since $\theta_{ax} = \theta_x \circ h_a$, one has

$$T(\theta_{ax}) = T(\theta_x) \circ T(h_a),$$

and consequently, for $y \in T(A_s)$,

$$T(\theta_{ax})(y) = T(\theta_x)\big(T(h_a)(y)\big) = T(\theta_x)(ya)$$

by the definition of the right A-module structure on $T(A_s)$. This proves the existence of t_M.

It is immediate that this homomorphism is functorial in M: for every homomorphism $w : M \to M'$ of left A-modules, the diagram

(7.2.2.2)

$$\begin{array}{ccc} T(A_s) \otimes_A M & \xrightarrow{t_M} & T(M) \\ {\scriptstyle 1\otimes w}\downarrow & & \downarrow{\scriptstyle T(w)} \\ T(A_s) \otimes_A M' & \xrightarrow[t_{M'}]{} & T(M') \end{array} \tag{7.2.2.2}$$

is commutative. The functoriality of (7.2.2.1) also shows that, when A is commutative, t_M is a homomorphism of A-modules (III.7.1.2).

(7.2.3) When A is commutative, one can more generally define a canonical homomorphism of A-modules

(7.2.3.1)

$$T(N) \otimes_A M \longrightarrow T(N \otimes_A M). \tag{7.2.3.1}$$

for every A-module N: in the construction of (III.7.2.2), replace θ_x by the homomorphism of A-modules

$$N \longrightarrow N \otimes_A M, \qquad y \longmapsto y \otimes x.$$

It is immediate that this homomorphism is functorial in both M and N.

In particular, if B is an A-algebra, not necessarily commutative, there is a homomorphism, functorial in M,

(7.2.3.2)

$$(T(M))_{(B)} = T(M) \otimes_A B \longrightarrow T(M \otimes_A B) = T_{(B)}(M_{(B)}). \tag{7.2.3.2}$$

By the functoriality of (7.2.3.1) in M, this is a homomorphism of B-modules.

There is moreover a commutative diagram

(7.2.3.3)

$$\begin{array}{ccc} T(A) \otimes_A M & \xrightarrow{t_M} & T(M) \\ \downarrow & & \downarrow \\ T_{(B)}(B_s) \otimes_B M_{(B)} & \xrightarrow[t_{M_{(B)}}]{} & T_{(B)}(M_{(B)}) \end{array} \tag{7.2.3.3}$$

The right vertical arrow is the composite

$$T(M) \longrightarrow T(M) \otimes_A B \longrightarrow T(M \otimes_A B) = T_{(B)}(M_{(B)})$$

of the canonical homomorphism and (7.2.3.2). The left vertical arrow in (7.2.3.3) is the homomorphism

$$T(A) \otimes_A M \longrightarrow T_{(B)}(B_s) \otimes_B (B \otimes_A M) = T_{(B)}(B_s) \otimes_A M,$$

where

$$T(A) \longrightarrow T_{(B)}(B_s) = T(B)$$

is $T(\rho)$, with ρ regarded as the homomorphism of A-modules $A \to B_{[\rho]}$.

Lemma (7.2.4). — *If T is an additive covariant functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums, then the canonical homomorphism t_L of (7.2.2.1) is an isomorphism for every free A-module L.*

Proof. — Write

$$L = \bigoplus_{\alpha \in I} L_\alpha,$$

where every L_α is isomorphic to A_s. The definition of t_M in (III.7.2.2) shows that

$$t_L = \bigoplus_{\alpha \in I} t_{L_\alpha},$$

because

$$T : \mathrm{Hom}_A(A_s, L) \longrightarrow \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(L))$$

is the direct sum of the $\mathbf{Z}$-linear maps

$$T_\alpha : \mathrm{Hom}_A(A_s, L_\alpha) \longrightarrow \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(L_\alpha)),$$

by the hypothesis on T. It therefore remains only to prove the lemma for $L = A_s$; but then t_L is precisely the canonical isomorphism

$$T(A_s) \otimes_A A_s \xrightarrow{\sim} T(A_s),$$

valid for every right A-module.

Proposition (7.2.5). — *Let T be an additive covariant functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. The following conditions are equivalent:*

(a) T is right exact;

(b) the canonical homomorphism t_M of (7.2.2.1) is an isomorphism for every left A-module M;

(b′) T is semi-exact and t_M is surjective for every left A-module M;

(c) T is isomorphic to a functor of M of the form $N \otimes_A M$, where N is a right A-module.

Proof. — It is clear that (b) implies (c), and that (c) implies (a). Let us prove that (a) implies (b). Put

$$T'(M) = T(A_s) \otimes_A M$$

for every left A-module M. There is an exact sequence

$$L' \longrightarrow L \longrightarrow M \longrightarrow 0,$$

where L and L' are free left A-modules. Since T and T' are right exact, there is a commutative diagram

$$\begin{array}{ccccccc} T'(L') & \longrightarrow & T'(L) & \longrightarrow & T'(M) & \longrightarrow & 0 \\ \downarrow{\scriptstyle t_{L'}} & & \downarrow{\scriptstyle t_L} & & \downarrow{\scriptstyle t_M} & & \\ T(L') & \longrightarrow & T(L) & \longrightarrow & T(M) & \longrightarrow & 0 \end{array}$$

whose rows are exact. By (III.7.2.4), t_L and $t_{L'}$ are isomorphisms; hence so is t_M, by the five lemma.

It is clear that (b) implies (b′). To show that (b′) implies (a), it is enough to prove the following lemma.

Lemma (7.2.5.1). — *Let* $\mathtt{K}$ *and* $\mathtt{K}'$ *be abelian categories, let* F *and* G *be additive covariant functors from* $\mathtt{K}$ *to* $\mathtt{K}'$*, and let*

$$f : F \longrightarrow G$$

be a natural transformation (T, I, 1.2) *such that*

$$f_E : F(E) \longrightarrow G(E)$$

is an epimorphism for every object E *of* $\mathtt{K}$*. If* F *is right exact and* G *is semi-exact, then* G *is right exact.*

Proof. — It is enough to show that, for every epimorphism $v : E' \to E$ in $\mathtt{K}$,

$$G(v) : G(E') \longrightarrow G(E)$$

is an epimorphism. There is a commutative diagram

$$\begin{array}{ccc} F(E') & \xrightarrow{F(v)} & F(E) \\ \downarrow{\scriptstyle f_{E'}} & & \downarrow{\scriptstyle f_E} \\ G(E') & \xrightarrow[G(v)]{} & G(E) \end{array}$$

Here $F(v)$, $f_{E'}$, and f_E are epimorphisms; consequently $G(v)$ is also an epimorphism.

Remark (7.2.6). — *For every right* A*-module* N*, put*

$$T_N(M) = N \otimes_A M$$

for every left A*-module* M*. Thus* T_N *is an additive covariant functor from* $\mathtt{Ab}_A$ *to* $\mathtt{Ab}$*, is right exact, and commutes with direct sums. If* $T_N(A_s)$ *is canonically identified with* N*, the corresponding homomorphism* (7.2.2.1) *is immediately seen to be the identity. It follows that the right* A*-module* N *in (III.7.2.5)(c) is determined up to unique isomorphism and is canonically isomorphic to* $T(A_s)$*.*

Equivalently, the assignments

$$T \longmapsto T(A_s) \qquad \text{and} \qquad N \longmapsto T_N$$

define an equivalence (T, I, 1.2) *between the category of right* A*-modules and the category of additive covariant functors*

$$\mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

that are right exact and commute with direct sums.

Proposition (7.2.7). — *Let* A *be a left Artinian ring whose quotient by its radical* $\mathfrak{m}$ *is a field* k*, and let* T *be an additive covariant functor from* $\mathtt{Ab}_A$ *to* $\mathtt{Ab}$ *that commutes with direct sums. The conditions of (III.7.2.5) are then also equivalent to the following condition:*

(d) T is semi-exact and the homomorphism

$$T(\varepsilon) : T(A_s) \longrightarrow T(k)$$

induced by the canonical homomorphism $\varepsilon : A_s \to k$ is surjective.

Proof. — It is clear that condition (b′) of (III.7.2.5) implies (d). We prove that (d) implies (b′). There is an integer n such that $\mathfrak{m}^n = 0$. For every A-module M, put **III** | 180

$$M_h = \mathfrak{m}^h M.$$

We shall prove, by descending induction on h, that t_{M_h} is surjective. The assertion is clear for $h = n$. If $h < n$, there is an exact sequence

$$0 \longrightarrow M_{h+1} \longrightarrow M_h \longrightarrow M_h/M_{h+1} \longrightarrow 0,$$

and the induction hypothesis says that $t_{M_{h+1}}$ is surjective. Moreover, M_h/M_{h+1} is annihilated by $\mathfrak{m}$, and is therefore an $(A/\mathfrak{m})$-module; in other words, it is a direct sum of A-modules isomorphic to k. Since T commutes with direct sums, it is therefore enough, in order to prove that $t_{M_h/M_{h+1}}$ is surjective, to prove that t_k is surjective. The diagram

$$\begin{array}{ccc} T(A_s) \otimes_A A_s & \xrightarrow{t_{A_s}} & T(A_s) \\ {\scriptstyle 1\otimes\varepsilon}\downarrow & & \downarrow{\scriptstyle T(\varepsilon)} \\ T(A_s) \otimes_A k & \xrightarrow[t_k]{} & T(k) \end{array}$$

is commutative. Condition (d), together with (III.7.2.4), therefore shows that t_k is surjective.

To finish the proof, it remains to show that if

$$0 \longrightarrow M' \longrightarrow M \xrightarrow{v} M'' \longrightarrow 0$$

is an exact sequence of A-modules for which $t_{M'}$ and $t_{M''}$ are surjective, then t_M is surjective. There is a commutative diagram

$$\begin{array}{ccccccc} T'(M') & \longrightarrow & T'(M) & \longrightarrow & T'(M'') & \longrightarrow & 0 \\ {\scriptstyle t_{M'}}\downarrow & & {\scriptstyle t_M}\downarrow & & {\scriptstyle t_{M''}}\downarrow & & \downarrow \\ T(M') & \longrightarrow & T(M) & \xrightarrow[T(v)]{} & T(M'') & \longrightarrow & \mathrm{Coker}(T(v)) \end{array}$$

whose two rows are exact, by the hypothesis that T is semi-exact. The induction hypothesis says that $t_{M'}$ and $t_{M''}$ are epimorphisms, while the last vertical arrow is a monomorphism. The five lemma *(M, I, 1.1)* therefore shows that t_M is an epimorphism.

7.3 Exactness criteria for homological functors of modules

Proposition (7.3.1). — *Let A be a ring, not necessarily commutative, and let $T_\bullet$ be a covariant homological functor* (T, II, 2.1) *from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Let p be an integer such that T_p and T_{p-1} are defined. The following conditions are equivalent:*

(a) T_p is right exact;

(b) T_{p-1} is left exact; **III** | 181

(c) for every left A-module M, the canonical functorial homomorphism (7.2.2.1)

$$T_p(A_s) \otimes_A M \longrightarrow T_p(M) \tag{7.3.1.1}$$

is an isomorphism;

(d) for every left A-module M, the homomorphism (7.3.1.1) *is an epimorphism;*

(e) T_p is isomorphic to a functor of the form

$$M \longmapsto N \otimes_A M,$$

where N is a right A-module.

If, in addition, the conditions of (III.7.2.7) on A and $\mathfrak{m}$ are satisfied, the preceding conditions are also equivalent to

(f) the canonical homomorphism

$$T_p(\varepsilon) : T_p(A_s) \longrightarrow T_p(k)$$

is an epimorphism.

Proof. — By definition of a homological functor, T_i is semi-exact for every i for which it is defined; moreover, for every exact sequence

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0,$$

one has

$$\mathrm{Ker}\big(T_{i-1}(u)\big) = \mathrm{Coker}\big(T_i(v)\big).$$

It is therefore clear that (a) and (b) are equivalent, and the other assertions follow immediately from (III.7.2.5) and (III.7.2.7).

Corollary (7.3.2). — *Let A be a commutative ring. With the notation of (III.7.3.1), suppose that T_p is right exact. If $f \in A$ annihilates no nonzero element of an A-module M, then it annihilates no nonzero element of $T_{p-1}(M)$. In particular, if A is an integral domain, the A-module $T_{p-1}(A)$ is torsion-free.*

Proof. — If h_f denotes the homothety $x \mapsto fx$ of M, the hypothesis says that h_f is injective. Condition (b) of (III.7.3.1) then shows that $T_{p-1}(h_f)$ is injective as well.

Proposition (7.3.3). — *Let A be a ring, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Let p be an integer such that T_{p-1}, T_p, and T_{p+1} are defined. The following conditions are equivalent:*

(a) T_p is exact;

(b) T_{p+1} and T_p are right exact;

(c) T_p and T_{p-1} are left exact;

(d) T_{p+1} is right exact and T_{p-1} is left exact;

(e) for every A-module M, the canonical homomorphisms

$$T_i(A_s) \otimes_A M \longrightarrow T_i(M) \tag{7.3.3.1}$$

are isomorphisms for $i = p$ and $i = p+1$;

(e′) for every A-module M, the homomorphisms (7.3.3.1) are epimorphisms for $i = p$ and $i = p+1$;

(f) for every A-module M, the homomorphism (7.3.3.1) is an isomorphism for $i = p$, and $T_p(A_s)$ is flat as a right A-module;

(f′) for every A-module M, the homomorphism (7.3.3.1) is an epimorphism for $i = p$, and $T_p(A_s)$ is flat as a right A-module.

Proof. — The equivalence of (a), (b), (c), and (d) follows from the equivalence of conditions (a) and (b) of (III.7.3.1). The equivalence of (b), (e), and (e′) follows from the equivalence of (a), (c), and (d) in (III.7.3.1). Finally, to say that $T_p(A_s)$ is flat means that the functor

$$M \longmapsto T_p(A_s) \otimes_A M$$

is left exact; the equivalence of (a), (f), and (f′) again follows from the equivalence of (a), (c), and (d) in (III.7.3.1).

Corollary (7.3.4). — *Suppose that A is commutative, that T_p is exact, and that $T_p(A)$ is an A-module of finite presentation. Then the function*

$$x \longmapsto \mathrm{rank}_{\kappa(x)}\big(T_p(\kappa(x))\big)$$

is locally constant on $X = \mathrm{Spec}(A)$, and is therefore constant if $\mathrm{Spec}(A)$ is connected.

Proof. — By condition (f) of (III.7.3.3), $T_p(A)$ is a flat A-module. It is consequently projective of finite type, so $\widetilde{T_p(A)}$ is a locally free $\mathscr{O}_X$-module (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 2, th. 1). Moreover,

$$T_p(\kappa(x)) = T_p(A) \otimes_A \kappa(x)$$

by condition (e) of (III.7.3.3), and the rank at x of the A-module $T_p(A)$ is locally constant *(loc. cit.)*. This proves the corollary.

Proposition (7.3.5). — *Suppose that A is a left Artinian ring whose quotient by its radical $\mathfrak{m}$ is a field k. The conditions of (III.7.3.3) are also equivalent to each of the following:*

(g) the canonical homomorphism

$$T_i(\varepsilon) : T_i(A_s) \longrightarrow T_i(k)$$

is an epimorphism for $i = p$ and $i = p + 1$;

(h) $T_p(\varepsilon)$ is an epimorphism and $T_p(A_s)$ is flat as a right A-module, or, equivalently, is a free right A-module (Bourbaki, Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5).*

Suppose in addition that A is commutative and that the A-module $T_p(k)$ has finite length d. The preceding conditions are then also equivalent to each of the following:

(i) for every A-module M of finite length,

$$\operatorname{length}\big(T_p(M)\big) = d \operatorname{length}(M); \tag{7.3.5.1}$$

(j)

$$\operatorname{length}\big(T_p(A)\big) = d \operatorname{length}(A). \tag{7.3.5.2}$$

The equivalence of (g) and (h) with the conditions of (III.7.3.3) follows at once from (III.7.2.7). For the remaining assertions we shall use the following lemma.

Lemma (7.3.5.3). — *Let $\mathtt{K}$ and $\mathtt{K}'$ be abelian categories, and let*

$$F : \mathtt{K} \longrightarrow \mathtt{K}'$$

be an additive covariant functor. Suppose that F is semi-exact and that $F(S)$ has finite length in $\mathtt{K}'$ for every simple object S of $\mathtt{K}$. Then $F(E)$ has finite length in $\mathtt{K}'$ for every object E of finite length in $\mathtt{K}$. For every exact sequence

$$0 \longrightarrow E' \xrightarrow{u} E \xrightarrow{v} E'' \longrightarrow 0$$

of objects of finite length in $\mathtt{K}$, one has

$$\operatorname{length} F(E) \leq \operatorname{length} F(E') + \operatorname{length} F(E''), \tag{7.3.5.4}$$

and equality in (7.3.5.4) holds if and only if

$$0 \longrightarrow F(E') \longrightarrow F(E) \longrightarrow F(E'') \longrightarrow 0$$

is exact.

Proof. — The sequence

$$F(E') \xrightarrow{F(u)} F(E) \xrightarrow{F(v)} F(E'')$$

is exact by hypothesis. If $F(E')$ and $F(E'')$ have finite length, so do $\Im(F(u))$ and $\Im(F(v))$. Since $\operatorname{Ker}(F(v)) = \Im(F(u))$, the object $F(E)$ has finite length, and

$$\begin{aligned} \operatorname{length} F(E) &= \operatorname{length} \Im(F(u)) + \operatorname{length} \Im(F(v)) \\ &\leq \operatorname{length} F(E') + \operatorname{length} F(E''). \end{aligned} \tag{7.3.5.5}$$

Induction on the length of E proves the first assertion. Moreover, equality in (7.3.5.5) can hold only when

$$\operatorname{length} \Im(F(u)) = \operatorname{length} F(E')$$

and

$$\operatorname{length} \Im(F(v)) = \operatorname{length} F(E'').$$

The first equality is equivalent to $\operatorname{length}\operatorname{Ker}(F(u)) = 0$, hence to $\operatorname{Ker}(F(u)) = 0$; the second is equivalent to $\operatorname{length}\operatorname{Coker}(F(v)) = 0$, hence to $\operatorname{Coker}(F(v)) = 0$. This proves the last assertion.

Now let M be an A-module of finite length, with A commutative. The quotients in a Jordan–Hölder series of M are necessarily isomorphic to the A-module k. Induction on the length of M, using (7.3.5.4), therefore gives

$$\operatorname{length} T_p(M) \leq d \operatorname{length}(M). \tag{7.3.5.6}$$

It also follows from (III.7.3.5.3) that if T_p is exact, then equality (7.3.5.1) holds. Thus condition (a) of (III.7.3.3) implies (i), and (i) plainly implies (j). It remains to prove the following lemma.

Lemma (7.3.5.7). — *The equality*

$$\operatorname{length} T_p(A) = d \operatorname{length} A$$

implies that $T_p(\varepsilon)$ is an epimorphism and that $T_p(A)$ is a flat A-module.

Proof. — Starting from the exact sequence

$$0 \longrightarrow \mathfrak{m} \longrightarrow A \longrightarrow k \longrightarrow 0,$$

formulas (7.3.5.4) and (7.3.5.6) give

$$\begin{aligned}\operatorname{length} T_p(A) &\leq \operatorname{length} T_p(\mathfrak{m}) + \operatorname{length} T_p(k)\\ &\leq d\big(\operatorname{length}\mathfrak{m} + \operatorname{length} k\big) = d \operatorname{length} A.\end{aligned}$$

By (III.7.3.5.3), equality can hold only if

$$0 \longrightarrow T_p(\mathfrak{m}) \longrightarrow T_p(A) \longrightarrow T_p(k) \longrightarrow 0 \tag{7.3.5.8}$$

is exact. By (III.7.2.7) and (III.7.2.5), T_p is isomorphic to a functor

$$M \longmapsto N \otimes_A M.$$

The exactness of (7.3.5.8), together with the exact sequence of Tor, shows that

$$\operatorname{Tor}_1^A(N, k) = 0.$$

It follows that $N = T_p(A)$ is a flat A-module *(0, 10.1.3)*.

Lemma (7.3.6). — *Let A be a ring, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Suppose that T_p and T_{p+1} are defined and that T_p is left exact. Then T_{p+1} is exact if and only if $T_{p+1}(A_s)$ is flat as a right A-module.*

Proof. — By (III.7.3.1), the canonical homomorphism

$$T_{p+1}(A_s) \otimes_A M \longrightarrow T_{p+1}(M)$$

is an isomorphism of functors. The assertion now follows from the definition of a flat A-module.

Proposition (7.3.7). — *Let A be a ring, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Suppose there is an i_0 such that T_i is exact for $i \leq i_0$. Then, for every integer $p > i_0$, the following conditions are equivalent:*

(a) T_q is exact for $q \leq p$;

(b) $T_q(A_s)$ is flat as a right A-module for $q \leq p$;

(c) for every A-module M, the canonical homomorphism

$$T_q(A_s) \otimes_A M \longrightarrow T_q(M)$$

is surjective for $q \leq p+1$.

Proof. — The equivalence of (a) and (b) follows by induction on q from (III.7.3.6), since T_{i_0} is exact by hypothesis. The equivalence of (a) and (c) follows from the equivalence of conditions (a) and (e′) in (III.7.3.3). III | 184

(7.3.8) Let A be a commutative ring, let B be an A-algebra, not necessarily commutative, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$. It follows from the definitions in (III.7.1.3) that the functor from $\mathtt{Ab}_B$ to $\mathtt{Ab}$ obtained by extension of scalars from A to B, denoted

$$T_\bullet^{(B)} = \big(T_i^{(B)}\big),$$

is again a homological functor.

Corollary (7.3.9). — *Suppose that $T_\bullet$ satisfies the general conditions of (III.7.3.7) and commutes with direct limits. Suppose further that A is an integral domain and that every $T_n(A)$ is an A-module of finite presentation. Then, for every integer N, there is an $f \in A - \{0\}$ such that*

$$T_p^{(A_f)} : \mathtt{Ab}_{A_f} \longrightarrow \mathtt{Ab}$$

is exact for $p \leq N$.

Proof. — By hypothesis, T_i is exact for $i \leq i_0$, so $T_i(A)$ is flat for these values of i. By condition (b) of (III.7.3.7), it is enough to choose f such that

$$T_i^{(A_f)}(A_f) = T_i(A_f)$$

is a free A_f-module for $i_0 < i \leq N$. Now

$$T_i(A_f) = \big(T_i(A)\big)_f,$$

since T_i commutes with direct limits (III.7.1.4). If x is the generic point of $\mathrm{Spec}(A)$, then $\big(T_i(A)\big)_x$ is a finite-dimensional vector space over the fraction field of A. Since each $T_i(A)$ is of finite presentation, an f with the desired property exists (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 1, cor. of prop. 2).

If only finitely many indices i satisfy $T_i \neq 0$, one may choose $f \in A - \{0\}$ such that every $T_p^{(A_f)}$ is exact.

Corollary (7.3.10). — *Suppose that $T_\bullet$ satisfies the general conditions of (III.7.3.7) and commutes with direct limits, that A is commutative and Noetherian, and that every $T_n(A)$ is an A-module of finite type. Then, for every integer N, there is a dense open subset U of* $\mathrm{Spec}(A)$ *such that, for every integer $p \leq N$, the function*

$$x \longmapsto \mathrm{rank}_{\kappa(x)}\big(T_p(\kappa(x))\big)$$

is constant on U.

Proof. — Let $\mathfrak{p}$ be a minimal prime ideal of A. The ring $B = A/\mathfrak{p}$ is an integral domain, and $\mathrm{Spec}(B)$ is identified with an irreducible component of the topological space $\mathrm{Spec}(A)$. We prove by induction on $p \leq N$ that there is an $f_p \in B - \{0\}$ such that, if $B' = B_{f_p}$, then $T_i^{(B')}$ is exact and $T_i(B')$ is a B'-module of finite type for $i \leq p$.

The assertion holds for $p \leq i_0$ by hypothesis, with $f_p = 1$, so that $B' = B = A/\mathfrak{p}$. Indeed, T_p is then exact, and $T_p(B)$ is isomorphic to

$$T_p(A)/T_p(\mathfrak{p});$$

it is therefore an A-module, and *a fortiori* a B-module, of finite type.

For the induction step, let f_p be the canonical image in B of an element $g_p \in A$, and put $A' = A_{g_p}$. Then

$$B' = A'/\mathfrak{p}',$$

where $\mathfrak{p}'$ is the minimal prime ideal $\mathfrak{p}A_{g_p}$ of A'. Since

$$T_i(A_{g_p}) = \big(T_i(A)\big)_{g_p},$$

the $T_i(A')$ are A'-modules of finite type. Thus $T_\bullet^{(A')}$ satisfies the same hypotheses as $T_\bullet$, with i_0 replaced by p.

We may therefore reduce to the case $A' = A$ in which T_p is exact. The exact sequence

$$0 \longrightarrow \mathfrak{p} \longrightarrow A \longrightarrow A/\mathfrak{p} \longrightarrow 0$$

then gives an exact sequence

$$T_{p+1}(A) \longrightarrow T_{p+1}(A/\mathfrak{p}) \xrightarrow{\partial} T_p(\mathfrak{p}) \longrightarrow T_p(A).$$

Since T_p is exact, the last arrow is injective. Consequently $T_{p+1}(A/\mathfrak{p})$ is a quotient of $T_{p+1}(A)$, and is therefore of finite type.

The proof of (III.7.3.9) used the finite-type hypothesis on $T_i(A)$ only for $i \leq N$. We may therefore apply that corollary to the integral domain B, the functor $T_\bullet^{(B)}$, and $N = p + 1$, which completes the induction.

There is consequently an $f_N \in B - \{0\}$ such that $\mathrm{Spec}(B_{f_N})$ is an open subset V, dense in $\mathrm{Spec}(B)$, that meets no other irreducible component of $\mathrm{Spec}(A)$. **III** | 185

If the assertion has been proved for B_{f_N}, there is a dense open subset W of V on which the functions in the statement are constant, since

$$A_x = (B_{f_N})_x \qquad (x \in W).$$

Repeating the same argument for every irreducible component of $\mathrm{Spec}(A)$ proves the corollary. We may therefore reduce to the case where A is an integral domain. The argument of (III.7.3.9) then gives an $f \in A - \{0\}$ such that the $T_p(A_f)$ are free A_f-modules of finite type for $p \leq N$. The conclusion follows from (III.7.3.4).

Proposition (7.3.11). — *Let A be a commutative local ring with residue field k, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Suppose there is an i_0 such that T_i is exact for $i \leq i_0$, and that every $T_n(A)$ is an A-module of finite presentation. The equivalent conditions (a), (b), and (c) of (III.7.3.7) imply the following two conditions, and are equivalent to them when A is also reduced:*

(d) for every $x \in \mathrm{Spec}(A)$,

$$\mathrm{rank}_{\kappa(x)}\, T_q(\kappa(x)) = \mathrm{rank}_k\, T_q(k) \qquad (q \leq p);$$

(d′) for every generic point x_j of an irreducible component of $\mathrm{Spec}(A)$*,*

$$\mathrm{rank}_{\kappa(x_j)}\, T_q(\kappa(x_j)) = \mathrm{rank}_k\, T_q(k) \qquad (q \leq p).$$

Proof. — Since $T_q(A)$ is an A-module of finite presentation, condition (b) of (III.7.3.7) is equivalent to saying that $T_q(A)$ is free for $q \leq p$ (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5). Condition (c) implies

$$T_q(\kappa(x)) = T_q(A) \otimes_A \kappa(x) \qquad (q \leq p).$$

Thus the equivalent conditions of (III.7.3.7) imply (d), and it is immediate that (d) implies (d′).

It remains to show that (d′) implies (a) when A is reduced. We argue by induction on $q \leq p$, since T_q is exact for $q \leq i_0$. Suppose that T_k is exact for $k \leq q < p$. We shall show that $T_{q+1}(A)$ is a free A-module. By the induction hypothesis, the canonical map

$$T_{q+1}(A) \otimes_A M \longrightarrow T_{q+1}(M)$$

is an isomorphism for every A-module M, by condition (c) of (III.7.3.7) and (III.7.3.3). Applying this to $M = \kappa(x_j)$ and $M = k$, condition (d′) gives

$$\mathrm{rank}_{\kappa(x_j)}\big(T_{q+1}(A) \otimes_A \kappa(x_j)\big) = \mathrm{rank}_k\, T_{q+1}(k)$$

for every generic-point index j. It follows that $T_{q+1}(A)$ is free (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, prop. 7), which completes the proof.

The preceding results will be considerably strengthened for the homological functors of the special type to be studied in (III.7.4): the resulting exactness criteria involve only a single one of the T_p.

7.4 Exactness criteria for the functors $H_\bullet(P_\bullet \otimes_A M)$

(7.4.1) Let A be a ring, not necessarily commutative, and let $P_\bullet$ be a complex of flat right A-modules. Since the functor

$$M \longmapsto P_k \otimes_A M$$

is then exact in $\mathtt{Ab}_A$ for every k, the ∂-functor

$$T_\bullet(M) = H_\bullet(P_\bullet \otimes_A M) \tag{7.4.1.1}$$

is a *homological functor* from $\mathtt{Ab}_A$ to $\mathtt{Ab}$; it is plainly A-linear when A is commutative (III.7.1.2), and it commutes with inductive limits. **III** | 186

If A is commutative, then, for every A-algebra B, the homological functor $T_\bullet^{(B)}$ (III.7.3.8) is, by definition, given by

$$T_i^{(B)}(N) = H_i(P_\bullet \otimes_A N_{[\rho]}), \tag{7.4.1.2}$$

where $\rho : A \to B$ is the homomorphism defining the A-algebra structure on B. Since one may also write

$$P_\bullet \otimes_A N_{[\rho]} = P_\bullet \otimes_A (B \otimes_B N)_{[\rho]} = (P_\bullet \otimes_A B) \otimes_B N,$$

it follows that

$$T_i^{(B)}(N) = H_i(P'_\bullet \otimes_B N) \tag{7.4.1.3}$$

for every B-module N, where $P'_\bullet$ is the complex $P_\bullet \otimes_A B$ of flat B-modules (0.6.2.1).

Proposition (7.4.2). — *Under the general hypotheses of (III.7.4.1), and for a given integer $p \in \mathbf{Z}$, the following conditions are equivalent:*

(a) T_p is left exact, or, equivalently, T_{p+1} is right exact;

(b)

$$Z'_p(P_\bullet) = \mathrm{Coker}(P_{p+1} \longrightarrow P_p)$$

is a flat right A-module;

(c) there exists a complex $P'_\bullet$ of flat right A-modules for which the differential

$$d'_{p+1} : P'_{p+1} \longrightarrow P'_p$$

is zero, together with an isomorphism of homological functors

$$H_\bullet(P_\bullet \otimes_A M) \simeq H_\bullet(P'_\bullet \otimes_A M).$$

Proof. — By definition there is a sequence, exact and functorial in M,

$$0 \longrightarrow T_p(M) \longrightarrow Z'_p(P_\bullet) \otimes_A M \longrightarrow P_{p-1} \otimes_A M.$$

Indeed, by right exactness of the tensor product,

$$Z'_p(P_\bullet \otimes_A M) = \mathrm{Coker}(P_{p+1} \otimes_A M \longrightarrow P_p \otimes_A M) = Z'_p(P_\bullet) \otimes_A M.$$

For every homomorphism $f : M \to N$ one therefore has the commutative diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & T_p(M) & \longrightarrow & Z'_p(P_\bullet) \otimes_A M & \longrightarrow & P_{p-1} \otimes_A M \\ & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle v} & & \downarrow{\scriptstyle w} \\ 0 & \longrightarrow & T_p(N) & \longrightarrow & Z'_p(P_\bullet) \otimes_A N & \longrightarrow & P_{p-1} \otimes_A N, \end{array} \tag{7.4.2.1}$$

whose rows are exact. If f is a monomorphism, then so is w, because P_{p-1} is flat. If T_p is left exact, u is likewise a monomorphism, and it follows that v is a monomorphism. Thus $Z'_p(P_\bullet)$ is flat. Conversely, if $Z'_p(P_\bullet)$ is flat, then v is a monomorphism for every monomorphism $f : M \to N$, and diagram (7.4.2.1) shows that u is a monomorphism. Hence T_p, which is already half-exact, is left exact. This proves the equivalence of (a) and (b).

It is immediate that (c) implies (a). Indeed, if $d'_{p+1} : P'_{p+1} \to P'_p$ is zero and

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A-modules, then the boundary map in the exact sequence

$$H_{p+1}(P'_\bullet \otimes_A M'') \xrightarrow{\partial} H_p(P'_\bullet \otimes_A M') \longrightarrow H_p(P'_\bullet \otimes_A M)$$

is zero by definition *(M, IV, 1)*, so T_p is left exact. **III** | 187

Conversely, let us prove that (b) implies (c). Put

$$C = Z'_p(P_\bullet), \qquad B = B_p(P_\bullet) = \Im(d_{p+1}), \qquad Z = Z_{p+1}(P_\bullet) = \mathrm{Ker}(d_{p+1}).$$

There are two short exact sequences

$$0 \longrightarrow B \longrightarrow P_p \longrightarrow C \longrightarrow 0, \qquad 0 \longrightarrow Z \longrightarrow P_{p+1} \longrightarrow B \longrightarrow 0.$$

Since C and P_p are flat, B is flat; since B and P_{p+1} are flat, Z is flat. Both sequences remain exact after tensoring with any left A-module M (0.6.1.2). In particular, the canonical maps identify $B \otimes_A M$ with the image of $d_{p+1} \otimes 1_M$ and $Z \otimes_A M$ with its kernel.

Define a complex $U_\bullet$ by $U_i = P_i$ for $i \geq p+2$, $U_{p+1} = Z$, and $U_i = 0$ for $i \leq p$; its differential in degree $p+2$ is d_{p+2} with target restricted to Z, and its other nonzero differentials are inherited from $P_\bullet$. Similarly,

define $L_\bullet$ by $L_i = P_i$ for $i \leq p-1$, $L_p = C$, and $L_i = 0$ for $i \geq p+1$; its differential in degree p is induced by d_p on the quotient C. Then $P'_\bullet = U_\bullet \oplus L_\bullet$ is a complex of flat modules, with $P'_i = P_i$ for $i \neq p, p+1$, $P'_p = C$, $P'_{p+1} = Z$, and $d'_{p+1} = 0$.

The inclusion in degree $p+1$ defines a map of complexes $a : U_\bullet \to P_\bullet$, and the quotient in degree p defines a map $b : P_\bullet \to L_\bullet$. After tensoring with M, the first induces homology isomorphisms in every degree $i \geq p+1$: in degree $p+1$ this follows from the kernel identification above; in degree $p+2$, from the injectivity of $Z \otimes_A M \to P_{p+1} \otimes_A M$; the higher degrees are unchanged. The second induces homology isomorphisms in every degree $i \leq p$: in degree p, one takes the quotient by the image $B \otimes_A M$; in degree $p-1$, one uses the surjectivity of $P_p \otimes_A M \to C \otimes_A M$; the lower degrees are unchanged. Taking the inverses of the first isomorphisms and retaining the second ones therefore gives natural isomorphisms, for every i,

$$H_i(P_\bullet \otimes_A M) \xrightarrow{\sim} H_i(P'_\bullet \otimes_A M).$$

These commute with the boundary maps associated to every short exact sequence $0 \to M' \to M \to M'' \to 0$: in degrees $i \geq p+2$ and $i \leq p$, this follows from naturality for the maps of complexes a and b. In degree $p+1$, every cycle in $P_\bullet \otimes_A M''$ comes from $Z \otimes_A M''$ and lifts, through the surjection $Z \otimes_A M \to Z \otimes_A M''$, to a cycle in $P_\bullet \otimes_A M$, so the boundary map is zero. The boundary map for $P'_\bullet$ is also zero because $d'_{p+1} = 0$. Thus these isomorphisms form the required isomorphism of ∂-functors.

We also note that the conditions of (III.7.4.2) imply that $B_p(P_\bullet)$ is flat, since there is an exact sequence

$$0 \longrightarrow B_p(P_\bullet) \longrightarrow P_p \longrightarrow Z'_p(P_\bullet) \longrightarrow 0$$

in which P_p and $Z'_p(P_\bullet)$ are flat (0.6.1.2).

Corollary (7.4.3). — *Suppose that A is a regular Noetherian ring of dimension 1, in other words, a product of Dedekind rings (0.17.1.3) and (0.17.3.7); for example, A may be a principal ideal ring. Then T_p is left exact if and only if $T_p(A)$ is a flat A-module. The functor T_p is exact if and only if both $T_p(A)$ and $T_{p-1}(A)$ are flat A-modules.*

Proof. — Recall that, for a module M over a Dedekind ring, being flat is equivalent to being torsion-free (0.6.3.3) and (0.6.3.4). Under the hypotheses of (III.7.4.3), every submodule of a flat A-module is therefore flat.

The second assertion follows from the first, since saying that T_p is exact amounts to saying that T_p and T_{p-1} are left exact. To prove the first assertion, observe that there is an exact sequence

$$0 \longrightarrow H_p(P_\bullet) \longrightarrow Z'_p(P_\bullet) \longrightarrow B_{p-1}(P_\bullet) \longrightarrow 0,$$

where $B_{p-1}(P_\bullet)$ is flat, being a submodule of the flat A-module P_{p-1}. Hence $H_p(P_\bullet)$ is flat if and only if $Z'_p(P_\bullet)$ is flat (0.6.1.2).

The most important applications of (III.7.4.2) are the following.

Proposition (7.4.4). — *Let A be a Noetherian ring and $P_\bullet$ a complex of flat A-modules. Suppose either that the P_i are of finite type, or that the $H_i(P_\bullet)$ are A-modules of finite type and there exists an i_0 such that $H_i(P_\bullet) = 0$ for $i < i_0$. Let $T_\bullet$ be the homological functor defined by (7.4.1.1). Then the set U of points $y \in \operatorname{Spec}(A)$ such that $(T_p)_y$ (III.7.1.4) is right exact, respectively left exact or exact, is open in $\operatorname{Spec}(A)$.*

Proof. — Under the second hypothesis on $P_\bullet$, one may replace $P_\bullet$ by a complex $P'_\bullet$ of finite free A-modules **III** | 188 such that the functor

$$H_\bullet(P'_\bullet \otimes_A M)$$

is isomorphic to $T_\bullet(M)$ as a ∂-functor (0.11.9.3). Thus one may always reduce to the first hypothesis. In that case the modules $Z'_i(P_\bullet)$ are of finite type. Moreover, it is enough to prove the assertions concerning left exactness (cf. (III.7.4.2), (a)).

Let $x \in U$. Since $M \mapsto M_x$ is exact,

$$\big(Z'_p(P_\bullet)\big)_x = Z'_p\big((P_\bullet)_x\big).$$

Taking (7.4.1.3) into account, the hypothesis and (III.7.4.2), (b), show that $\big(Z'_p(P_\bullet)\big)_x$ is a flat A_x-module. It is therefore free, since it is of finite type and A_x is a Noetherian local ring (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5). It follows that there is an $f \in A$ for which $\big(Z'_p(P_\bullet)\big)_f$ is free over A_f (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 1, cor. of prop. 2). A fortiori, $\big(Z'_p(P_\bullet)\big)_y$ is free over A_y for every $y \in D(f)$, and (III.7.4.2), (b), completes the proof.

Corollary (7.4.5). — *Under the hypotheses of (III.7.4.4), suppose in addition that A is an integral domain. Then the set U of $x \in \operatorname{Spec}(A)$ such that $(T_p)_x$ is exact is open and nonempty.*

Proof. — It is enough to show that $(T_p)_x$ is exact at the generic point x of $\mathrm{Spec}(A)$. This is immediate, since A_x is a field and every additive functor on $\mathtt{Ab}_{A_x}$ is exact.

Proposition (7.4.6). — *Under the general hypotheses of (III.7.4.4), conditions (a), (b), and (c) of (III.7.4.2) are also equivalent to:*

(d) there exists an A-module Q and a functorial isomorphism

$$T_p(M) \simeq \mathrm{Hom}_A(Q, M). \tag{7.4.6.1}$$

Moreover, this property determines the A-module Q up to a unique isomorphism, and Q is of finite type.

Proof. — The uniqueness of Q is a special case of the uniqueness of a representing object for a representable functor (0.8.1.5). It is clear that the right-hand side of (7.4.6.1) is left exact.

Conversely, suppose that T_p is left exact. As in (III.7.4.4), one may first reduce to the case where the P_i are flat and of finite type, and hence, since A is Noetherian, projective of finite type (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 2, cor. of th. 1). The dual $\check{P}_i$ of P_i is then also a projective A-module of finite type, P_i is canonically isomorphic to the dual of $\check{P}_i$, and the canonical homomorphism

$$P_i \otimes_A M \longrightarrow \mathrm{Hom}_A(\check{P}_i, M)$$

is bijective (Bourbaki, *Algèbre*, chap. II, 3rd ed., § 4, no. 2, prop. 2).

By (III.7.4.2), (c), one may furthermore suppose that $d_{p+1} : P_{p+1} \to P_p$ is zero. There is then an exact sequence

$$0 \longrightarrow T_p(M) \xrightarrow{u} P_p \otimes_A M \xrightarrow{v} P_{p-1} \otimes_A M,$$

where $v = d_p \otimes 1$. Put $Q' = \mathrm{Ker}(d_p)$, so that

$$0 \longrightarrow Q' \longrightarrow P_p \xrightarrow{d_p} P_{p-1}$$

is exact. Transposition gives the exact sequence

$$\check{P}_{p-1} \xrightarrow{{}^t d_p} \check{P}_p \longrightarrow \check{Q}' \longrightarrow 0.$$

We claim that

$$Q = \check{Q}' = \mathrm{Coker}({}^t d_p)$$

has the required property. Indeed, there is an exact sequence

$$0 \longrightarrow \mathrm{Hom}_A(Q, M) \longrightarrow \mathrm{Hom}_A(\check{P}_p, M) \xrightarrow{v'} \mathrm{Hom}_A(\check{P}_{p-1}, M),$$

where $v' = \mathrm{Hom}_A({}^t d_p, 1)$. Under the canonical identifications

$$P_i \otimes_A M \simeq \mathrm{Hom}_A(\check{P}_i, M),$$

the homomorphism v' identifies with $v = d_p \otimes 1$. Thus one obtains the required functorial isomorphism

$$T_p(M) \xrightarrow{\sim} \mathrm{Hom}_A(Q, M).$$

Finally, Q is a quotient of $\check{P}_p$, and is therefore of finite type.

Proposition (7.4.7). — *Suppose that the general hypotheses of (III.7.4.4) hold. For every A-module M of finite type:*

(i) the $T_i(M)$ are A-modules of finite type;

(ii) for every ideal $\mathfrak{m}$ of A, the canonical homomorphism

$$\widehat{T_i(M)} \longrightarrow \varprojlim_n T_i\big(M \otimes_A (A/\mathfrak{m}^{n+1})\big) \tag{7.4.7.1}$$

is bijective, where the left-hand side denotes the separated completion of $T_i(M)$ for the $\mathfrak{m}$-adic topology.

Proof. — As in (III.7.4.4), one reduces first to the case where the P_i are of finite type. Since A is Noetherian, the submodules of the $P_i \otimes_A M$ are of finite type, which immediately proves (i). Assertion (ii) follows from the more general lemma below.

Lemma (7.4.7.2). — *Let A be a Noetherian ring and $u : E \to F$ a homomorphism of finite-type A-modules. For every finite-type A-module M, put*

$$K(M) = \mathrm{Ker}(u \otimes 1_M), \qquad C(M) = \mathrm{Coker}(u \otimes 1_M).$$

Then, for every ideal $\mathfrak{m}$ of A, the canonical homomorphisms

$$\widehat{K(M)} \longrightarrow \varprojlim_n K(M_n), \qquad \widehat{C(M)} \longrightarrow \varprojlim_n C(M_n) \tag{7.4.7.3}$$

are bijective, where

$$M_n = M \otimes_A (A/\mathfrak{m}^{n+1}) = M/\mathfrak{m}^{n+1}M.$$

Proof. — The modules $E \otimes_A M$ and $F \otimes_A M$ are of finite type, and the functor $M \mapsto \widehat{M}$ is exact on the category of finite-type A-modules (0.7.3.3). Consequently, $\widehat{K(M)}$ and $\widehat{C(M)}$ are, respectively, the kernel and the cokernel of

$$\widehat{u \otimes 1} : \widehat{E \otimes_A M} \longrightarrow \widehat{F \otimes_A M}.$$

Left exactness of $\varprojlim$ therefore gives

$$\widehat{K(M)} = \varprojlim_n K(M_n).$$

On the other hand, right exactness of the tensor product gives

$$C(M_n) = C(M) \otimes_A (A/\mathfrak{m}^{n+1}),$$

and hence

$$\widehat{C(M)} = \varprojlim_n C(M_n)$$

by definition.

Remark (7.4.8). — *Taking (III.6.10.5) and (III.6.10.6) into account, one sees that, with an additional flatness hypothesis, (III.7.4.7) recovers the fact that (4.1.7.1) is an isomorphism, which is the essential content of the "first comparison theorem" for proper morphisms. Moreover, the statement applies not only to a coherent $\mathscr{O}_X$-module but to a* complex *of such modules. It would be interesting to obtain a statement that contains both (III.7.4.7) and (4.1.7.1) as special cases.*

Notice also that, when the P_i are of finite type, the proof of (III.7.4.7) does not use their flatness. It would be worth examining whether the conclusion of (III.7.4.7) continues to hold when the P_i are assumed neither flat nor of finite type, but the $H_i(P_\bullet)$ are assumed to be of finite type for every i and zero for $i < i_0$. Can one then replace $P_\bullet$ by a complex $P'_\bullet$ of finite-type A-modules in such a way that the functors

$$H_\bullet(P_\bullet \otimes_A M) \quad \textit{and} \quad H_\bullet(P'_\bullet \otimes_A M),$$

which are no longer homological, are still isomorphic?

7.5 The case of Noetherian local rings

(7.5.1) Let A be a Noetherian local ring and $\mathfrak{m}$ its maximal ideal. For every A-module M, denote by $\widehat{M}$ its separated completion for the $\mathfrak{m}$-adic topology, which is isomorphic to

$$\varprojlim_n \big(M \otimes_A (A/\mathfrak{m}^{n+1})\big) = \varprojlim_n \big(M/\mathfrak{m}^{n+1}M\big).$$

Let T be a covariant additive functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$. The canonical homomorphisms (III.7.2.3.1)

$$T(M) \otimes_A (A/\mathfrak{m}^{n+1}) \longrightarrow T\big(M \otimes_A (A/\mathfrak{m}^{n+1})\big)$$

plainly form a projective system of A-homomorphisms. On passing to the limit they therefore give an $\widehat{A}$-homomorphism, functorial in M,

$$\widehat{T(M)} \longrightarrow \varprojlim_n T(M_n), \tag{7.5.1.1}$$

where

$$M_n = M \otimes_A (A/\mathfrak{m}^{n+1}), \qquad A_n = A/\mathfrak{m}^{n+1}.$$

Proposition (7.5.2). — *Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. Let T be a covariant additive functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that is half-exact and commutes with inductive limits. Suppose, moreover, that for every finite-type A-module M, the module $T(M)$ is of finite type and the canonical homomorphism (7.5.1.1) is an isomorphism. Then the following conditions are equivalent:*

(a) T is right exact;

(b) for every n, the functor $N \mapsto T(N)$ is right exact in the category of finite-type A_n-modules; equivalently, T is right exact in the category of finite-length A-modules;

(c) the canonical homomorphism

$$T(\varepsilon) : T(A) \longrightarrow T(k)$$

is surjective;

(d) for every sufficiently large n, the canonical homomorphism

$$T(A_n) \longrightarrow T(k)$$

is surjective.

Proof. — It is clear that (a) implies (b). Let us prove the converse. Suppose that $u : M \to N$ is an epimorphism of A-modules; we must show that $T(u)$ is an epimorphism. Since T commutes with inductive limits and filtered inductive limits are exact in the category of modules, it is enough to consider the case where M and N are of finite type.

The modules $T(M)$ and $T(N)$ are then of finite type. Since A is Noetherian and local, it is enough to prove that

$$\widehat{T(u)} : \widehat{T(M)} \longrightarrow \widehat{T(N)}$$

is surjective (0.7.3.5) and (0.6.4.1). By hypothesis, $\widehat{T(M)}$ and $\widehat{T(N)}$ are, respectively,

$$\varprojlim_n T(M_n) \quad \text{and} \quad \varprojlim_n T(N_n).$$

Thus $\widehat{T(u)}$ is the inverse limit of the projective system of homomorphisms

$$T(u \otimes 1_{A_n}) : T(M_n) \longrightarrow T(N_n).$$

Condition (b) says precisely that these homomorphisms are surjective. Moreover, $T(M_n)$ is a finite-type A_n-module, and A_n is Artinian. It follows that $\widehat{T(u)}$ is surjective (0.13.1.2) and (0.13.2.2).

Clearly (a) implies (c), and (c) implies (d). Finally, (III.7.2.7) shows that (b) and (d) are equivalent, because T is half-exact in $\mathtt{Ab}_{A_n}$. This completes the proof.

Corollary (7.5.3). — *Under the general hypotheses of (III.7.5.2), if $T(k) = 0$, then $T(M) = 0$ for every A-module M.*

Proof. — Since k is the only simple A-module, (7.3.5.4) shows that $T(E) = 0$ for every finite-length A-module E. If M is of finite type, then

$$\widehat{T(M)} \simeq \varprojlim_n T(M_n).$$

Each M_n has finite length, so $\widehat{T(M)} = 0$. By hypothesis $T(M)$ is of finite type, and it embeds in $\widehat{T(M)}$ (0.7.3.5); hence $T(M) = 0$. Finally, for an arbitrary A-module M, the module $T(M)$ is the inductive limit of the $T(N_\alpha)$, where the N_α run through the finite-type submodules of M. The proof is complete.

Proposition (7.5.4). — *Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. Let $T_\bullet$ be a homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with inductive limits. Suppose, moreover, that for every i and every finite-type A-module M, the module $T_i(M)$ is of finite type and the canonical homomorphism*

$$\widehat{T_i(M)} \longrightarrow \varprojlim_n T_i(M_n)$$

is bijective. For a given integer p, the following conditions are equivalent:

(a) *T_p is exact;*

(b) *T_p is right exact and $T_p(A)$ is a free A-module;*

(c) *the canonical homomorphisms*

$$T_{p+1}(A) \longrightarrow T_{p+1}(k), \qquad T_p(A) \longrightarrow T_p(k)$$

are surjective;

(d) *for every n, the canonical homomorphisms*

$$T_{p+1}(A_n) \longrightarrow T_{p+1}(k), \qquad T_p(A_n) \longrightarrow T_p(k)$$

are surjective;

(e) *for every n, the functor $N \mapsto T_p(N)$ is exact in the category of finite-type A_n-modules.*

Proof. — By (III.7.3.3), condition (a) is equivalent to saying that T_{p+1} and T_p are right exact. Since $T_\bullet$ is a homological functor in $\mathtt{Ab}_{A_n}$, the same argument as in (III.7.3.1) shows that (e) is equivalent to saying that T_p and T_{p+1} are right exact on the category of finite-type A_n-modules. It follows from (III.7.5.2) that (a) and (e) are equivalent; the same proposition also gives the equivalence of (a), (c), and (d).

Finally, every finite-type flat module over the local ring A is free (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5). The equivalence of (a) and (b) therefore follows from (III.7.3.1) and (III.7.3.3).

Corollary (7.5.5). — *Suppose that the general hypotheses of (III.7.5.4) hold.*

(i) *If $T_p(k) = 0$, then $T_p = 0$, the functor T_{p+1} is right exact, and T_{p-1} is left exact.*

(ii) *If $T_{p-1}(k) = T_{p+1}(k) = 0$, then T_p is exact, the canonical homomorphism*

$$T_p(A) \otimes_A M \longrightarrow T_p(M)$$

is bijective, and $T_p(A)$ is a free A-module.

Proof. — Assertion (i) follows at once from (III.7.5.3), since T_p is half-exact; the last assertion follows from the definition of a homological functor. Taking (III.7.3.3) into account, (i) immediately gives the first two assertions of (ii), and the freeness of $T_p(A)$ follows from (III.7.5.4).

Corollary (7.5.6). — *Suppose that the general hypotheses of (III.7.5.4) hold, and suppose in addition that A is a discrete valuation ring.*

(i) *The functor T_p is right exact if and only if $T_{p-1}(A)$ is a free A-module.*

(ii) *The functor T_p is exact if and only if $T_p(A)$ and $T_{p-1}(A)$ are free A-modules.*

Proof. — It is clear that (ii) implies (i) (cf. (III.7.3.3)). To prove (i), let f be a generator of the maximal ideal of A, that is, a uniformizer of A. A finite-type A-module M is free, equivalently flat, if and only if the homothety

$$h_f : M \longrightarrow M, \qquad x \longmapsto fx,$$

is injective, since in the present situation this amounts to saying that M is torsion-free (0.6.3.4). The exact sequence

$$0 \longrightarrow A \xrightarrow{h_f} A \longrightarrow k \longrightarrow 0$$

gives the exact homology sequence

$$T_p(A) \longrightarrow T_p(k) \longrightarrow T_{p-1}(A) \xrightarrow{h_f} T_{p-1}(A).$$

Thus $T_{p-1}(A)$ is free if and only if $T_p(A) \to T_p(k)$ is surjective, and the conclusion follows from (III.7.5.2).

Remark (7.5.7). — *By (III.7.4.7), the general hypotheses of (III.7.4.4) imply that the homological functor $T_\bullet$ defined by (7.4.1.1) satisfies the general hypotheses of (III.7.5.4). In that case, (III.7.5.6) is therefore contained in (III.7.4.3).*

7.6 Descent of exactness properties. Semicontinuity theorem and Grauert's exactness criterion

Proposition (7.6.1). — *Under the hypotheses of (III.7.4.1), let B be a commutative A-algebra. If T_p is right exact (respectively left exact, exact), then the same is true of $T_p^{(B)}$; the converse is true when B is a faithfully flat A-module.*

Proof. — The first assertion is a special case of a trivial assertion of (III.7.1.3). Conversely, suppose first that B is a flat A-module. Then, for every A-module M,

$$H_\bullet\big(P_\bullet \otimes_A (M \otimes_A B)\big) = H_\bullet(P_\bullet \otimes_A M) \otimes_A B,$$

which may also be written, for every p,

$$T_p(M) \otimes_A B \simeq T_p^{(B)}(M_{(B)}) \tag{7.6.1.1}$$

up to canonical isomorphism. Suppose that $T_p^{(B)}$ is right exact (respectively left exact, exact). Since $M \mapsto M_{(B)}$ is an exact functor, the left-hand side of (7.6.1.1) is a right exact (respectively left exact, exact) functor of M. If B is now faithfully flat over A, it follows that T_p has the same exactness property (0.6.4.1).

Proposition (7.6.2). — *Under the hypotheses of (III.7.4.1), suppose in addition that A is a reduced Noetherian ring and that the P_i are finite-type A-modules. For T_p to be right exact (respectively left exact, exact), it is necessary and sufficient that $T_p^{(B)}$ have the same property for every A-algebra B that is a discrete valuation ring.*

Proof. — By (III.7.3.1) and (III.7.3.3), we may restrict attention to right exactness, and only sufficiency has to be proved. By (III.7.4.2), it is enough to show that $Z'_{p-1}(P_\bullet)$ is a flat A-module. Since P_{p-1} is of finite type, $Z'_{p-1}(P_\bullet)$ is also of finite type; the criterion of (0.10.2.8) therefore shows that it is enough for

$$Z'_{p-1}(P_\bullet) \otimes_A B$$

to be a flat B-module for every A-algebra B that is a discrete valuation ring. Since $P_\bullet$ is a complex of flat A-modules, we have

$$Z'_{p-1}(P_\bullet) \otimes_A B = Z'_{p-1}(P_\bullet \otimes_A B).$$

The complex $P_\bullet \otimes_A B$ consists of flat B-modules (0.6.2.1), and for every B-module N we have

$$H_\bullet(P_\bullet \otimes_A N) = H_\bullet\big((P_\bullet \otimes_A B) \otimes_B N\big),$$

and hence

$$T_p^{(B)}(N) = H_p\big((P_\bullet \otimes_A B) \otimes_B N\big).$$

Applying (III.7.4.2) to $T_p^{(B)}$, we see that the hypothesis that $T_p^{(B)}$ is right exact is equivalent to $Z'_{p-1}(P_\bullet \otimes_A B)$ being a flat B-module.

The preceding criterion leads us to examine the case of discrete valuation rings more closely.

Proposition (7.6.3). — *Under the hypotheses of (III.7.4.1), suppose that A is a regular Noetherian ring of dimension 1 (in other words, that A is Noetherian and, for every $x \in \mathrm{Spec}(A)$, the ring A_x is a field or a discrete valuation ring). Then, for every integer p and every A-module M, there is a canonical exact sequence, functorial in M,*

$$0 \longrightarrow T_p(A) \otimes_A M \xrightarrow{t_M} T_p(M) \longrightarrow \mathrm{Tor}_1^A\big(T_{p-1}(A), M\big) \longrightarrow 0. \tag{7.6.3.1}$$

Proof. — In what follows, to simplify the notation we omit the complex $P_\bullet$ from the usual homological notations $H_p(P_\bullet)$, $B_p(P_\bullet)$, $Z_p(P_\bullet)$, and $Z'_p(P_\bullet)$. We have the three exact sequences

$$\begin{aligned} 0 &\longrightarrow H_p \longrightarrow Z'_p \longrightarrow B_{p-1} \longrightarrow 0, \\ 0 &\longrightarrow B_{p-1} \longrightarrow Z_{p-1} \longrightarrow H_{p-1} \longrightarrow 0, \\ 0 &\longrightarrow Z_{p-1} \longrightarrow P_{p-1} \longrightarrow B_{p-2} \longrightarrow 0. \end{aligned}$$

Since P_{p-1} and P_{p-2} are flat, so are their respective submodules B_{p-1}, Z_{p-1}, and B_{p-2}: for every $x \in \mathrm{Spec}(A)$, flat A_x-modules and torsion-free A_x-modules are the same. Tensoring with M therefore gives the exact sequences

$$0 = \mathrm{Tor}_1^A(B_{p-1}, M) \longrightarrow H_p \otimes_A M \longrightarrow Z'_p \otimes_A M \xrightarrow{u} B_{p-1} \otimes_A M \longrightarrow 0, \tag{7.6.3.2}$$

$$0 = \mathrm{Tor}_1^A(Z_{p-1}, M) \longrightarrow \mathrm{Tor}_1^A(H_{p-1}, M) \longrightarrow B_{p-1} \otimes_A M \xrightarrow{v} Z_{p-1} \otimes_A M, \tag{7.6.3.3}$$

and

$$0 = \mathrm{Tor}_1^A(B_{p-2}, M) \longrightarrow Z_{p-1} \otimes_A M \xrightarrow{w} P_{p-1} \otimes_A M. \tag{7.6.3.4}$$

By definition,

$$T_p(M) = \mathrm{Ker}(d_p \otimes 1)/\Im(d_{p+1} \otimes 1).$$

It is therefore the kernel of the homomorphism

$$(P_p \otimes_A M)/\Im(d_{p+1} \otimes 1) \longrightarrow P_{p-1} \otimes_A M$$

induced by $d_p \otimes 1$. By the definition $Z'_p = P_p/B_p$, this homomorphism may also be written $Z'_p \otimes_A M \to P_{p-1} \otimes_A M$; it is the composite

$$Z'_p \otimes_A M \xrightarrow{u} B_{p-1} \otimes_A M \xrightarrow{v} Z_{p-1} \otimes_A M \xrightarrow{w} P_{p-1} \otimes_A M.$$

Since w is injective by (7.6.3.4), there is an exact sequence

$$0 \longrightarrow \mathrm{Ker}\, u \longrightarrow T_p(M) \longrightarrow \mathrm{Ker}\, v \longrightarrow 0.$$

In view of (7.6.3.2) and (7.6.3.3), and since $H_p = T_p(A)$ by definition, this is precisely (7.6.3.1).

Remarks (7.6.4). — (i) *The group $H_\bullet(P_\bullet \otimes_A M)$ is the homology of the bicomplex $P_\bullet \otimes_A M$, where M is regarded as a complex concentrated in degree 0. It is consequently (III.6.3.6)(III.6.3.2) the abutment of the regular spectral sequence whose E_2-term is*

$$E^2_{pq} = \mathrm{Tor}_p^A\big(H_q(P_\bullet), M\big) = \mathrm{Tor}_p^A\big(T_q(A), M\big).$$

III | 194

The hypothesis on A implies that $\mathrm{Tor}_p^A(E, F) = 0$ for $p \geq 2$ and arbitrary A-modules E, F (0.17.2.2). It is known (M, XV) that this implies the exactness of

$$0 \longrightarrow E^2_{0,q} \longrightarrow H_q(P_\bullet \otimes_A M) \longrightarrow E^2_{1,q-1} \longrightarrow 0,$$

which is precisely (7.6.3.1).

(ii) *Taking (III.7.3.1) into account, the exact sequence* (7.6.3.1) *recovers (III.7.4.3) as a special case.*

Corollary (7.6.5). — *Under the hypotheses of (III.7.4.1), suppose that A is a discrete valuation ring with field of fractions K and residue field k, and that the $T_i(A)$ are finite-type A-modules. Then*

$$\mathrm{rank}_k\, T_p(k) \geq \mathrm{rank}_k\big(T_p(A) \otimes_A k\big) \geq \mathrm{rank}_A\, T_p(A) = \mathrm{rank}_K\, T_p(K). \tag{7.6.5.1}$$

Moreover, equality of the two extreme terms of this inequality holds if and only if T_p is exact, or, equivalently, if and only if $T_p(A)$ and $T_{p-1}(A)$ are free A-modules.

Proof. — Set $M = k$ in the exact sequence (7.6.3.1). Since the terms are k-vector spaces, this gives

$$\mathrm{rank}_k\, T_p(k) = \mathrm{rank}_k\big(T_p(A) \otimes_A k\big) + \mathrm{rank}_k\, \mathrm{Tor}_1^A\big(T_{p-1}(A), k\big).$$

On the other hand, since $T_p(A)$ is a finite-type module over the local integral domain A, we have (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 1 of prop. 4)

$$\mathrm{rank}_k\big(T_p(A) \otimes_A k\big) \geq \mathrm{rank}_A\, T_p(A) = \mathrm{rank}_K\big(T_p(A) \otimes_A K\big), \tag{7.6.5.2}$$

and the two sides of (7.6.5.2) are equal if and only if $T_p(A)$ is a free A-module (*loc. cit.*, prop. 7). Since K is a flat A-module, we also have, by definition,

$$T_p(A) \otimes_A K = H_p(P_\bullet) \otimes_A K = H_p(P_\bullet \otimes_A K) = T_p(K).$$

This proves (7.6.5.1). It also shows that equality can hold only if, first, $T_p(A)$ is free, and, second,

$$\mathrm{Tor}_1^A\big(T_{p-1}(A), k\big) = 0,$$

the latter condition being equivalent to $T_{p-1}(A)$ being free (0.10.1.3). Finally, for finite-type modules over A, flatness and freeness are equivalent (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5), and the result follows from (III.7.4.3).

(7.6.6) The hypotheses remaining those of (III.7.4.1), for every $x \in \mathrm{Spec}(A)$ set

$$d_p(x) = d_p^T(x) = \mathrm{rank}_{\kappa(x)}\, T_p(\kappa(x)). \tag{7.6.6.1}$$

Lemma (7.6.7). — *Let $\varphi : A \to A'$ be a ring homomorphism, and let*

$$f = {}^a\varphi : \mathrm{Spec}(A') \longrightarrow \mathrm{Spec}(A)$$

be the corresponding map (I.1.2.1). If $T'_\bullet = T_\bullet^{(A')}$ (III.7.1.3), then

$$d_p^{T'} = d_p^T \circ f. \tag{7.6.7.1}$$

III | 195

Proof. — For $x' \in \mathrm{Spec}(A')$, put $x = f(x')$. Then

$$\begin{aligned} H_\bullet(P_\bullet \otimes_A \kappa(x')) &= H_\bullet\big((P_\bullet \otimes_A \kappa(x)) \otimes_{\kappa(x)} \kappa(x')\big) \\ &= H_\bullet(P_\bullet \otimes_A \kappa(x)) \otimes_{\kappa(x)} \kappa(x'), \end{aligned}$$

because $\kappa(x')$ is flat over $\kappa(x)$. This proves (7.6.7.1).

Lemma (7.6.8). — *If A is Noetherian and $P_\bullet$ consists of finite-type A-modules, the function $x \mapsto d_p^T(x)$ on* $\mathrm{Spec}(A)$ *is constructible.*

Proof. — For every irreducible closed subset Y of $X = \mathrm{Spec}(A)$, we must show that there is a nonempty open subset U of Y on which d_p is constant (0.9.2.2). Write $Y = \mathrm{Spec}(A/\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of A such that $A/\mathfrak{a}$ is reduced. By (III.7.6.7), we may reduce to the case $Y = X$ and A a Noetherian integral domain; the assertion then follows from (III.7.4.5).

Theorem (7.6.9). — *Let A be a Noetherian ring, let $P_\bullet$ be a complex of finite-type flat A-modules, and let*

$$T_\bullet(M) = H_\bullet(P_\bullet \otimes_A M)$$

be the homological functor defined by $P_\bullet$. For $x \in \mathrm{Spec}(A)$, set

$$d_p(x) = \mathrm{rank}_{\kappa(x)}\, T_p(\kappa(x)).$$

Then:

(i) The function d_p is constructible and upper semicontinuous on $\mathrm{Spec}(A)$.

(ii) If T_p is exact, d_p is continuous (and hence locally constant) on $\mathrm{Spec}(A)$. *The converse is true when A is reduced.*

Proof. — (i) Constructibility was proved in (III.7.6.8). To prove upper semicontinuity, it is therefore enough (0.9.2.4) to show that if $x' \neq x$ is a generalization of x in $\mathrm{Spec}(A)$, then $d_p(x') \leq d_p(x)$. There is a discrete valuation ring B and a morphism

$$f : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

such that, if a is the closed point of $\mathrm{Spec}(B)$ and b its generic point, then $f(a) = x$ and $f(b) = x'$ (II.7.1.9). By (7.6.7.1), we are reduced to proving $d_p(a) \geq d_p(b)$ on $\mathrm{Spec}(B)$, which is exactly (7.6.5.1).[1]

(ii) The first assertion was already proved in (III.7.3.4). For the converse, use the valuative criterion (III.7.6.2). In view of (7.6.7.1), we are reduced to the case in which A is a discrete valuation ring. Since $\mathrm{Spec}(A)$ then has only two points, the hypothesis that d_p is constant implies that T_p is exact, by (III.7.6.5).

Corollary (7.6.10). — *Let A be a Noetherian ring, let $\mathfrak{p}_i$ $(1 \leq i \leq r)$ be its minimal prime ideals, and let k_i be the residue field of $A_{\mathfrak{p}_i}$.*

(i) For every $x \in \mathrm{Spec}(A)$, there is an index i such that

$$d_p(x) \geq \mathrm{rank}_{k_i}\, T_p(k_i). \tag{7.6.10.1}$$

In particular, if A is an integral domain and K is its field of fractions, then

$$d_p(x) \geq \mathrm{rank}_K\, T_p(K) \tag{7.6.10.2}$$

for every $x \in \mathrm{Spec}(A)$.

III | 196

[1] The principle of proving (i) by reduction to the case of a discrete valuation ring was communicated to us orally by Hironaka.

(ii) Suppose in addition that A is local and reduced, and let k be its residue field. Then T_p is exact if and only if

$$\operatorname{rank}_k T_p(k) = \operatorname{rank}_{k_i} T_p(k_i) \qquad (1 \leq i \leq r). \tag{7.6.10.3}$$

Proof. — Assertion (i) is immediate: every neighbourhood of x contains one of the $\mathfrak{p}_i$, and it is enough to apply the definition of semicontinuity. If A is local, the only neighbourhood of the maximal ideal $\mathfrak{m}$ in $\operatorname{Spec}(A)$ is all of $\operatorname{Spec}(A)$; hence

$$d_p(x) \leq \operatorname{rank}_k T_p(k)$$

for every $x \in \operatorname{Spec}(A)$. Together with (i), this shows that (7.6.10.3) implies that d_p is constant on $\operatorname{Spec}(A)$, and hence that T_p is exact by (III.7.6.9)[(ii)]. The converse follows at once from the same result.

Remarks (7.6.11). — *One may ask whether (III.7.6.9)[(i)] can be strengthened to the inequality*

$$\operatorname{rank}_{\kappa(x)} T_p(\kappa(x)) \geq \operatorname{rank}_{\kappa(x)} \big(T_p(A) \otimes_A \kappa(x)\big) \tag{7.6.11.1}$$

for every $x \in \operatorname{Spec}(A)$. This inequality does hold when A is a discrete valuation ring and x is its maximal ideal (III.7.6.5). Restrict now to the case in which A is a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field k. The following conditions are equivalent:

a) For every complex $P_\bullet$ of finite-type flat A-modules,

$$\operatorname{rank}_k T_p(k) \geq \operatorname{rank}_k \big(T_p(A) \otimes_A k\big) \qquad \text{for every integer } p. \tag{7.6.11.2}$$

b) For every finite-type A-module M,

$$\operatorname{rank}_k (M \otimes_A k) \geq \operatorname{rank}_k (\check{M} \otimes_A k). \tag{7.6.11.3}$$

c) For every finite-type A-module N,

$$\operatorname{rank}_k \operatorname{Tor}_1^A(N, k) \geq \operatorname{rank}_k \operatorname{Tor}_2^A(N, k). \tag{7.6.11.4}$$

Equivalently, by dimension shifting (M, V, 7.2), one may require, for every $i \geq 1$,

$$\operatorname{rank}_k \operatorname{Tor}_i^A(N, k) \geq \operatorname{rank}_k \operatorname{Tor}_{i+1}^A(N, k). \tag{7.6.11.5}$$

Here are brief indications of the proof. To see that a) *implies* b)*, consider an exact sequence*

$$L_1 \xrightarrow{d} L_0 \longrightarrow M \longrightarrow 0,$$

where L_0 and L_1 are finite free, and apply a) *to the complex*

$$P_1 \xrightarrow{{}^t d} P_0, \qquad P_0 = \check{L}_1, \quad P_1 = \check{L}_0,$$

all its other terms being zero. Then

$$T_1(A) = \check{M}$$

and

$$T_1(k) = \operatorname{Hom}_A(M, k) = \operatorname{Hom}_k(M/\mathfrak{m}M, k).$$

Thus $T_1(k)$ is the dual of the vector space $M \otimes_A k$ and has the same rank. To prove that b) *implies* c)*, we first establish the following lemma.*

Lemma (7.6.11.6). — *Given a complex of flat A-modules*

$$\cdots \longrightarrow 0 \longrightarrow P_1 \xrightarrow{d} P_0 \longrightarrow 0 \longrightarrow \cdots,$$

there is an exact sequence

$$0 \longrightarrow \operatorname{Tor}_2^A(Z_0', k) \longrightarrow T_1(A) \otimes_A k \longrightarrow T_1(k) \longrightarrow \operatorname{Tor}_1^A(Z_0', k) \longrightarrow 0. \tag{7.6.11.7}$$

Proof. — *From the exact sequence*

$$0 \longrightarrow Z_1 \longrightarrow P_1 \longrightarrow B_0 \longrightarrow 0$$

we obtain

$$0 \longrightarrow \mathrm{Tor}_1^A(B_0, k) \longrightarrow Z_1 \otimes_A k \longrightarrow P_1 \otimes_A k \xrightarrow{u} B_0 \otimes_A k \longrightarrow 0.$$

The exact sequence

$$0 \longrightarrow B_0 \longrightarrow Z_0 \longrightarrow Z_0' \longrightarrow 0$$

and the flatness of $Z_0 = P_0$ *give*

$$\mathrm{Tor}_1^A(B_0, k) = \mathrm{Tor}_2^A(Z_0', k).$$

By definition, $Z_1 = T_1(A)$. *Moreover,* $T_1(k) = \mathrm{Ker}(d \otimes 1)$, *and* $d \otimes 1$ *factors as*

$$P_1 \otimes_A k \xrightarrow{u} B_0 \otimes_A k \xrightarrow{v} Z_0 \otimes_A k = P_0 \otimes_A k.$$

Thus $T_1(k) = u^{-1}(R)$, *where* $R = \mathrm{Ker}\, v$. *Since* u *is surjective,* $R = u(T_1(k))$; *finally,* $R = \mathrm{Tor}_1^A(Z_0', k)$ *by the definition of* v. *This proves* (7.6.11.7).

To deduce c) *from* b), *take an exact sequence*

$$L_1 \xrightarrow{d} L_0 \longrightarrow N \longrightarrow 0$$

with L_0, L_1 *finite free, and consider the functor* T *associated with the complex formed by* L_1 *and* L_0. *Since* L_0 *and* L_1 *are free, they identify with their biduals. If*

$$M = \mathrm{Coker}({}^t d),$$

then

$$T_1(A) = \mathrm{Ker}\, d = \check{M}.$$

Furthermore,

$$M \otimes_A k = \mathrm{Coker}({}^t d \otimes 1_k)$$

has the same rank over k *as* $\mathrm{Ker}(d \otimes 1_k)$. *Hypothesis* b) *therefore gives*

$$\mathrm{rank}_k\big(T_1(A) \otimes_A k\big) \leq \mathrm{rank}_k\, T_1(k).$$

Since $Z_0' = N$, *inequality* (7.6.11.4) *follows from the exact sequence* (7.6.11.7).

Finally, to prove that c) *implies* a), *apply (III.7.6.11.6) to the two-term complex*

$$P_p \longrightarrow P_{p-1}.$$

Hypothesis c), *applied to* Z_{p-1}', *gives*

$$\mathrm{rank}_k\, R \geq \mathrm{rank}_k\, S,$$

where

$$R = \mathrm{Ker}(P_p \otimes_A k \longrightarrow P_{p-1} \otimes_A k), \qquad S = Z_p \otimes_A k.$$

Factor $d_{p+1} : P_{p+1} \to P_p$ *as*

$$P_{p+1} \xrightarrow{v} Z_p \xrightarrow{j} P_p.$$

Then

$$\Im(d_{p+1} \otimes 1) = (j \otimes 1)\big(\Im(v \otimes 1)\big).$$

Since

$$T_p(A) \otimes_A k = (Z_p/B_p) \otimes_A k = (Z_p \otimes_A k)/\Im(v \otimes 1)$$

and

$$T_p(k) = R/\Im(d_{p+1} \otimes 1),$$

inequality (7.6.11.2) *follows.*

Now suppose that the local ring A *is regular of dimension* n. *It is known [17] that the* A*-module* $\mathrm{Tor}_i^A(k, k)$ *is isomorphic to the exterior power*

$$\bigwedge^i (\mathfrak{m}/\mathfrak{m}^2).$$

Thus (7.6.11.4) *fails for* $N = k$ *as soon as* $n \geq 4$. *On the other hand, suppose that* A *is a local integral domain such that every finite-type reflexive* A*-module is free, as happens when* A *is a regular ring of dimension* 2. *Then* (7.6.11.3) *holds. Indeed, the dual* $\check{M}$ *of a finite-type* A*-module* M *is reflexive, hence free, and consequently*

$$\mathrm{rank}_k(\check{M} \otimes_A k) = \mathrm{rank}_K(\check{M}) = \mathrm{rank}_K(M),$$

where K *is the field of fractions of* A. *Every basis of* $M \otimes_A k$ *over* k *consists of the images of a system of generators of* M *(Bourbaki,* Algèbre commutative, *chap. II, § 3, no. 2, cor. 2 of prop. 4). Hence*

$$\mathrm{rank}_K(M) \leq \mathrm{rank}_k(M \otimes_A k),$$

which proves the assertion.

7.7 Application to proper morphisms: I. The exchange property

The next three paragraphs are, for the most part, translations into the language of morphisms of preschemes of the results of the preceding paragraphs.

(7.7.1) Let $f : X \to Y$ be a quasi-compact separated morphism of preschemes, and let $\mathscr{P}_\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-Modules whose differential has degree -1. Suppose in addition that the $\mathscr{O}_X$-Modules $\mathscr{P}_i$ are Y-flat (0.6.7.1). **III** | 198

We shall consider the ∂-functor

$$\mathscr{M} \longmapsto \mathscr{T}_\bullet(\mathscr{M}),$$

also denoted $\mathscr{T}_\bullet(\mathscr{P}_\bullet, \mathscr{M})$, from the category of quasi-coherent $\mathscr{O}_Y$-Modules to itself (by (III.6.2.3)), defined by

$$\mathscr{T}_n(\mathscr{P}_\bullet, \mathscr{M}) = \mathscr{T}_n(\mathscr{M}) = \mathscr{H}^{-n}\big(f, \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}\big) \qquad (n \in \mathbf{Z}), \tag{7.7.1.1}$$

where $\mathscr{P}^\bullet$ is the complex whose term of degree j is $\mathscr{P}_{-j}$, so that its differential has degree $+1$. The functor $\mathscr{T}_\bullet$ thus defined is a *homological functor* in $\mathscr{M}$ (III.6.2.6).

(7.7.2) Let $g : Y' \to Y$ be a morphism, and set

$$X' = X_{(Y')} = X \times_Y Y', \qquad f' = f_{(Y')} : X' \longrightarrow Y'.$$

Thus f' is quasi-compact and separated. Put also

$$\mathscr{P}'_\bullet = \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}.$$

This is a complex of quasi-coherent $\mathscr{O}_{X'}$-Modules that are Y'-flat, by (I.9.1.12) and (0.6.2.1). With the same conventions on degrees, set

$$\mathscr{T}_\bullet^{Y'}(\mathscr{M}') = \mathscr{H}^\bullet\big(f', \mathscr{P}'^\bullet \otimes_{\mathscr{O}_{Y'}} \mathscr{M}'\big) = \mathscr{H}^\bullet\big(f', \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}'\big). \tag{7.7.2.1}$$

This is a homological functor of the quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{M}'$. When Y' is an affine scheme with ring A', we write $\mathscr{T}_\bullet^{A'}$ in place of $\mathscr{T}_\bullet^{Y'}$. For every A'-module M',

$$\mathscr{T}_\bullet^{A'}(\widetilde{M'}) = \big(\Gamma(Y', \mathscr{T}_\bullet^{Y'}(\widetilde{M'}))\big)^\sim.$$

We set

$$T_\bullet^{A'}(M') = \Gamma(Y', \mathscr{T}_\bullet^{Y'}(\widetilde{M'}));$$

this is a homological functor from A'-modules to A'-modules.

If $Y = \mathrm{Spec}(A)$ is also affine, the functor $T_\bullet^{A'}$ on A'-modules is obtained from the homological functor $T_\bullet^A$ on A-modules by extension of scalars from A to A' (III.7.1.3). Indeed, let $g : Y' \to Y$ correspond to the ring homomorphism $A \to A'$, and let $g' : X' \to X$ be the corresponding morphism, which is affine (II.1.6.2). If $\mathfrak{U}$ is an affine open covering of X, then $\mathfrak{U}' = g'^{-1}(\mathfrak{U})$ is an affine open covering of X'. By (III.6.2.2), it is enough to check

$$C^\bullet\big(\mathfrak{U}, \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} g_*(\mathscr{M}')\big) = C^\bullet\big(\mathfrak{U}', \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}'\big).$$

Ultimately, if U is any affine open subset of X and $U' = g'^{-1}(U)$, this reduces to

$$\Gamma\big(U, \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} g_*(\mathscr{M}')\big) = \Gamma\big(U', \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}'\big),$$

which is immediate from (I.1.3) and (III.3.2).

In particular, if U is an open subset of Y, then for every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,

$$\mathscr{T}_\bullet^U(\mathscr{M}|U) = \mathscr{T}_\bullet(\mathscr{M})|U. \tag{7.7.2.2}$$

(7.7.3) For every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$ there is a canonical homomorphism, functorial in $\mathscr{M}$,

$$\mathscr{T}_p(\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{M} \longrightarrow \mathscr{T}_p(\mathscr{M}). \tag{7.7.3.1}$$

Indeed, when Y is affine this homomorphism was defined in (III.7.2.2). The definition extends immediately to the general case: if U and V are affine open subsets of Y with $V \subset U$, the diagram **III** | 199

$$\begin{array}{ccc} \big(\mathscr{T}_p(\mathscr{O}_Y)\otimes_{\mathscr{O}_Y}\mathscr{M}\big)|U & \longrightarrow & \mathscr{T}_p^U(\mathscr{M}|U) \\ = \mathscr{T}_p^U(\mathscr{O}_Y|U)\otimes_{\mathscr{O}_Y|U}(\mathscr{M}|U) & & = \mathscr{T}_p(\mathscr{M})|U \\ \downarrow & & \downarrow \\ \big(\mathscr{T}_p(\mathscr{O}_Y)\otimes_{\mathscr{O}_Y}\mathscr{M}\big)|V & \longrightarrow & \mathscr{T}_p^V(\mathscr{M}|V) \\ = \mathscr{T}_p^V(\mathscr{O}_Y|V)\otimes_{\mathscr{O}_Y|V}(\mathscr{M}|V) & & = \mathscr{T}_p(\mathscr{M})|V \end{array}$$

commutes by (III.7.2.3.3).

For every morphism $g : Y' \to Y$ there is a canonical homomorphism

$$\mathscr{T}_p(\mathscr{O}_Y)\otimes_{\mathscr{O}_Y}\mathscr{O}_{Y'} \longrightarrow \mathscr{T}_p^{Y'}(\mathscr{O}_{Y'}), \tag{7.7.3.2}$$

which is precisely the special case of (III.6.7.11.2), for the abutments, in which $S = Y$, $v_1 = f$, $v_2 = 1_Y$, and $\mathscr{P}_\bullet^{(2)}$ is concentrated in degree 0 with sole term $\mathscr{M}$.

When $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(A')$ are affine, (7.7.3.2) is the homomorphism of sheaves corresponding to the canonical homomorphism of A'-modules defined in (III.7.2.2),

$$T_p^A(A)\otimes_A A' \longrightarrow T_p^{A'}(A') = T_p^A(A').$$

This follows immediately from (III.6.7.11), since in the situation at hand one may take $\mathfrak{U}'^{(i)} = u_i^{-1}(\mathfrak{U}^{(i)})$ there.

(7.7.4) Suppose that f is proper, that $Y = \mathrm{Spec}(A)$ is an affine Noetherian scheme, and that $\mathscr{P}_\bullet$ is a bounded-below complex of coherent, Y-flat $\mathscr{O}_X$-Modules. We saw in (III.6.10.5) that, up to isomorphism, one may write

$$\mathscr{T}_p(\mathscr{M}) = \mathscr{H}_p\big(\mathscr{L}_\bullet\otimes_{\mathscr{O}_Y}\mathscr{M}\big), \qquad \mathscr{L}_\bullet = \widetilde{L_\bullet},$$

where $L_\bullet$ is a bounded-below complex of finite-type free A-modules. Thus $\mathscr{T}_\bullet$ is of the kind studied in detail in (III.7.4) and (III.7.6). We shall now translate the results of that study.

Theorem (7.7.5). — Let Y be a locally Noetherian prescheme, (U_α) a covering of Y by affine open subsets, $f : X \to Y$ a proper morphism, and $\mathscr{P}_\bullet$ a bounded-below complex of coherent, Y-flat $\mathscr{O}_X$-Modules. The homological functor $\mathscr{T}_\bullet(\mathscr{M})$ defined by (7.7.1.1) then has the following properties.

I) (The semicontinuity property).[1] The function

$$y \longmapsto d_p(y) = \mathrm{rank}_{\kappa(y)}\, T_p^{\kappa(y)}(\kappa(y)) \tag{7.7.5.1}$$

is upper semicontinuous. III | 200

II) (The exchange property). For a fixed integer p, the following conditions are equivalent:

a) $\mathscr{T}_p$ *is right exact.*

a′) $\mathscr{T}_p(\mathscr{M})$ *is isomorphic to a functor of the form* $\mathscr{N}\otimes_{\mathscr{O}_Y}\mathscr{M}$, *where necessarily*

$$\mathscr{N} \simeq \mathscr{T}_p(\mathscr{O}_Y) = \mathscr{H}^{-p}(f, \mathscr{P}^\bullet).$$

a″) *The canonical functorial homomorphism* (7.7.3.1) *is an isomorphism.*

b) $\mathscr{T}_{p-1}$ *is left exact.*

b′) *There is an $\mathscr{O}_Y$-Module $\mathscr{Q}$, necessarily coherent and determined up to unique isomorphism, together with an isomorphism of functors*

$$\mathscr{T}_{p-1}(\mathscr{M}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}). \tag{7.7.5.2}$$

c) If A_α denotes the ring of the affine open subset U_α, then for every α the functor $T_p^{A_\alpha}$ on A_α-modules is right exact.

[1] A special case of this theorem already occurs in note [3] of Chow–Igusa. The semicontinuity property was discovered, for analytic spaces and under rather special hypotheses, by Kodaira–Spencer, *On the variations of almost-complex structures*, in *Algebraic Geometry and Topology: A Symposium in Honor of S. Lefschetz*, Princeton Series no. 12, pp. 139–150, Princeton, 1957; the general version was proved by Grauert [5].

d) For every morphism $g : Y' \to Y$, the canonical homomorphism

$$\mathscr{T}_p(\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{T}_p^{Y'}(\mathscr{O}_{Y'}) \tag{7.7.5.3}$$

is an isomorphism.

Proof. — The semicontinuity property is local on Y, and therefore follows from (III.7.7.4) and (III.7.6.9). It is clear that a″) implies a′), and that a′) implies a). When Y is affine, the equivalence of a), a″), b), and b′) was proved in (III.7.3.1) and (III.7.4.6), in view of (III.7.7.4).

To pass to the general case, first observe that a) is equivalent to c); this also proves that a) is local on Y. The same argument proves the locality of a″) and b). Condition c) plainly implies a), so it remains to prove the converse. It is enough to show that, for every affine open subset U of Y and every exact sequence

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

of quasi-coherent $(\mathscr{O}_Y|U)$-Modules, there is an exact sequence

$$0 \longrightarrow \mathscr{G}' \longrightarrow \mathscr{G} \longrightarrow \mathscr{G}'' \longrightarrow 0$$

of quasi-coherent $\mathscr{O}_Y$-Modules such that

$$\mathscr{F}' = \mathscr{G}'|U, \qquad \mathscr{F} = \mathscr{G}|U, \qquad \mathscr{F}'' = \mathscr{G}''|U.$$

This follows immediately from the hypothesis that Y is locally Noetherian and from (I.9.4.2): extend $\mathscr{F}$ to a quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$, extend $\mathscr{F}'$ to a quasi-coherent submodule $\mathscr{G}'$ of $\mathscr{G}$, and take $\mathscr{G}'' = \mathscr{G}/\mathscr{G}'$.

To prove the equivalence of b) and b′) in the general case, note first that, when Y is affine, $\mathscr{Q}$ is determined up to unique isomorphism. If U is an affine open subset of the affine scheme Y, it follows that there is a functorial isomorphism

$$\mathscr{T}_{p-1}^U(\mathscr{M}|U) \simeq \mathscr{H}om_{\mathscr{O}_Y|U}(\mathscr{Q}|U, \mathscr{M}|U).$$

For arbitrary Y and each affine open subset U, there is a coherent $(\mathscr{O}_Y|U)$-Module $\mathscr{Q}_U$ and a functorial isomorphism

$$\mathscr{T}_{p-1}^U(\mathscr{M}|U) \simeq \mathscr{H}om_{\mathscr{O}_Y|U}(\mathscr{Q}_U, \mathscr{M}|U).$$

The preceding observation shows that, whenever $V \subset U$ is affine, $\mathscr{Q}_U|V$ is canonically identified with $\mathscr{Q}_V$. Hence there exists a unique $\mathscr{O}_Y$-Module $\mathscr{Q}$ satisfying (7.7.5.2).

It remains to prove the equivalence of a) and d). Condition d) is local on Y, and we have just seen that the same is true of a); moreover, d) is also local on Y'. When $Y = \operatorname{Spec}(A)$ and $Y' = \operatorname{Spec}(A')$, the functor $T_\bullet^{A'}$ is obtained from $T_\bullet^A$ by extension of scalars to A'. It is then clear that a′) implies that (7.7.5.3) is an isomorphism. **III** | 201

Conversely, continue to suppose that $Y = \operatorname{Spec}(A)$ is affine and take for A' the A-algebra $A \oplus M$, where M is an arbitrary A-module and multiplication is given by

$$(a_1, m_1)(a_2, m_2) = (a_1a_2, a_1m_2 + a_2m_1).$$

Then

$$T_p^{A'}(A') = T_p(A \oplus M) = T_p(A) \oplus T_p(M).$$

If (7.7.5.3) is bijective, the same is therefore true of the canonical homomorphism

$$T_p(A) \otimes_A M \longrightarrow T_p(M).$$

Thus d) implies a″), and the proof is complete.

Theorem (7.7.6). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be proper, and let $\mathscr{F}$ be a coherent, Y-flat $\mathscr{O}_X$-Module. There is a coherent $\mathscr{O}_Y$-Module $\mathscr{Q}$, determined up to unique isomorphism, and an isomorphism of functors in the quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,*

$$f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}\big) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}). \tag{7.7.6.1}$$

Consequently there is an isomorphism of functors

$$\Gamma\big(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}\big) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}). \tag{7.7.6.2}$$

Proof. — The functor $\mathscr{M} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}$ is exact (0.6.7.4), while f_* is left exact. Hence

$$\mathscr{M} \longmapsto f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}\big)$$

is left exact. Apply the equivalence of b) and b′) in (III.7.7.5) with $p = 1$.

Corollary (7.7.7). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be proper, let $\mathscr{F}$ and $\mathscr{F}'$ be coherent, Y-flat $\mathscr{O}_X$-Modules, and let $u : \mathscr{F} \to \mathscr{F}'$ be a homomorphism. Consider the two functors of the quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,*

$$\begin{aligned}\mathscr{T}(\mathscr{M}) &= \operatorname{Ker}\left(f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \longrightarrow f_*(\mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M})\right),\\ T(\mathscr{M}) &= \Gamma(Y, \mathscr{T}(\mathscr{M}))\\ &= \operatorname{Ker}\left(\Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \longrightarrow \Gamma(X, \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M})\right).\end{aligned}$$

There is a coherent $\mathscr{O}_Y$-Module $\mathscr{R}$, determined up to unique isomorphism, and isomorphisms of functors

$$\mathscr{T}(\mathscr{M}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{R}, \mathscr{M}), \tag{7.7.7.1}$$

$$T(\mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{R}, \mathscr{M}). \tag{7.7.7.2}$$

Proof. — It is enough to prove (7.7.7.2). This proves (7.7.7.1) when Y is affine, and the passage to the general case proceeds as in the proof of the equivalence of b) and b′) in (III.7.7.5), using the uniqueness, up to unique isomorphism, of a representing object of a representable functor (0.8.1.8).

By (III.7.7.6), there are coherent $\mathscr{O}_Y$-Modules $\mathscr{Q}$ and $\mathscr{Q}'$ and functorial isomorphisms

$$\Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}),$$

$$\Gamma(X, \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}', \mathscr{M}).$$

The homomorphism $u : \mathscr{F} \to \mathscr{F}'$ canonically defines a morphism of functors between the two left-hand sides.

It corresponds to a unique homomorphism $v : \mathscr{Q}' \to \mathscr{Q}$ of $\mathscr{O}_Y$-Modules for which

$$\begin{array}{ccc}\Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) & \longrightarrow & \Gamma(X, \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M})\\ \sim\downarrow & & \downarrow\sim\\ \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}) & \longrightarrow & \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}', \mathscr{M})\end{array}$$

commutes (0.8.1.4). The contravariant functor

$$\mathscr{N} \longmapsto \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{N}, \mathscr{M})$$

is left exact in the category of $\mathscr{O}_Y$-Modules. Taking $\mathscr{R} = \operatorname{Coker}(v)$ therefore gives (7.7.7.2).

Corollary (7.7.8). — *Under the hypotheses of (III.7.7.6) on X, Y, and f, let $\mathscr{F}$ and $\mathscr{G}$ be coherent $\mathscr{O}_X$-Modules satisfying:*

(i) $\mathscr{F}$ is Y-flat;

(ii) $\mathscr{G}$ is isomorphic to the cokernel of a homomorphism $\mathscr{E}_1 \to \mathscr{E}_0$ between finite-type locally free $\mathscr{O}_X$-Modules.

Consider the two functors of the quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,

$$\begin{aligned}\mathscr{T}(\mathscr{M}) &= f_*\mathscr{H}om_{\mathscr{O}_X}\big(\mathscr{G}, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}\big),\\ T(\mathscr{M}) &= \Gamma(Y, \mathscr{T}(\mathscr{M}))\\ &= \operatorname{Hom}_{\mathscr{O}_X}\big(\mathscr{G}, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}\big).\end{aligned}$$

There is a coherent $\mathscr{O}_Y$-Module $\mathscr{N}$, determined up to unique isomorphism, and isomorphisms of functors

$$\mathscr{T}(\mathscr{M}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{N}, \mathscr{M}), \tag{7.7.8.1}$$

$$T(\mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{N}, \mathscr{M}). \tag{7.7.8.2}$$

Proof. — The functorial isomorphism (0.5.4.2.1) gives, functorially in $\mathscr{M}$,

$$\begin{aligned}\mathscr{H}om_{\mathscr{O}_X}\big(\mathscr{E}_i, \mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}\big) &\simeq \mathscr{E}_i^\vee\otimes_{\mathscr{O}_X}\big(\mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}\big)\\ &\simeq \big(\mathscr{E}_i^\vee\otimes_{\mathscr{O}_X}\mathscr{F}\big)\otimes_{\mathscr{O}_Y}\mathscr{M}\\ &\simeq \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}_i,\mathscr{F})\otimes_{\mathscr{O}_Y}\mathscr{M}\end{aligned}$$

for $i = 0, 1$. Put

$$\mathscr{F}_i = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}_i, \mathscr{F}) \qquad (i = 0, 1).$$

These are coherent (0.5.3.5) and Y-flat (0.5.4.2). If $v : \mathscr{E}_1 \to \mathscr{E}_0$ is the given homomorphism, put

$$u = \mathscr{H}om(v, 1_{\mathscr{F}}) : \mathscr{F}_0 \longrightarrow \mathscr{F}_1.$$

Since

$$\mathscr{H} \longmapsto \mathscr{H}om_{\mathscr{O}_X}\big(\mathscr{H}, \mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}\big)$$

is left exact, there are functorial isomorphisms

$$\begin{aligned}\mathscr{H}om_{\mathscr{O}_X}\big(\mathscr{G}, \mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}\big) &\simeq \operatorname{Ker}\big(\mathscr{H}om_{\mathscr{O}_X}\big(\mathscr{E}_0, \mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}\big) \longrightarrow \mathscr{H}om_{\mathscr{O}_X}\big(\mathscr{E}_1, \mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}\big)\big)\\ &\simeq \operatorname{Ker}\left(\mathscr{F}_0\otimes_{\mathscr{O}_Y}\mathscr{M} \longrightarrow \mathscr{F}_1\otimes_{\mathscr{O}_Y}\mathscr{M}\right).\end{aligned}$$

As f_* is left exact, it follows that

$$f_*\mathscr{H}om_{\mathscr{O}_X}\big(\mathscr{G}, \mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}\big) \simeq \operatorname{Ker}\left(f_*(\mathscr{F}_0\otimes_{\mathscr{O}_Y}\mathscr{M}) \longrightarrow f_*(\mathscr{F}_1\otimes_{\mathscr{O}_Y}\mathscr{M})\right).$$

The assertion now follows from (III.7.7.7). **III** | 203

Remarks (7.7.9). — (i) *In (III.7.7.6), (III.7.7.7), and (III.7.7.8), the formation of the $\mathscr{O}_Y$-Modules $\mathscr{Q}$, $\mathscr{R}$, and $\mathscr{N}$ commutes with base change. For example, retaining the notation of (III.7.7.2), in the situation of (III.7.7.6) one has, for every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{M}'$,*

$$f'_*\big(\mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}'\big) \simeq \mathscr{H}om_{\mathscr{O}_{Y'}}\big(g^*(\mathscr{Q}), \mathscr{M}'\big).$$

Indeed, by the observation in (III.7.7.2), it is enough to use

$$\operatorname{Hom}_{\mathscr{O}_Y}\big(\mathscr{Q}, g_*(\mathscr{M}')\big) = \operatorname{Hom}_{\mathscr{O}_{Y'}}\big(g^*(\mathscr{Q}), \mathscr{M}'\big),$$

which is (0.4.4.3.1).

Likewise, if in (III.7.7.7) one replaces $Y, f, \mathscr{M}, \mathscr{F}, \mathscr{F}'$ by

$$Y', f', \mathscr{M}', \mathscr{F}\otimes_{\mathscr{O}_X}\mathscr{O}_{X'}, \mathscr{F}'\otimes_{\mathscr{O}_X}\mathscr{O}_{X'},$$

then one must replace $\mathscr{R}$ by $g^*(\mathscr{R})$. Finally, in (III.7.7.8), if $X, Y, f, \mathscr{M}, \mathscr{F}$ are replaced by

$$X', Y', f', \mathscr{M}', \mathscr{F}\otimes_{\mathscr{O}_X}\mathscr{O}_{X'},$$

and $\mathscr{G}$ by

$$\mathscr{G}' = \mathscr{G}\otimes_{\mathscr{O}_X}\mathscr{O}_{X'},$$

then $\mathscr{N}$ must be replaced by $g^*(\mathscr{N})$. This follows because there remains an exact sequence

$$\mathscr{E}'_1 \longrightarrow \mathscr{E}'_0 \longrightarrow \mathscr{G}' \longrightarrow 0, \qquad \mathscr{E}'_i = \mathscr{E}_i\otimes_{\mathscr{O}_X}\mathscr{O}_{X'},$$

and because, for $i = 0, 1$,

$$(\mathscr{E}'_i)^\vee\otimes_{\mathscr{O}_{X'}}\big(\mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}'\big) = \mathscr{E}_i^\vee\otimes_{\mathscr{O}_X}\big(\mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}'\big).$$

(ii) *Condition (ii) in (III.7.7.8) is satisfied by every coherent $\mathscr{O}_X$-Module $\mathscr{G}$ whenever there exists a Y-ample invertible $\mathscr{O}_X$-Module; this holds, for example, when Y is affine and $f : X \to Y$ is projective. Indeed, by (II.5.5.1) and (II.2.7.10), there is a finite-type locally free $\mathscr{O}_X$-Module $\mathscr{E}_0$ with $\mathscr{G}$ isomorphic to a quotient of $\mathscr{E}_0$. Since $\mathscr{E}_0$ and $\mathscr{G}$ are coherent, so is the kernel $\mathscr{G}_1$ of $\mathscr{E}_0 \to \mathscr{G}$. Applying the same result to $\mathscr{G}_1$ gives an exact sequence*

$$\mathscr{E}_1 \longrightarrow \mathscr{E}_0 \longrightarrow \mathscr{G} \longrightarrow 0$$

with $\mathscr{E}_0$ and $\mathscr{E}_1$ finite-type locally free.

(iii) *We shall prove in Chapter V that the restrictive hypothesis (ii) of (III.7.7.8) is redundant.*

Proposition (7.7.10). — (Local criteria for the exchange property). *Under the general hypotheses of (III.7.7.5), let $y \in Y$ and let p be an integer. The following conditions are equivalent:*

a) The functor $T_p^{\mathscr{O}_y}$ is right exact.

b) The canonical homomorphism

$$T_p^{\mathscr{O}_y}(\mathscr{O}_y) \longrightarrow T_p^{\mathscr{O}_y}(\kappa(y))$$

is surjective.

c) For every integer n, the canonical homomorphism

$$T_p^{\mathscr{O}_y}(\mathscr{O}_y/\mathfrak{m}_y^{n+1}) \longrightarrow T_p^{\mathscr{O}_y}(\kappa(y))$$

is surjective.

Moreover, the set of points $y \in Y$ satisfying these conditions is the largest open subset U of Y for which $\mathscr{T}_p^U$ is right exact.

Proof. — In view of (III.7.7.4), the equivalence of a), b), and c) follows from (III.7.4.7) and (III.7.5.2). That the locus U on which $T_p^{\mathscr{O}_y}$ is right exact is open follows from (III.7.4.4). Conversely, if $\mathscr{T}_p^V$ is right exact, then $T_p^{\mathscr{O}_y}$ is right exact for every $y \in V$, by condition c) of (III.7.7.5) and (III.7.6.1).

Corollary (7.7.11). — *If $\mathscr{T}_p$ is right exact (respectively left exact), then for every morphism $g : Y' \to Y$ the functor $\mathscr{T}_p^{Y'}$ is right exact (respectively left exact). The converse holds when g is faithfully flat.*

Proof. — The first assertion follows immediately from (III.7.6.1) and from the fact that the question is local on Y and Y', by conditions c) and b) of (III.7.7.5). For the converse, it is enough, by (III.7.7.10), to show that $T_p^{\mathscr{O}_y}$ is right exact (respectively left exact) for every $y \in Y$. By hypothesis there is a point $y' \in Y'$ with $g(y') = y$, and $\mathscr{O}_{y'}$ is faithfully flat over $\mathscr{O}_y$. The conclusion follows from the hypothesis and (III.7.6.1). III | 204

Remarks (7.7.12). — (i) *Under the hypotheses of (III.7.7.4), suppose in addition that $\mathscr{P}_\bullet$ is a finite complex. Since one may choose a finite affine open covering $\mathfrak{U}$ of X, it follows from (III.7.7.1) that the bicomplex*

$$C^\bullet\big(\mathfrak{U}, \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}\big)$$

is finite as well. More precisely, there is a finite set E of pairs, independent of $\mathscr{M}$, such that

$$C^h\big(\mathfrak{U}, \mathscr{P}^k \otimes_{\mathscr{O}_Y} \mathscr{M}\big) = 0 \qquad \text{for } (h,k) \notin E.$$

Consequently there is an integer i_1 such that

$$\mathscr{T}_i(\mathscr{M}) = 0 \qquad (i \geq i_1)$$

for every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$. Thus $\mathscr{T}_i$ is trivially exact for these values of i, and (III.7.4.1) shows that $Z_i'(L_\bullet)$ is a flat finite-type A-module, hence a finite-type projective A-module since A is Noetherian.

Define a complex $L'_\bullet$ of A-modules by

$$L'_i = L_i \quad (i < i_1), \qquad L'_{i_1} = Z'_{i_1}(L_\bullet), \qquad L'_i = 0 \quad (i > i_1),$$

and put $\mathscr{L}'_\bullet = \widetilde{L'_\bullet}$. Plainly

$$\mathscr{H}_i(\mathscr{L}'_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}) = \mathscr{H}_i(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M})$$

for $i < i_1 - 1$ and for $i \geq i_1$, the two sides being zero in the latter range. Since

$$\mathrm{Im}(Z'_{i_1} \otimes_A M) = \mathrm{Im}(L_{i_1} \otimes_A M)$$

by definition, the same equality holds for $i = i_1 - 1$. Thus in (III.7.7.4) one may also take $\mathscr{L}_\bullet$ to be finite, provided only that its terms are locally free $\mathscr{O}_Y$-Modules associated to finite-type projective A-modules.

This applies in particular when $\mathscr{P}_\bullet$ has only one nonzero term $\mathscr{F}$, in degree 0. In that case

$$\mathscr{T}_n(\mathscr{M}) = R^{-n}f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}\big),$$

and one may take $\mathscr{L}_i = 0$ for $i > 0$. In this case it is preferable to use cohomological notation, writing $\mathscr{T}^{-p}$ in place of $\mathscr{T}_p$.

(ii) *If the terms $\mathscr{P}_i$ in (III.7.7.5) are no longer assumed Y-flat, the conclusions remain valid provided one defines*

$$\mathscr{T}_p(\mathscr{M}) = \mathbf{Tor}_p^Y\big(f, 1_Y; \mathscr{P}_\bullet, \mathscr{M}\big). \tag{7.7.12.1}$$

Indeed,

$$\mathbf{Tor}_n^Y\big(f, 1_Y; \mathscr{P}_\bullet, \mathscr{O}_Y\big)$$

is then a coherent $\mathscr{O}_Y$-Module by (III.6.7.9). The proof of (III.6.10.5) applies without change, in view of (III.6.10.1), and also shows that, when $Y = \mathrm{Spec}(A)$ is affine,

$$\mathscr{T}_p(\mathscr{M}) = \mathscr{H}_p\big(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}\big), \qquad \mathscr{L}_\bullet = \widetilde{L_\bullet},$$

where $L_\bullet$ is a complex of finite-type free A-modules. This proves the assertion.

7.8 Application to proper morphisms: II. Cohomological flatness criteria

Definition (7.8.1). — *Let X and Y be two preschemes, let $f : X \to Y$ be a quasi-compact separated morphism, let $\mathscr{P}_\bullet$ be a complex of quasi-coherent, Y-flat $\mathscr{O}_X$-Modules, and let $\mathscr{T}_\bullet$ be the homological functor on quasi-coherent $\mathscr{O}_Y$-Modules defined by (7.7.1.1). Let y be a point of Y.* III | 205

We say that $\mathscr{P}_\bullet$ is homologically flat over Y at y, in dimension p *(or* cohomologically flat over Y at y, in dimension $-p$*) if there is an open neighbourhood U of y in Y such that*

$$\mathscr{T}_p^U = \mathscr{T}_U^{-p}$$

is exact.

We say that $\mathscr{P}_\bullet$ is homologically flat in dimension p over Y *(or* cohomologically flat in dimension $-p$ over Y*) if it is homologically flat over Y at every point $y \in Y$, in dimension p.*

When $\mathscr{P}_\bullet$ is homologically flat over Y (respectively over Y at y) in every *dimension p, we say simply that $\mathscr{P}_\bullet$ is* homologically flat over Y *(respectively over Y at y), or* cohomologically flat over Y *(respectively over Y at y).*

(7.8.2) By definition, the notion of homological flatness over Y is *local on Y*.

If Y is locally Noetherian, or if Y is a scheme, then for $\mathscr{P}_\bullet$ to be homologically flat over Y in dimension p, it is necessary and sufficient that the functor $\mathscr{T}_p$ be exact. The proof for locally Noetherian Y was given in the course of the proof of (III.7.7.5); when Y is a scheme the argument is the same, based on (I.9.4.2) applied to an affine open subset of a quasi-compact scheme.

Proposition (7.8.3). — *With the notation and hypotheses of (III.7.8.1), the following conditions are equivalent:*

a) $\mathscr{P}_\bullet$ is homologically flat over Y at y, in dimension p.

b) There is an open neighbourhood U of y in Y such that $\mathscr{T}_p^U$ and $\mathscr{T}_{p+1}^U$ are right exact.

c) There is an open neighbourhood U of y in Y such that $\mathscr{T}_p^U$ and $\mathscr{T}_{p-1}^U$ are left exact.

d) There is an open neighbourhood U of y in Y such that $\mathscr{T}_{p+1}^U$ is right exact and $\mathscr{T}_{p-1}^U$ is left exact.

Proof. — In view of the interpretation of $\mathscr{T}_p$ when Y is affine, this is merely a translation of part of (III.7.3.3).

Proposition (7.8.4). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, let $\mathscr{P}_\bullet$ be a bounded-below complex of coherent, Y-flat $\mathscr{O}_X$-Modules, and let $\mathscr{T}_\bullet$ be the functor defined by (7.7.1.1). For every $y \in Y$, the following conditions are equivalent:*

a) $\mathscr{P}_\bullet$ is homologically flat over Y at y, in dimension p.

b) The functor $\mathscr{T}_p^{\mathscr{O}_y}$ is exact.

c) There is an integer n_0 such that, for $n \geq n_0$,

$$\operatorname{length} \mathscr{T}_p^{\mathscr{O}_y}\big(\mathscr{O}_y/\mathfrak{m}_y^{n+1}\big) = \operatorname{length} \mathscr{T}_p^{\mathscr{O}_y}\big(\kappa(y)\big) \cdot \operatorname{length}\big(\mathscr{O}_y/\mathfrak{m}_y^{n+1}\big). \tag{7.8.4.1}$$

Here all lengths are lengths of $\mathscr{O}_y$-modules.

d) There is an open neighbourhood U of y such that

$$\mathscr{H}^{-p}(f, \mathscr{P}^\bullet)|U$$

is isomorphic to an $(\mathscr{O}_Y|U)$-Module of the form $(\mathscr{O}_Y|U)^m$ and, for every quasi-coherent $(\mathscr{O}_Y|U)$-Module $\mathscr{M}$, the canonical homomorphism

$$\mathscr{H}^{-p}(f, \mathscr{P}^\bullet)|U \otimes_{\mathscr{O}_Y|U} \mathscr{M} \longrightarrow \mathscr{H}^{-p}\big(f, (\mathscr{P}^\bullet|U) \otimes_{\mathscr{O}_Y|U} \mathscr{M}\big) \tag{7.8.4.2}$$

is bijective.

When these conditions hold, one also has:

e) There is a neighbourhood of y on which the function $z \mapsto d_p(z)$, defined in (7.7.5.1), is constant.

Moreover, if Y is reduced at y (0.4.1.4), condition e) is equivalent to the other conditions.

Proof. — Indeed, condition b) is equivalent to saying that $\mathscr{T}_p^{\mathscr{O}_y}$ and $\mathscr{T}_{p+1}^{\mathscr{O}_y}$ are right exact (III.7.3.3). The **III** | 206
equivalence of a) and b) then follows from (III.7.7.10) and (III.7.8.3).

Since $\mathscr{O}_y/\mathfrak{m}_y^{n+1}$ is Artinian, and

$$\mathscr{T}_p^{\mathscr{O}_y}\big(\mathscr{O}_y/\mathfrak{m}_y^{n+1}\big) \quad \text{and} \quad \mathscr{T}_p^{\mathscr{O}_y}\big(\kappa(y)\big)$$

are finite-type $(\mathscr{O}_y/\mathfrak{m}_y^{n+1})$-modules (III.7.7.4), hence have finite length, the equivalence of b) and c) follows again from (III.7.7.10) and (III.7.3.5.7).

That a) implies e), and that a) and e) are equivalent when $\mathscr{O}_y$ is reduced, follows from (III.7.6.9). Finally, a) implies that (7.8.4.2) is bijective, by the definition (III.7.8.1) and (III.7.7.5). It also implies that

$$\big(\mathscr{H}^{-p}(f, \mathscr{P}^\bullet)\big)_y$$

is a flat $\mathscr{O}_y$-module by condition (f) of (III.7.3.3), hence a free $\mathscr{O}_y$-module of finite type (0.10.1.3), since it is of finite type by (III.7.7.4). As $\mathscr{H}^{-p}(f, \mathscr{P}^\bullet)$ is a coherent $\mathscr{O}_Y$-Module (III.7.7.4), it is locally free in a neighbourhood of y (0.5.2.7). Conversely, d) plainly implies a), by the definition of the functor $\mathscr{T}_p^U$ (7.7.2.2).

Proposition (7.8.5). — *Under the hypotheses of (III.7.8.4), the following conditions are equivalent:*

a) $\mathscr{P}_\bullet$ is homologically flat over Y in every dimension $i \leq p$.

b) For $i \leq p+1$, the functors $\mathscr{T}_i$ are right exact.

c) For $i \geq -p$, the $\mathscr{O}_Y$-Modules $\mathscr{H}^i(f, \mathscr{P}^\bullet)$ are locally free.

Proof. — The equivalence of a) and b) is immediate from (III.7.8.3), and a) implies c) by (III.7.8.4).

Conversely, suppose that c) holds. Recall also that $\mathscr{L}_i = 0$ for $i < i_0$ (III.7.7.4), and hence $\mathscr{T}_i = 0$ for $i \leq i_0$. Every point $y \in Y$ therefore has an affine neighbourhood $U = \mathrm{Spec}(A)$ such that $T_i^A(A)$ is a free A-module for $i \leq p$. It follows from (III.7.3.7) that $\mathscr{T}_i^U = T_i^A$ is exact for $i \leq p$.

We shall chiefly apply the criteria for cohomological flatness to the case where the complex $\mathscr{P}_\bullet$ is concentrated in a *single* coherent $\mathscr{O}_X$-Module $\mathscr{F}$, flat over Y, taken as $\mathscr{P}_0$. Recall that in this case

$$\mathscr{T}_p(\mathscr{M}) = R^{-p} f_*\big(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}\big).$$

Proposition (7.8.6). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper flat morphism, and let y be a point of Y. Denote the fibre $f^{-1}(y) = X \otimes_Y \kappa(y)$ by X_y. Suppose that*

$$\Gamma(X_y, \mathscr{O}_{X_y}) = R$$

is a separable $\kappa(y)$-algebra (Bourbaki, Algèbre*, chap. VIII, § 7, no. 5), that is, a direct sum of finitely many finite-degree separable extensions of $\kappa(y)$. Then $\mathscr{O}_X$ is cohomologically flat over Y at y, in dimension* 0.

Proof. — By (III.7.8.4), we may restrict to the case where Y is the spectrum of the local ring $A = \mathscr{O}_y$. The flatness of f implies $\mathscr{T}_{-1} = 0$, so that T_0^A is already left exact and it remains only to prove that it is right exact. By condition c) of (III.7.7.10), we may even reduce to the case where $A = \mathscr{O}_y$ is Artinian.

Let k' be a finite extension of $\kappa(y)$ which splits R, so that

$$R \otimes_{\kappa(y)} k'$$

is a direct sum of finitely many fields isomorphic to k'. There is a local homomorphism from A to a local ring A' such that A' is a finite free A-algebra and its residue field is isomorphic to k' (0.10.3.2). By (III.7.6.1), it is therefore enough to prove that $T_0^{A'}$ is right exact; in other words, we may suppose that R is a direct sum of m fields isomorphic to $\kappa(y)$.

We shall use the following elementary lemma.

Lemma (7.8.6.1). — *Let Z be a locally ringed space. For Z to be connected, it is necessary and sufficient* **III** | 207
that the ring $\Gamma(Z, \mathscr{O}_Z)$ not be a product of two nonzero rings.

Proof. — It is clear that if Z is the union of two disjoint nonempty open subsets Z_1 and Z_2, then $\Gamma(Z, \mathscr{O}_Z)$ is isomorphic to the product of the two nonzero rings $\Gamma(Z_1, \mathscr{O}_Z)$ and $\Gamma(Z_2, \mathscr{O}_Z)$.

Conversely, to say that $\Gamma(Z, \mathscr{O}_Z)$ is such a product is equivalent to saying that it contains an idempotent s distinct from 0 and 1. For every $z \in Z$, the germ s_z is then an idempotent in the local ring $\mathscr{O}_z$, and hence

equals 0 or 1. The set of z for which $s_z = 0$ is open. On the other hand, $s_z = 1$ implies, by definition, that $s(z) \neq 0$, and the set of z for which $s_z = 1$ is open as well (0.5.5.2). The conclusion follows.

It follows from the lemma that X_y has exactly m connected components X'_i and that

$$\Gamma(X'_i, \mathscr{O}_{X'_i}) = \kappa(y)$$

for every i. Since A is local and Artinian, its spectrum consists of one point, and consequently X and X_y have the same underlying space. Thus X has m connected components X_i such that

$$X'_i = X_i \otimes_Y \kappa(y).$$

We are finally reduced to the case $R = \kappa(y)$. By condition b) of (III.7.7.10), it remains to prove that the canonical homomorphism

$$\Gamma(X, \mathscr{O}_X) \longrightarrow \Gamma(X_y, \mathscr{O}_{X_y})$$

is surjective. This is immediate, since the composite

$$\Gamma(Y, \mathscr{O}_Y) = A \longrightarrow \Gamma(X, \mathscr{O}_X) \longrightarrow \Gamma(X_y, \mathscr{O}_{X_y}) = \kappa(y)$$

is already surjective.

Corollary (7.8.7). — *Under the hypotheses of (III.7.8.6), there is an open neighbourhood U of y such that:*

(i) $f_*(\mathscr{O}_X)|U$ is isomorphic to an $(\mathscr{O}_Y|U)$-Module of the form $(\mathscr{O}_Y|U)^m$;

(ii) for every $z \in U$, the canonical homomorphism

$$\big(f_*(\mathscr{O}_X)\big)_z \otimes_{\mathscr{O}_z} \kappa(z) \longrightarrow \Gamma(X_z, \mathscr{O}_{X_z})$$

is bijective.

Proof. — Assertion (i) follows from (III.7.8.6) and (III.7.8.4). Assertion (ii) follows from the exactness of $\mathscr{T}_0^U$, for a suitable choice of U, and from (7.7.5.3).

Corollary (7.8.8). — *Suppose that the hypotheses of (III.7.8.6) hold and, in addition, that*

$$\Gamma(X_y, \mathscr{O}_{X_y}) = \kappa(y).$$

Then there is an open neighbourhood U of y such that the canonical homomorphism

$$\mathscr{O}_Y|U \longrightarrow f_*(\mathscr{O}_X)|U$$

is bijective.

Proof. — Indeed, it follows from assertion (ii) of (III.7.8.7) that the integer m occurring in assertion (i) of that corollary is necessarily equal to 1.

Corollary (7.8.9). — *Under the hypotheses of (III.7.8.6), there are an open neighbourhood U of y, a coherent $\mathscr{O}_U$-Module $\mathscr{Q}$, determined up to unique isomorphism, and an isomorphism of functors in the quasi-coherent $\mathscr{O}_U$-Module $\mathscr{M}$,*

$$R^1 f_*\big(f^*(\mathscr{M})\big) \simeq \mathscr{H}om_{\mathscr{O}_U}(\mathscr{Q}, \mathscr{M}). \tag{7.8.9.1}$$

Proof. — The hypothesis implies that $\mathscr{T}_0^U$ is exact for a suitable U. It is therefore enough to apply the equivalence of conditions a) and b') in (III.7.7.5), with $p = 0$ and with $\mathscr{P}_\bullet$ the complex concentrated in degree 0 with term $\mathscr{O}_X$.

Remarks (7.8.10). — (i) *Under the hypotheses of (III.7.8.6), consider the Stein factorisation of f (III.4.3.3),*

$$X \xrightarrow{f'} Y' \xrightarrow{g} Y.$$

Here

$$Y' = \mathrm{Spec}\big(f_*(\mathscr{O}_X)\big).$$

The finite morphism g is such that

$$g_*(\mathscr{O}_{Y'}) = f_*(\mathscr{O}_X)$$

is locally free near y, and its fibre over y is the spectrum of a separable algebra over $\kappa(y)$ (II.1.5.1). We shall deduce in Chapter IV that there is an open neighbourhood U of y in Y such that, for the restriction $g^{-1}(U) \to U$ of g, every *fibre $g^{-1}(z)$, $z \in U$, is the spectrum of a separable algebra over $\kappa(z)$. This is what we shall call an* étale covering *of U. It will then follow from assertion (ii) of (III.7.8.7) that the hypothesis on the point y in (III.7.8.6) also holds at every point of a neighbourhood of y.*

(ii) *We shall see in Chapter V that, even if X is projective over Y (and even if it is also "simple" over Y, a property to be defined in Chapter IV), the $\mathscr{O}_U$-Module $\mathscr{Q}$ of (III.7.8.9) need not be locally free. In other words, under these hypotheses $\mathscr{O}_X$ need not be cohomologically flat in* dimension 1 *over Y at y. In Chapter V we shall interpret $\mathscr{Q}$ as the sheaf of 1-differentials of the Picard scheme of X relative to Y, along the unit section.* **III** | 208

7.9 Application to proper morphisms: III. Invariance of the Euler–Poincaré characteristic and the Hilbert polynomial

(7.9.1) Let A be a ring and let M be a projective A-module of finite type. Recall (Bourbaki, *Alg. comm.*, chap. II, §5, no. 2) that this is equivalent to saying that the $\mathscr{O}_X$-Module $\widetilde{M}$ associated with M on $X = \operatorname{Spec}(A)$ is locally free of finite type. For every $\mathfrak{p} \in \operatorname{Spec}(A)$, the *rank of M at $\mathfrak{p}$*, denoted $\operatorname{rank}_{\mathfrak{p}}(M)$, is the rank of the free $A_{\mathfrak{p}}$-module $M_{\mathfrak{p}}$ (or, equivalently, the rank at $\mathfrak{p}$ of the locally free $\mathscr{O}_X$-Module $\widetilde{M}$). Thus

$$\operatorname{rank}_{\mathfrak{p}} M = \operatorname{rank}_{\mathfrak{p}}(M_{\mathfrak{p}}) = \operatorname{rank}_{\kappa(\mathfrak{p})}\big(M \otimes_A \kappa(\mathfrak{p})\big). \tag{7.9.1.1}$$

Proposition (7.9.2). — *Let $P_\bullet$ be a finite complex of projective A-modules of finite type, and, for every A-module M, set*

$$T_\bullet(M) = H_\bullet(P_\bullet \otimes_A M).$$

Then, for every $\mathfrak{p} \in \operatorname{Spec}(A)$,

$$\sum_i (-1)^i \operatorname{rank}_{\kappa(\mathfrak{p})} T_i\big(\kappa(\mathfrak{p})\big) = \sum_i (-1)^i \operatorname{rank}_{\mathfrak{p}}(P_i). \tag{7.9.2.1}$$

Proof. — Indeed, $T_i(\kappa(\mathfrak{p})) = H_i(P_\bullet \otimes_A \kappa(\mathfrak{p}))$ by definition. In view of (7.9.1.1), formula (7.9.2.1) is simply the invariance of the Euler–Poincaré characteristic of a finite complex of finite-dimensional vector spaces upon passage to homology (0.11.10.2).

Corollary (7.9.3). — *The function*

$$\mathfrak{p} \longmapsto \sum_i (-1)^i \operatorname{rank}_{\kappa(\mathfrak{p})} T_i\big(\kappa(\mathfrak{p})\big)$$

is locally constant on $\operatorname{Spec}(A)$.

Theorem (7.9.4). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{P}_\bullet$ be a finite complex of coherent, Y-flat $\mathscr{O}_X$-Modules. If*

$$\mathscr{T}_\bullet(\mathscr{M}) = \mathscr{H}^{-\bullet}\big(f, \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}\big) \qquad \text{(cf. (7.7.1.1))},$$

then the function **III** | 209

$$y \longmapsto \sum_i (-1)^i \operatorname{rank}_{\kappa(y)} \mathscr{T}_i\big(\kappa(y)\big) \tag{7.9.4.1}$$

is locally constant on Y.

Proof. — We may restrict ourselves to the case where $Y = \operatorname{Spec}(A)$ is affine and A is Noetherian. Since the complex $\mathscr{P}_\bullet$ is finite, (III.7.7.12)(i) shows that

$$\mathscr{T}_p(\mathscr{M}) = \mathscr{H}_p(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}),$$

where $\mathscr{L}_\bullet = \widetilde{L}_\bullet$ and $L_\bullet$ is a finite complex of projective A-modules of finite type. The theorem now follows from (III.7.9.3).

(7.9.5) Under the hypotheses of (III.7.9.4), the function (7.9.4.1) is *constant* when Y is *connected*. When Y is connected and nonempty, its unique (integral) value is denoted

$$\operatorname{EP}(f, \mathscr{P}_\bullet), \qquad \operatorname{EP}(Y, \mathscr{P}_\bullet), \qquad \text{or simply} \qquad \operatorname{EP}(\mathscr{P}_\bullet)$$

when no confusion is possible; this integer is called the *Euler–Poincaré characteristic* of $\mathscr{P}_\bullet$ relative to f (or to Y). In general, we likewise write

$$\mathrm{EP}(f, \mathscr{P}_\bullet; y), \qquad \mathrm{EP}(Y, \mathscr{P}_\bullet; y), \qquad \mathrm{EP}(\mathscr{P}_\bullet; y)$$

for the right-hand side of (7.9.4.1).

(7.9.6) Under the hypotheses of (III.7.9.4) concerning X, Y, and f, let

$$0 \longrightarrow \mathscr{P}'_\bullet \xrightarrow{u} \mathscr{P}_\bullet \xrightarrow{v} \mathscr{P}''_\bullet \longrightarrow 0$$

be an exact sequence of finite complexes of coherent, Y-flat $\mathscr{O}_X$-Modules, where the homomorphisms u and v have even degrees $2d$ and $2d'$, respectively. Since $\mathscr{T}_\bullet$ is a homological functor (III.7.7.1), there is an exact homology sequence

$$\cdots \longrightarrow \mathscr{T}_i(\mathscr{P}'_\bullet, \kappa(y)) \longrightarrow \mathscr{T}_{i+2d}(\mathscr{P}_\bullet, \kappa(y)) \longrightarrow \mathscr{T}_{i+2d+2d'}(\mathscr{P}''_\bullet, \kappa(y)) \longrightarrow \mathscr{T}_{i-1}(\mathscr{P}'_\bullet, \kappa(y)) \longrightarrow \cdots,$$

which moreover has only finitely many terms. On writing that the Euler–Poincaré characteristic of this complex is zero (0.11.10.1), we obtain at once

$$\mathrm{EP}(\mathscr{P}_\bullet; y) = \mathrm{EP}(\mathscr{P}'_\bullet; y) + \mathrm{EP}(\mathscr{P}''_\bullet; y) \tag{7.9.6.1}$$

for every $y \in Y$.

Suppose, for example, that $\mathscr{P}_\bullet = (\mathscr{P}_i)$ and that $\mathscr{P}_i = 0$ for $i < 0$. We then have the following exact sequence of complexes:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathscr{P}_1 & \longrightarrow & \mathscr{P}_2 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & \mathscr{P}_0 & \longrightarrow & \mathscr{P}_1 & \longrightarrow & \mathscr{P}_2 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & \mathscr{P}_0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

the nonzero vertical arrows being identity automorphisms. Applying (7.9.6.1) to this exact sequence and arguing by induction on the length of $\mathscr{P}_\bullet$ gives

$$\mathrm{EP}(\mathscr{P}_\bullet; y) = \sum_i (-1)^i \mathrm{EP}(\mathscr{P}_i; y). \tag{7.9.6.2}$$

Here, for each coherent $\mathscr{O}_X$-Module $\mathscr{F}$ that is flat over Y, the notation $\mathrm{EP}(\mathscr{F}; y)$ (or $\mathrm{EP}(f, \mathscr{F}; y)$, or $\mathrm{EP}(Y, \mathscr{F}; y)$) means $\mathrm{EP}(\mathscr{Q}_\bullet; y)$ (with the corresponding relative notation) for the complex $\mathscr{Q}_\bullet$ whose only nonzero term is $\mathscr{F}$ in degree 0. Thus the study of Euler–Poincaré characteristics of complexes reduces to that of complexes with a single term.

Proposition (7.9.7). — *Under the hypotheses of (III.7.9.4), let Y' be a locally Noetherian prescheme, let $g : Y' \to Y$ be a morphism, put*

$$X' = X \times_Y Y', \qquad f' = f_{(Y')} : X' \longrightarrow Y',$$

and let

$$\mathscr{P}'_\bullet = \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

be the resulting finite complex of $\mathscr{O}_{X'}$-Modules. Then $\mathscr{P}'_\bullet$ consists of coherent, Y'-flat $\mathscr{O}_{X'}$-Modules, and, for every $y' \in Y'$,

$$\mathrm{EP}(\mathscr{P}'_\bullet; y') = \mathrm{EP}(\mathscr{P}_\bullet; g(y')). \tag{7.9.7.1}$$

Proof. — The $\mathscr{O}_{X'}$-Modules $\mathscr{P}'_i$, being inverse images of the $\mathscr{P}_i$ under the projection $X' \to X$, are coherent. They are Y'-flat by (0.6.2.1) and (I.4.14.5), the question being local on X, Y, and Y'. Finally, f' is proper

by (II.5.4.2), so the left-hand side of (7.9.7.1) is defined. Formula (7.9.7.1) now follows from (6.10.4.2), (III.7.7.2), and (III.7.6.7), after reducing, as one may always do, to the case where Y and Y' are affine.

Proposition (7.9.8). — *Suppose that the hypotheses of (III.7.9.4) hold and, in addition, that there is an integer i_0 such that*

$$\mathscr{T}_i(\kappa(y)) = 0 \qquad \text{for } i \neq i_0 \text{ and every } y \in Y.$$

Then

$$\mathscr{T}_{i_0}(\mathscr{O}_Y) = \mathscr{H}^{-i_0}(f, \mathscr{P}^\bullet)$$

is a locally free $\mathscr{O}_Y$-Module, and its rank at $y \in Y$ is

$$(-1)^{i_0}\,\mathrm{EP}(f, \mathscr{P}_\bullet; y).$$

Proof. — First observe that the hypotheses of (III.7.4.4) are satisfied by the functors $\mathscr{T}_i^{\mathscr{O}_y}$, so that (III.7.4.7) applies to them. The hypothesis and (III.7.5.3) imply that $\mathscr{T}_i^{\mathscr{O}_y}$ is zero for $i \neq i_0$. Consequently $\mathscr{T}_{i_0}$ is exact by (III.7.3.3), and (III.7.8.4) shows that $\mathscr{H}^{-i_0}(f, \mathscr{P}^\bullet)$ is locally free. Its rank at $y \in Y$ is

$$\mathrm{rank}_{\kappa(y)}\,\mathscr{T}_{i_0}(\kappa(y)) = (-1)^{i_0}\,\mathrm{EP}(f, \mathscr{P}_\bullet; y),$$

by definition, since $\mathscr{T}_i(\kappa(y)) = 0$ for $i \neq i_0$.[1]

Corollary (7.9.9). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{F}$ be a coherent, Y-flat $\mathscr{O}_X$-Module. Suppose that there is an integer i_0 such that*

$$H^i\big(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} \kappa(y)\big) = 0$$

for every $i \neq i_0$ and every $y \in Y$. Then $R^{i_0}f_*(\mathscr{F})$ is a locally free $\mathscr{O}_Y$-Module, and its rank at y is $(-1)^{i_0}\,\mathrm{EP}(f, \mathscr{F}; y)$.

In particular:

Corollary (7.9.10). — *Under the preliminary hypotheses of (III.7.9.9) concerning X, Y, and $\mathscr{F}$, suppose that $R^i f_*(\mathscr{F}) = 0$ for every $i > 0$. Then $f_*(\mathscr{F})$ is a locally free $\mathscr{O}_Y$-Module, and its rank at y is $\mathrm{EP}(f, \mathscr{F}; y)$.*

Proof. — In view of (III.7.9.9), it is enough to prove the following lemma.

Lemma (7.9.10.1). — *Under the hypotheses of (III.7.9.10),*

$$H^i\big(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} \kappa(y)\big) = 0$$

for every $i > 0$ and every $y \in Y$.

Proof. — We may restrict ourselves to the case where $Y = \mathrm{Spec}(A)$ is affine. With the notation of (III.7.9.4), and with $\mathscr{P}_\bullet$ reduced to its only nonzero term $\mathscr{F}$ in degree 0, the hypothesis gives $\mathscr{T}_p(\mathscr{O}_Y) = 0$ for $p < 0$. It follows from (III.7.3.7) that $\mathscr{T}_p$ is exact for $p < 0$, and the lemma then follows from the equivalence of conditions a) and d) in (III.7.7.5).

Proposition (7.9.11). — *Under the hypotheses of (III.7.9.4), let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module that is very ample over Y, and, for every $n \in \mathbf{Z}$, put*

$$\mathscr{P}_\bullet(n) = \mathscr{P}_\bullet \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}.$$

Then, for every $y \in Y$, the function

$$n \longmapsto \mathrm{EP}(f, \mathscr{P}_\bullet(n); y) \tag{7.9.11.1}$$

is a polynomial with rational coefficients, and is the same for all points in any one connected component of Y.

Proof. — It is clear that $\mathscr{P}_\bullet(n)$ is a complex of Y-flat $\mathscr{O}_X$-Modules. By (7.9.6.2), we may reduce to the case where $\mathscr{P}_\bullet$ has a single nonzero term $\mathscr{F} \neq 0$ in degree 0. Since the questions are local on Y, we may moreover suppose that Y is affine and that f is projective (II.5.5.3). Put

$$X_y = f^{-1}(y), \qquad \mathscr{L}_y = \mathscr{L} \otimes_{\mathscr{O}_Y} \kappa(y);$$

[1] The printed French proof omits the factor $(-1)^{i_0}$ in this displayed equality; it is required both by the definition and by the statement of the proposition.

then $\mathscr{L}_y$ is a very ample $\mathscr{O}_{X_y}$-Module (II.4.4.10). By (III.7.7.2), for the functor $\mathscr{T}_\bullet$ associated with $\mathscr{P}_\bullet(n)$ we have

$$\mathscr{T}_i(\kappa(y)) = H^{-i}\big(X_y, \mathscr{F}_y \otimes \mathscr{L}_y^{\otimes n}\big), \qquad \mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} \kappa(y).$$

Thus $\mathrm{EP}(f, \mathscr{F}(n); y)$ is precisely the Euler–Poincaré characteristic $\chi_{\kappa(y)}(\mathscr{F}_y(n))$ defined in (III.2.5.1). It follows from (III.2.5.3) that (7.9.11.1) is a polynomial. For each fixed n, its value is constant on each connected component of Y by (III.7.9.4); this completes the proof.

(7.9.12) The polynomial (7.9.11.1), with rational coefficients, will be denoted

$$\mathrm{PH}(f, \mathscr{P}_\bullet; y) \qquad \text{or} \qquad \mathrm{PH}(\mathscr{P}_\bullet; y),$$

and will be called the *Hilbert polynomial at y* relative to $\mathscr{P}_\bullet$, f, and $\mathscr{L}$ (or simply the *Hilbert polynomial at y* of $\mathscr{P}_\bullet$, or of f, when no confusion is possible). When Y is connected and nonempty, the reference to y is omitted from the notation and terminology. This invariant will play an essential role in Chapter V, in the theory of "moduli" of coherent quotient sheaves of a given coherent sheaf.

With the notation of (III.7.9.6) and (III.7.9.11), we have

$$\mathrm{PH}(\mathscr{P}_\bullet; y) = \mathrm{PH}(\mathscr{P}'_\bullet; y) + \mathrm{PH}(\mathscr{P}''_\bullet; y), \tag{7.9.12.1}$$

and in particular

$$\mathrm{PH}(\mathscr{P}_\bullet; y) = \sum_i (-1)^i \,\mathrm{PH}(\mathscr{P}_i; y). \tag{7.9.12.2}$$

These formulas follow immediately from (7.9.6.1) and (7.9.6.2). Likewise, with the notation and hypotheses of (III.7.9.7),

$$\mathrm{PH}(\mathscr{P}'_\bullet; y') = \mathrm{PH}(\mathscr{P}_\bullet; g(y')). \tag{7.9.12.3}$$

Formula (7.9.12.2) reduces the study of the Hilbert polynomials of a complex to that of the Hilbert polynomial of a single Y-flat $\mathscr{O}_X$-Module. Such polynomials have the following remarkable interpretation, independent of homological considerations.

Corollary (7.9.13). — *Let Y be a Noetherian prescheme, let $f : X \to Y$ be a proper morphism, let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module that is very ample over Y, and let $\mathscr{F}$ be a coherent, Y-flat $\mathscr{O}_X$-Module. There is an integer n_0 such that, for $n \geq n_0$, $f_*(\mathscr{F}(n))$ is a locally free $\mathscr{O}_Y$-Module whose rank at $y \in Y$ is*

$$\mathrm{PH}(f, \mathscr{F}; y)(n).$$

Proof. — Since f is projective (II.5.5.3), there is an integer n_0 such that, for $n \geq n_0$, **III** | 212

$$R^i f_*(\mathscr{F}(n)) = 0 \qquad \text{for every } i > 0$$

by (III.2.2.1). The conclusion follows from (III.7.9.10).

The following flatness criterion will be important in Chapter V, in the theory of "moduli" of coherent sheaves.

Proposition (7.9.14). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a projective morphism, and let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module that is ample for f. For every $\mathscr{O}_X$-Module $\mathscr{F}$ and every $n \in \mathbf{Z}$, put*

$$\mathscr{F}(n) = \mathscr{F} \otimes \mathscr{L}^{\otimes n}.$$

A coherent $\mathscr{O}_X$-Module $\mathscr{F}$ is Y-flat if and only if there is an integer n_0 such that, for every $n \geq n_0$, $f_*(\mathscr{F}(n))$ is a locally free $\mathscr{O}_Y$-Module.

Proof. — The necessity of the condition is proved as in (III.7.9.13); the result of (III.2.2.1) applies to an ample sheaf because f is projective.

Conversely, we may reduce to the case where $Y = \mathrm{Spec}(A)$ is affine. By the hypothesis and (III.2.2.2)(i), the A-modules $\Gamma(X, \mathscr{F}(n))$ are projective and of finite type (Bourbaki, *Alg. comm.*, chap. II, §5, no. 2, th. 1). Let

$$S = \bigoplus_{n \geq 0} \Gamma(X, \mathscr{L}^{\otimes n})$$

be the associated graded ring. We know that X is canonically identified with $\mathrm{Proj}(S)$ by (II.4.5.2)(b) and (III.5.4.4). Let

$$M = \bigoplus_{n \geq n_0} \Gamma(X, \mathscr{F}(n)).$$

After replacing $\mathscr{L}$, if necessary, by a power $\mathscr{L}^{\otimes d}$, we may suppose that S is generated by finitely many elements of degree 1 (III.2.3.5.1); it then follows from (II.2.7.5) and (II.2.7.2) that $\mathscr{F}$ is identified with $\mathscr{P}roj_0(M)$.

For every homogeneous element $g \in S$ of positive degree, we therefore have

$$\Gamma(X_g, \mathscr{F}) = M_{(g)}.$$

Now M, being a direct sum of projective A-modules, is a flat A-module. Hence M_g is flat by (0.6.3.2), and so is $M_{(g)}$, which is a direct factor of M_g by (0.6.1.2). We conclude from (I.4.14.5) that $\mathscr{F}$ is Y-flat at every point of X_g. Since the open subsets X_g cover X, the proposition is proved.

(To be continued.)

INDEX OF NOTATION

$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$: 6.3.1.

$\mathscr{T}or_n^{\mathscr{O}_S}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$, $\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$: 6.4.1 and 6.5.1.

$\mathscr{T}or_n^S(\mathscr{F}, \mathscr{G})$: 6.5.1.

$\mathscr{T}or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}, \ldots, \mathscr{P}_\bullet^{(m)})$: 6.5.15.

$\mathbf{H}^{-n}(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)})$, $\mathbf{H}^{-n}(X^{(i)}, \mathscr{P}_\bullet^{(i)})$: 6.6.1.

$\mathbf{Tor}_n^A(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})$, $\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$: 6.6.2.

$\mathscr{T}or_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)})$, $\mathscr{T}or_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$: 6.6.2.

$\mathscr{T}or_n^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$: 6.6.8, 6.7.1, and 6.7.3.

$\mathscr{T}or_n^S((f_i)_{i\in I}; (\mathscr{P}_\bullet^{(i)})_{i\in I})$, $\mathscr{T}or_n^S(f_1, \ldots, f_m; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(m)})$: 6.7.11.

$\mathbf{Ab}$, $\mathbf{Ab}_A$: 7.1.1.

$T_{(B)}$, $T^{(B)}$, $T \otimes_A B$ (T a functor from $\mathbf{Ab}_A$ to $\mathbf{Ab}$): 7.1.3.

t_M: 7.2.2.

$T_{\mathfrak{p}}$: 7.1.4.

$\mathscr{T}_\bullet^{Y'}(\mathscr{M}')$, $\mathscr{T}_\bullet^U(\mathscr{M}|U)$: 7.7.2.

$\mathrm{EP}(f, \mathscr{P}_\bullet; y)$, $\mathrm{EP}(Y, \mathscr{P}_\bullet; y)$, $\mathrm{EP}(\mathscr{P}_\bullet; y)$: 7.9.5.

$\mathrm{PH}(f, \mathscr{P}_\bullet; y)$, $\mathrm{PH}(\mathscr{P}_\bullet; y)$: 7.9.12.

TERMINOLOGICAL INDEX

Euler–Poincaré characteristic of a complex of Modules: 7.9.5.

Cohomologically flat at a point $y \in Y$ in dimension p, cohomologically flat over Y in dimension p, cohomologically flat over Y at the point y, cohomologically flat over Y (complex of $\mathscr{O}_X$-Modules flat over Y): 7.8.1.

Homotopic complexes: 6.1.4.

Ringed space of cohomological dimension $\leq n$: 6.5.5.

Extension of scalars in an additive covariant functor: 7.1.3.

Sheaf of rings of cohomological dimension $\leq n$: 6.5.5.

Künneth formula: 6.7.8.

Homologically flat at a point $y \in Y$ in dimension p, homologically flat over Y in dimension p, homologically flat over Y at the point y, homologically flat over Y (complex of $\mathscr{O}_X$-Modules flat over Y): 7.8.1.

Homotopy: 6.1.4.

Hypertor of two complexes of A-modules: 6.3.1.

Local hypertor of two complexes of Modules: 6.4.1.

Global hypertor of two complexes of Modules: 6.6.2, 6.6.8, and 6.7.3.

Hilbert polynomial relative to a complex of Modules: 7.9.12.

Künneth spectral sequences: 6.7.3.

TABLE OF CONTENTS

Chapter III. — **Cohomological study of coherent sheaves (continued)** 5

§ 6. Local and global Tor functors; Künneth formula 5
6.1. Introduction 5
6.2. Hypercohomology of complexes of Modules on a prescheme 6
6.3. Hypertor of two complexes of modules 9
6.4. Local hypertor functors of complexes of quasi-coherent Modules; the case of affine schemes 14
6.5. Local hypertor functors of complexes of quasi-coherent Modules: the general case 16
6.6. Global hypertor functors of complexes of quasi-coherent Modules and Künneth spectral sequences: the case of an affine base 21
6.7. Global hypertor functors of complexes of quasi-coherent Modules and Künneth spectral sequences: the general case 25
6.8. Associativity spectral sequences for global hypertor 32
6.9. Base-change spectral sequences in global hypertor 34
6.10. Local structure of certain cohomological functors 39

§ 7. Study of base change in covariant homological functors of Modules 43
7.1. Functors of A-modules 43
7.2. Characterization of the tensor-product functor 44
7.3. Exactness criteria for homological functors of modules 48
7.4. Exactness criteria for the functors $\mathrm{H}_i(P \otimes_A M)$ 53
7.5. The case of Noetherian local rings 58
7.6. Descent of exactness properties. Semicontinuity theorem and Grauert's exactness criterion 60
7.7. Application to proper morphisms: I. The exchange property 65
7.8. Application to proper morphisms: II. Cohomological flatness criteria 72
7.9. Application to proper morphisms: III. Invariance of the Euler–Poincaré characteristic and the Hilbert polynomial 76

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ERRATA AND ADDENDA

(List 2)

A) *Typographical errors*

(0_{I}, 3.2.1) On line 2 of p. 27, replace X by U.
(0_{I}, 3.2.6) On line 1 from the bottom of p. 27 and line 1 of p. 28, replace (3 times) $\mathfrak{S}_\lambda$ by $\mathscr{H}_\lambda$.
(0_{I}, 3.4.5) Add a parenthesis before 3.4.5).
(0_{I}, 4.2.1) On line 10 of p. 39, replace $\psi_*(A)$ by $\psi_*(\mathscr{A})$.
(0_{I}, 5.1.3) On line 16 of p. 45, replace $\mathscr{F}|V$ by $\mathscr{F}|U$.
(0_{I}, 5.3.9) On line 12 of p. 48, before "such that", add: over an open neighborhood U of x.
(0_{I}, 7.6.9) On line 3 from the bottom of p. 73, replace I_λ by J_λ.
(I, 1.1.15) On line 3 of p. 83, replace $\mathfrak{a}$ by $\mathfrak{a} \neq A$.
(I, 1.4.1) On line 2 from the bottom of p. 90, replace $\widetilde{N}$ by N.
(I, 2.3.1) On line 3 from the bottom of p. 101, replace (0, 4.1.6) by (0, 4.1.7).
(I, 2.3.2) On line 11 of p. 101, replace B by B_s and C by C_t.
(I, 2.5.5) On line 19 of p. 104, replace S-morphism by S-prescheme.
(I, 3.3.7) On line 17 of p. 109, replace $Y_{(S)}$ by $Y_{(S')}$.
(I, 3.4.5) On line 9 of p. 113, replace s by s.
(I, 3.4.8) On line 4 of p. 114, replace $p(x)$ by $f(x)$.
(I, 3.5.1) On line 3 of p. 115, replace $1_X \times g$ by $1_{X'} \times g$.
(I, 3.7.2) On line 8 of p. 119, replace the first X by X'.
(I, 4.2.2) On line 8 of p. 123, in the left vertical arrow of the diagram, replace $\alpha_{\psi(y)}$ by $\phi_{\psi(y)}$.
(I, 4.4.3) On line 16 of p. 126, replace *isomorphée* by *isomorphes*.
(I, 4.5.5) On line 17 from the bottom of p. 127, replace (4.2.4) by (4.2.5).
(I, 5.1.4) On line 14 from the bottom of p. 128, replace (2.1.7) by (2.1.8).
(I, 5.3.13) On line 20 of p. 134, replace (4.2.4) by (4.2.5).
(I, 6.4.2) On line 1 from the bottom of p. 147, replace *covering* by *finite covering*.
(I, 9.1.13) On line 15 of p. 171, replace $p^{-1}(\mathscr{F})$ by $p^*(\mathscr{F})$.
(I, 9.5.11) On line 7 of p. 179, replace Y by Y'.
(I, 9.6.5) On line 17 from the bottom of p. 180, replace (0, 4.1.4) by (0, 4.1.3).
(I, 10.12.2) On line 14 from the bottom of p. 206, replace $(X_n)_{(S_n)}$ by $(X_n)_{(S_m)}$.

(I), Terminological index: On line 6 from the bottom of p. 219, replace 0, 4.1.4 by 0, 4.1.5. On lines 16, 17, and 18 from the bottom of p. 222, replace I, 2.1.7 by I, 2.1.8; on line 19 from the bottom of p. 222, replace I, 2.1.8 by I, 2.1.7. **III** | 218
(II, 1.5.2) On line 17 of p. 12, insert an f beside the left vertical arrow of the diagram.
(II, 4.2.7) On line 16 from the bottom of p. 75, replace (III, 2.1.14) by (III, 2.1.13).
(II, 5.2.1) On line 16 from the bottom of p. 97, replace $X - U$ by U. On line 8 from the bottom of p. 97, replace X' by X_f.
(II, 5.3.6) On line 16 of p. 100, replace (4.6.18) by (4.6.17).
(II, 6.2.7) On line 9 from the bottom of p. 116, replace chap. V by chap. IV.
(II, 6.6.5) On line 15 from the bottom of p. 132, replace chap. V by chap. IV.
(II, 7.4.12) On line 1 of p. 151, replace chap. V by chap. III.
(II, 8.1.4) On line 15 of p. 154, replace (III, 2.3.8) by (III, 2.3.7).
(II, 8.10.5) On line 17 from the bottom of p. 187, replace $g_{(j)x}$ by $g_{j(x)}$.
(0_{III}, 10.2.7) On line 1 of p. 20, replace u-adic by $\mathfrak{m}$-adic. On line 13 from the bottom of p. 20, replace k by k_i.
(0_{III}, 11.1.1) On line 18 of p. 23, replace $\text{Im}(d_r^{p+r,q-r+1})$ by $\text{Im}(d_r^{p-r,q+r-1})$.
(0_{III}, 11.8.6) On line 14 from the bottom of p. 45, replace the arrows $\to$ by $\rightsquigarrow$ (4 times).
(0_{III}, 12.3.4) On line 4 of p. 61, replace $\rightsquigarrow$ by $\to$.
(0_{III}, 13.2.4) On line 2 from the bottom of p. 67, replace $\varinjlim$ by $\varprojlim$ (2 times).
(III, 1.1.7) On line 15 of p. 85, replace A^r by $\bigwedge(A^r)$ (2 times).
(III, 1.2.4) On line 7 of p. 87, replace $\Gamma(U, \mathscr{F})$ by $\Gamma(U, \mathscr{O}_X)$ and replace $\mathscr{G}$ by $\mathscr{F}$. On line 8 of p. 87, replace $\text{H}^p(\mathfrak{U}, \mathscr{G})$ by $\text{H}^p(\mathfrak{U}, \mathscr{F})$.
(III, 2.5.3.1) On line 3 of p. 111, replace $n > 0$ by $n \geq 0$. On line 8 of p. 111, replace $-r \leq n \leq 0$ by $-r \leq n < 0$.
(III, 3.1.2) On line 13 of p. 116, replace $\mathscr{G}$ by $\mathscr{G} \in \mathbf{K}'$.
(III, 4.3.6) On line 7 of p. 133, replace $K' = K \otimes_A \widehat{A}$ by $K' \supset K \otimes_A \widehat{A}$. On line 16 of p. 133, replace $R(X)$ by $R(Y)$.
(III, 4.4.12) On line 10 of p. 138, replace chap. V by chap. IV.
(III, 4.6.6) On line 19 of p. 142, replace "chap. V" by "in a later section".

B) *Textual modifications*[1]

(**Err**$_{\text{III}}$, 1) In (0_{I}, 5.1.3), on line 18 of p. 45, replace "direct sum" by "finite direct sum".
(**Err**$_{\text{III}}$, 2) In (0_{I}, 5.4.3), on lines 8 to 4 from the bottom of p. 49, replace the text beginning "In the general case..." by the following:

In the general case where X is a ringed space such that $\mathscr{O}_x$ is a *local* ring for every $x \in X$, if $\mathscr{L}$ is an $\mathscr{O}_X$-Module *of finite type* for which there exists an $\mathscr{O}_X$-Module $\mathscr{F}$ such that $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{F}$ is *isomorphic to* $\mathscr{O}_X$, the preceding argument shows that, for every $x \in X$, $\mathscr{L}_x$ is isomorphic to $\mathscr{O}_x$. It follows that $\mathscr{L}$ is then *invertible*. Indeed, for every $x \in X$, let U be an open neighborhood of x such that $\mathscr{L}|U$ is generated by n sections s_i ($1 \leq i \leq n$) over U (5.2.1). We may suppose, for example, that $s = s_1$ is such that $s_x \neq 0$; and since $\mathscr{L}_x$ is isomorphic to $\mathscr{O}_x$, for each i there exists a section t_i of $\mathscr{O}_X$ over an open neighborhood $V_i \subset U$ of x such that $(t_i)_x s_x = (s_i)_x$. Consequently there is an open neighborhood $V \subset \bigcap_{i=1}^n V_i$ of x such that $t_i s = s_i$ in V; in other words, $\mathscr{L}|V$ is generated by the *single* section $s|V$. Moreover, if z is a section over an open subset $W \subset V$ of the kernel of the homomorphism $\mathscr{O}_X|V \to \mathscr{L}|V$ defined by s (5.1.1), then z_y annihilates $\mathscr{L}_y$ for every $y \in W$, so $z_y = 0$ by hypothesis and hence $z = 0$, which completes the proof of our assertion. Finally, consideration of the tensor product $\mathscr{L}^{-1} \otimes \mathscr{L} \otimes \mathscr{F}$ immediately shows that $\mathscr{F}$ is isomorphic to $\mathscr{L}^{-1}$. **III** | 219
(**Err**$_{\text{III}}$, 3) In (0_{I}, 7.2.4), one must impose the condition that the powers $\mathfrak{J}^n$ are *closed* ideals in the admissible ring A in order for the proposition to be correct. The proof given is incorrect, and the corrected statement follows from Bourbaki, *Top. gén.*, chap. III, 3rd ed., § 3, no. 5, cor. 1 of prop. 9. The $\mathfrak{J}^n$ must likewise be assumed *closed* in A in statements (0_{I}, 7.2.5) and (0_{I}, 7.2.6).
(**Err**$_{\text{III}}$, 4) After proposition (I, 2.2.5), add: with the notation of (I, 2.2.5), if $\mathfrak{J}$ is an ideal of A, we denote by $\widetilde{\mathfrak{J}\mathscr{F}}$ the $\mathscr{O}_X$-submodule $\mathfrak{J}\widetilde{\mathscr{F}}$ defined in (0, 4.3.5).
(**Err**$_{\text{III}}$, 5) In (I, 3.4.5), on line 1 from the bottom of p. 112, replace "*a field K*" by "*an algebraically closed field K*". On line 1 of p. 113, replace "*extension*" by "*algebraically closed extension*". On lines 1 to 4 of p. 113, replace the text beginning "K will be called..." by the following:

[1] To facilitate the search for references, modifications belonging to the errata list inserted in chapter N will henceforth be designated by **Err**$_{\text{N}}$ followed by a number.

For every point of X with values in a field K, K will be called the *field of values* of the corresponding point; and if x is the locality of this point, the latter is also said to be *localized at* x. This defines a map $X(K) \to X$ that associates to a point with values in K its locality.

On lines 7 and 16 of p. 113, delete "geometric"; on lines 10 and 13 of p. 113, delete "geometric". On lines 10 and 11 from the bottom of p. 115, delete "*geometric*".

(**Err**$_{\text{III}}$, 6) At the end of the proof of (I, 3.6.1), on line 12 from the bottom of p. 117, add: If we set $X' = X \times_Y \mathrm{Spec}(\mathscr{O}_y/\mathfrak{a}_y)$, then, for every point $x \in X'$, identified by p with a point of X, we have $\mathscr{O}_{X',x} = \mathscr{O}_{X,x}/\mathfrak{a}_y\mathscr{O}_{X,x}$. Since the question is local on X and Y, we may suppose that $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$; then $(B/\mathfrak{a}_y B)_x = B_x/\mathfrak{a}_y B_x$ by flatness (0, 1.3.2).

(**Err**$_{\text{III}}$, 7) In (I, 5.1.1), on line 3 from the bottom of p. 127, replace "quasi-coherent $\mathscr{O}_X$-Module" by "*quasi-coherent ideal of* $\mathscr{B}$".

(**Err**$_{\text{III}}$, 8) After (I, 5.1.10), on line 16 of p. 131, add:

Remark (5.1.11). — A slight adaptation of the argument used in (5.1.9) shows that the conclusion remains valid *without assuming a priori that* X *is a prescheme*, but **III** | 220

assuming only that $(X, \mathscr{O}_X)$ is a ringed space in local rings, that $\mathscr{J}$ is an ideal of $\mathscr{O}_X$ such that $\mathscr{J}^n = 0$, that the ringed space $X_0 = (X, \mathscr{O}_X/\mathscr{J})$ is an affine scheme, and finally that the ideals $\mathscr{J}^k/\mathscr{J}^{k+1}$ are quasi-coherent $\mathscr{O}_{X_0}$-Modules. It suffices to use (1.8.1) (cf. (**Err**$_{\text{II}}$)) instead of (2.2.4) in the argument.

(**Err**$_{\text{III}}$, 9) In (I, 5.3.1), on line 11 of p. 132, after "or Δ_X", insert "or Δ_ϕ if $\phi : X \to S$ is the structural morphism".

(**Err**$_{\text{III}}$, 10) In (I, 5.3.9), the proof is insufficient, because it does not prove that $\Delta_X(X)$ is locally closed in $X \times_S X$. To obtain a correct proof, it suffices to use (4.2.4, *a*)): for every $x \in X$ and every affine neighborhood U of x in X, $U \times_S U$ is an affine neighborhood of $\Delta_X(x)$; taking (5.3.16) into account (whose proof uses only definition (5.3.1.1)), we are reduced to proving (5.3.9) when $S = \mathrm{Spec}(B)$ and $X = \mathrm{Spec}(A)$ are affine schemes. It is then clear from (5.3.1.1) that Δ_X corresponds to the canonical homomorphism $A \otimes_B A \to A$ that sends $x \otimes y$ to xy; since this homomorphism is surjective, Δ_X is then a closed immersion (4.2.3), which completes the proof.

(**Err**$_{\text{III}}$, 11) In (I, 7.1.15) and (I, 7.1.16), on lines 7 and 15 from the bottom of p. 158, delete "*geometric*".

(**Err**$_{\text{III}}$, 12) Replace (I, 7.4.7) by: We extend (by abuse of language) the definitions of (7.4.1) to the case where X is a *reduced* prescheme every point of which admits an open neighborhood having only a *finite* number of irreducible components. It then follows from (7.3.4) and (7.4.6) that, for a quasi-coherent $\mathscr{O}_X$-Module of finite type $\mathscr{F}$, to say that $\mathscr{F}$ is a *torsion sheaf* is equivalent to saying that $\mathrm{Supp}(\mathscr{F})$ *contains no irreducible component of* X.

(**Err**$_{\text{III}}$, 13) In (I, 10.11.7), on lines 17 to 23 of p. 205, the end of the proof beginning "It remains to prove..." is unnecessary by (10.11.6), since the sheaves in question are coherent by (0, 5.3.5).

(**Err**$_{\text{III}}$, 14) In (II, 1.7.8), after line 10 from the bottom of p. 16, add: when $\mathscr{E} = \mathscr{O}_S^n$, we also write $\mathbf{V}_S^n$ in place of $\mathbf{V}(\mathscr{E})$; if moreover $S = \mathrm{Spec}(A)$ is affine, we write $\mathbf{V}_A^n$ in place of $\mathbf{V}_S^n$; we have $\mathbf{V}_A^n = \mathrm{Spec}(A[T_1, \ldots, T_n])$, where the T_i are indeterminates.

(**Err**$_{\text{III}}$, 15) In (II, 1.7.10), on line 10 of p. 17, delete "*geometric*"; on lines 12 and 16–17 of p. 17, delete "geometric".

(**Err**$_{\text{III}}$, 16) In (II, 1.7.12), add: We set similarly:

$$\mathbf{V}(\mathscr{O}_S^n) = \mathbf{V}_S^n = S[T_1, \ldots, T_n].$$

(**Err**$_{\text{III}}$, 17) In (II, 4.2.6), on line 7 of p. 75, delete "*geometric*"; on line 8 of p. 75, delete "*geometric*".

(**Err**$_{\text{III}}$, 18) In (II, 4.6.13), on line 7 of p. 92, the argument must be made more precise, because with the notation of (4.4.10) one must prove here that the immersion Γ_f is *quasi-compact*, in order to apply (i bis). By (4.6.4), to prove that $\mathscr{L}$ is ample relative to f, we may restrict to the case where Y is affine. Observe also that in each of the hypotheses of (v), f is quasi-compact (I, 6.6.4). On the other hand, if g is separated, Γ_f **III** | 221 is a closed immersion (I, 5.4.3), hence quasi-compact (I, 6.6.4). If, on the contrary, X is locally Noetherian, then since Y is affine and f is quasi-compact, the underlying space of X is quasi-compact, hence Noetherian; and (I, 6.6.4) may again be applied to prove that Γ_f is quasi-compact.

(**Err**$_{\text{III}}$, 19) The statement of (II, 5.2.2) is incorrect: conditions *b*), *c*), and *c*′) are not local on Y when it is not known whether, for an open subset U of Y, a quasi-coherent $(\mathscr{O}_X|f^{-1}(U))$-Module is the restriction to $f^{-1}(U)$ of a quasi-coherent $\mathscr{O}_X$-Module. Thus, on line 16 of p. 98, replace "*Let* $f : X \to Y$ *be a separated quasi-compact morphism*" by: "*Let* X *and* Y *be two preschemes such that* X *is a scheme or the underlying space of* X *is locally Noetherian, and let* $f : X \to Y$ *be a quasi-compact morphism.*" In the proof, observe that the hypotheses imply that f is *separated* when X is assumed to be a scheme, by (I, 5.5.5 and 5.5.8); hence (I, 9.2.2) may be applied in both cases.

(**Err**$_{\text{III}}$, 20) In (II, 6.2.3), after line 18 from the bottom of p. 115, add:

A morphism $f : X \to Y$ *is said to be quasi-finite at a point* $x \in X$ *if there exist an affine open neighborhood* V *of* $y = f(x)$ *and an affine open neighborhood* U *of* x *such that* $f(U) \subset V$ *and the morphism* $U \to V$ *induced*

by restricting f is quasi-finite. A morphism $f : X \to Y$ is said to be locally quasi-finite if it is quasi-finite at every point of X.

(**Err**$_{\mathrm{III}}$, 21) In (II, 6.4.3), the proof is incorrect, since the elements of a finite-type A-submodule of K are not necessarily integral over A. Replace the last 9 lines of p. 121 by the following text: Since we may suppose that E is torsion-free, hence faithful, E is also a faithful $A[u]$-module (A being embedded in the ring of endomorphisms of E). Since E is an A-module of finite type, it follows that u is *integral* over A (Bourbaki, *Alg. comm.*, chap. V, § 1, no. 1, lemma 1). We immediately conclude that the eigenvalues of $u \otimes 1$ (in an algebraic closure of K) are elements *integral* over A, and consequently so are the $\sigma_i(u)$.

(**Err**$_{\mathrm{III}}$, 22) In (0_{III}, 9.1.1), on line 18 from the bottom of p. 12, delete "open".

(**Err**$_{\mathrm{III}}$, 23) In (0_{III}, 11.7.3), on line 10 of p. 43, after C'', add: such that for every projective object P of C' (resp. every projective object P' of C''), the functor $A' \rightsquigarrow T(P, A')$ (resp. $A \rightsquigarrow T(A, P')$) is exact in C' (resp. C).

(**Err**$_{\mathrm{III}}$, 24) The statement of (0_{III}, 13.7.7) is inaccurate and must be modified as follows: on line 6 of p. 78, delete "and $(\mathrm{R}^{n+1}T(A_k))_{k\in\mathbf{Z}}$"; on line 7 of p. 78, replace "$\mathrm{R}'^nT(A)$ is an S-module of finite type" by "$\mathrm{R}'^nT(A)$ and $\mathrm{R}'^{n+1}T(A)$ are S-modules of finite type"; on lines 12–13 of p. 78, delete "and $p + q = n + 1$". In the proof, on line 23 of p. 78, delete "and for $n + 1$".

(**Err**$_{\mathrm{III}}$, 25) In (III, 1.4.15), on line 6 from the bottom of p. 92, replace "*of finite type*" by "*quasi-compact*".

(**Err**$_{\mathrm{III}}$, 26) In (III, 2.2.4), on line 1 of p. 102, replace "*the supports of $\mathscr{F}$ and $\mathscr{H}$ are proper over Y*" by "*the support of* $\mathrm{Im}(\mathscr{F} \to \mathscr{G})$ *is proper over Y*". In the proof, replace the text of lines 10 to 16, beginning "This being so…", by: **III** | 222

Set $\mathscr{G}_1 = \mathrm{Im}(\mathscr{F} \to \mathscr{G}) = \mathrm{Ker}(\mathscr{G} \to \mathscr{H})$, which is coherent. Since we have the exact sequence $0 \to \mathscr{G}_1(n) \to \mathscr{G}(n) \to \mathscr{H}(n)$ and the functor f_* is left exact, the sequence $0 \to f_*(\mathscr{G}_1(n)) \to f_*(\mathscr{G}(n)) \to f_*(\mathscr{H}(n))$ is exact for every n; it therefore suffices to show that, for sufficiently large n, the sequence $f_*(\mathscr{F}(n)) \to f_*(\mathscr{G}_1(n)) \to 0$ is exact. In other words, we may restrict to the case where $\mathscr{H} = 0$ and the support of $\mathscr{G}$ is proper over Y, hence *closed in Z* (II, 5.4.10). Consequently $\mathscr{G}' = i_*(\mathscr{G})$ is a coherent $\mathscr{O}_Z$-Module such that $\mathscr{G}'|X = \mathscr{G}$. We know from (I, 9.4.3) that there exists a coherent $\mathscr{O}_Z$-Module $\mathscr{F}'$ such that $\mathscr{F}'|X = \mathscr{F}$. Moreover, since X is open in Z and $\mathrm{Supp}(\mathscr{G}) \subset X$ is closed in Z, it is immediate that one defines a surjective homomorphism $u' : \mathscr{F}' \to \mathscr{G}'$ of sheaves such that $u'|X$ is the given surjective homomorphism $u : \mathscr{F} \to \mathscr{G}$, by taking $u'|U = 0$ for every open subset U of Z not meeting $\mathrm{Supp}(\mathscr{G})$ and $u'|U = u|U$ for every open subset $U \subset X$. This being so, as at the beginning of the proof of (2.2.2), we see that we may reduce to the case where Y is affine; it is therefore enough to show that, for sufficiently large n, the homomorphism $\Gamma(u) : \Gamma(X, \mathscr{F}(n)) \to \Gamma(X, \mathscr{G}(n))$ is surjective. We have the commutative diagram

$$\begin{array}{ccc} \Gamma(Z, \mathscr{F}'(n)) & \xrightarrow{\Gamma(u')} & \Gamma(Z, \mathscr{G}'(n)) \\ \downarrow{\scriptstyle v} & & \downarrow{\scriptstyle w} \\ \Gamma(X, \mathscr{F}(n)) & \xrightarrow[\Gamma(u)]{} & \Gamma(X, \mathscr{G}(n)) \end{array}$$

where v and w are the restriction homomorphisms. The definition of $\mathscr{G}'$ moreover shows that w is *bijective*; and by (2.2.3), $\Gamma(u')$ is *surjective* for sufficiently large n, so the same is true of $\Gamma(u)$.

(**Err**$_{\mathrm{III}}$, 27) In (III, 2.2.5), after line 7 from the bottom of p. 102, add:

(iii) Under the hypotheses of (2.2.4) concerning X, Y, f, and $\mathscr{L}$, it is immediate that if $\mathscr{H}$ is a coherent $\mathscr{O}_X$-Module whose support is *proper over Y*, an argument analogous to that of (2.2.4) shows that there exists an integer N such that, for $n \geq N$, $\mathrm{R}^i f_*(\mathscr{H}(n)) = 0$ for every $i > 0$. It follows from the exact cohomology sequence that if $u : \mathscr{F} \to \mathscr{G}$ is a homomorphism of coherent $\mathscr{O}_X$-Modules such that $\mathrm{Ker}(u)$ and $\mathrm{Coker}(u)$ have supports *proper over Y*, then there exists N such that for $n \geq N$, the corresponding homomorphism $\mathrm{R}^i f_*(\mathscr{F}(n)) \to \mathrm{R}^i f_*(\mathscr{G}(n))$ is *bijective* for every $i > 0$.

(**Err**$_{\mathrm{III}}$, 28) In (III, 4.4.9), on line 17 from the bottom of p. 137, replace "*Let Y be a locally Noetherian integral prescheme*" by "*Let X and Y be two locally Noetherian integral preschemes*"; on line 16 from the bottom of p. 137, replace "*of finite type*" by "*locally of finite type*". The proof is essentially unchanged, since $f^{-1}(y)$ is locally of finite type over $k(y)$ and is therefore still discrete.

(**Err**$_{\mathrm{III}}$, 29) In (I, 9.3.4), on line 19 from the bottom of p. 173, replace "*Noetherian*" by "*quasi-compact*"; on lines 18 and 19 from the bottom of p. 173, replace "*coherent*" by "*quasi-coherent of finite type*". In the proof, we reduce to the case where **III** | 223

$X = \mathrm{Spec}(A)$ and $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type, and $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of finite type in A; the rest of the argument is then unchanged.

(**Err**$_{\mathrm{III}}$, 30) In (I, 9.3.5), replace lines 1 to 7 from the bottom of p. 173 by the following text:

Proposition (9.3.5). — *Let X be a prescheme and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite type. Then there exist a closed subprescheme Y of X, whose underlying space is equal to* $\mathrm{Supp}(\mathscr{F})$*, and a quasi-coherent $\mathscr{O}_Y$-Module of finite type $\mathscr{G}$ such that, if $j : Y \to X$ is the canonical injection, $\mathscr{F}$ is isomorphic to $j_*(\mathscr{G})$.*

It will suffice to show that the ideal $\mathscr{J}$ of $\mathscr{O}_X$, *annihilator of* $\mathscr{F}$, is *quasi-coherent*. We then take for Y the closed subprescheme of X defined by $\mathscr{J}$ (I, 4.1.2), and since $\mathscr{J}\mathscr{F} = 0$, $\mathscr{F}$ is an $(\mathscr{O}_X/\mathscr{J})$-Module and we answer the question by taking $\mathscr{G} = j^*(\mathscr{F})$. To see that $\mathscr{J}$ is quasi-coherent, we may (the question being local) restrict to the case where $X = \mathrm{Spec}(A)$ and $\mathscr{F} = \widetilde{M}$, where M is an A-module generated by finitely many elements x_i $(1 \leq i \leq r)$. The ideal $\mathscr{J}$ is then the intersection of the annihilators of the x_i. But the annihilator of x_i is the kernel of the homomorphism $\mathscr{O}_X \to \mathscr{F}$ corresponding to the homomorphism $s \mapsto sx_i$ from A to M; it is therefore indeed a quasi-coherent ideal (I, 4.1.1), and every finite intersection of such ideals is also a quasi-coherent ideal (I, 1.3.10).